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{
"data":{
"id":"1",
"type":"Ticket",
"attributes":{
"subject":"Problem with Ticket",
"description":"Hello,\r\n\r\nI have a problem with my ticket.\r\n\r\nCan you please check it?\r\n\r\nThanks\r\n\r\n\r\n
```

Mr. Gabriel Sanguinetti
Director de Ingeniería de Ventas Para Norteamérica

Mr. Juan Gabriel
Mr. Gabriel Sanguinetti
Mr. Juan Gabriel
Mr. Gabriel
Mr. Juan Gabriel
Mr. Gabriel
Mr. Juan Gabriel
Mr. Gabriel

```
{
  "_id" : ObjectId("[illegible]"),
  "_class" : "com.[illegible].[illegible]",
  "[illegible]" : [illegible],
  "[illegible]" : [illegible],
  "[illegible]" : [illegible],
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"_id":"0000-00-00",
"_rev":"1-00000000000000000000000000000000",
"_attachments":{},
"_conflicts":["0000-00-00T00:00:00.000+00:00"],
"_deleted_conflicts":["0000-00-00T00:00:00.000+00:00"],
"_local_seq":"0000-00-00T00:00:00.000+00:00",
"_revisions":{"start":1,"ids":["00000000000000000000000000000000"]},
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"_rev_tree":[["1-00000000000000000000000000000000",{"status":"available","children":{}}]],
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"_attachment_data":{"0000-00-00":"AAAA
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Your cart is empty

Home / Cart / Checkout

Home

Cart

Checkout

Home / Cart / Checkout

Home

Cart

Checkout

Home / Cart / Checkout

Home

Cart

Checkout

Generic Medical Consent

**Author: Dr. Ashish Gupta, Dr. Ankur Gupta, Dr. Ankur Gupta, Dr. Ankur

```
<!-- ~~~
```

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<div class="page">

<div>
<span id="1" />
<span id="2" />
<span id="3" />
<span id="4" />
<span id="5" />
<span id="6" />
<span id="7" />
<span id="8" />
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<span id="10" />
<span id="11" />
<span id="12" />
<span id="13" />
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<PLOT_SPECIFICATION
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  VERSION "1.0"
  DATE "Wed Jan 13 00:00:00 2010"
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  TERMINAL "unknown"
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>

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  ALIGNMENT "CENTER"
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>
```

Gnuplot is a command-line driven interactive function plotting program. Originally intended as a means to stimulate students' interest in mathematics, it has been used in scientific fields for over a decade. Gnuplot is capable of plotting 2D functions and data in various styles, and 3D functions and data in the form of surfaces and contour maps. Gnuplot can output to a variety of formats, including PostScript, PNG, and SVG.

Gnuplot is a portable command-line driven interactive data and function plotting utility for UNIX, IBM OS/2, MS Windows, DOS, and many other platforms. It was originally intended for students to explore mathematical functions and data in a friendly way, but has been used for many scientific and engineering applications since then. Gnuplot supports many types of plots in either 2D or 3D. It can plot functions and data read from ASCII files. Gnuplot can output to a variety of formats, including PNG, PostScript, and SVG.

```
<?xml version="1.0"?>
```

```
<div class="page">

<div>
<div>
<div>
```

Ad Clash Report

Ad Clash Report

Ad Clash Report

{
"id": "1707.07187",
"title": "Adversarial Examples for Semantic Segmentation and Object Detection",
"abstract": "Numerous works have recently shown that state-of-the-art classifiers are vulnerable to adversarial perturbations. We extend the study of adversarial examples to more complex vision tasks, namely semantic segmentation and object detection. We show that adversarial examples can also be easily generated for these tasks, and that they transfer across datasets and models. We also show that adversarial training can be used to defend against such examples, and that it leads to improved robustness.",
"authors": "Cihang Xie, Jianyu Wang, Zhishuai Zhang, Yuyin Zhou, Lingxi Xie, Alan Yuille",
"subjects": "Computer Vision and Pattern Recognition (cs.CV); Machine Learning (cs.LG); Machine Learning (stat.ML)",
"comments": "Technical report. Revised version of arXiv:1703.08603 and arXiv:1704.08784",
"doi": "10.48550/arXiv.1707.07187",
"licenses": "Creative Commons Attribution Non-Commercial Share Alike 4.0"
}

Adversarial Examples for Semantic Segmentation and Object Detection

Cihang Xie[1], Jianyu Wang[2], Zhishuai Zhang[1], Yuyin Zhou[1], Lingxi Xie[1], Alan Yuille[1]
[1]Johns Hopkins University, [2]Huazhong University of Science and Technology
{cihang.xie, zhishuai.zhang, yuyin.zhou, lingxi.xie, alan.yuille@jhu.edu, jianyuwang@hust.edu.cn}

Numerous works have recently shown that state-of-the-art classifiers are vulnerable to adversarial perturbations. We extend the study of adversarial examples to more complex vision tasks, namely semantic segmentation and object detection. We show that adversarial examples can also be easily generated for these tasks, and that they transfer across datasets and models. We also show that adversarial training can be used to defend against such examples, and that it leads to improved robustness.

Intro

Cross-Sectional Study of Dietary Patterns and Anemia Among Adolescent Girls in Rural Tribal Hamlets of Gadchiroli, Maharashtra

Dr. Ananya Bhattacharjee, Dr. Sudhir Kumar Tripathi, Dr. Meenakshi Rao, Dr. Vikram Singh Rathore, Dr. Nandini Sen, Dr. Ashok Kumar Jain, Dr. Sunita Devi Yadav, Dr. Rajesh Kumar Mishra, Dr. Pooja Sharma, Dr. Anil Kumar Verma, Dr. Sanjay Kumar Gupta, Dr. Ramesh Chandra Pandey, Dr. Kavita Singh, Dr. Mohan Lal Sharma, Dr. Geeta Devi Kushwaha, Dr. Vinod Kumar Singh, Dr. Ashok Kumar Mishra

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Cardi B Wallpaper Iphone 6

Home / Cardi B Wallpaper Iphone 6

Home
Cardi B Wallpaper Iphone 6 [2019] Wallpaper Desktop - Tumblr Wallpaper
Home
Home
Home
Home

Cardi B Wallpaper Iphone 6

Cardi B Wallpaper Iphone 6 - Cardi B Wallpaper -

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Home
Home

ISSN : 2454-9150 Vol-11, Issue-01,
Jan-2021 Impact Factor: 4.101 (SJIF) Peer Reviewed Journal International Advanced Research in Engineering and Technology
Page 1

Design of Electrical Generator operated on Low speed Wind Mill for Village Electrification

Prof. M. Ashok Kumar, Prof. Ch. Ramesh, Prof. Ch. Ramesh, Prof. Ch. Ramesh

```
<?xml version="1.0" ?>
<rss version="2.0" xmlns:dc="http://purl.org/dc/elements/1.1/" xmlns:content="http://purl.org/rss/1.0/modules/content/" xmlns:wfw="http://wellformedweb.org/CommentAPI/" xmlns:atom="http://www.w3.org/2005/Atom" xmlns:sy="http://purl.org/rss/1.0/modules/syndication/" xmlns:slash="http://purl.org/rss/1.0/modules/slash/" xmlns:media="http://search.yahoo.com/mrss/" xmlns:gr="http://www.google.com/schemas/reader/" xmlns:thr="http://purl.org/rss/1.0/modules/threading/" xmlns:feedburner="http://rssnamespace.org/feedburner/extension/1.0" xmlns:feedburner:origLink="http://www.nytimes.com/2022/12/07/opinion/sunday/ai-chatbot-chatgpt.html" xmlns:media:thumbnail="http://www.google.com/schemas/reader/media/thumbnail/1.0" xmlns:media:keywords="http://www.google.com/schemas/reader/media/keywords/1.0" xmlns:media:text="http://www.google.com/schemas/reader/media/text/1.0" xmlns:media:player="http://www.google.com/schemas/reader/media/player/1.0" xmlns:media:restriction="http://www.google.com/schemas/reader/media/restriction/1.0" xmlns:media:credit="http://www.google.com/schemas/reader/media/credit/1.0" xmlns:media:category="http://www.google.com/schemas/reader/media/category/1.0" xmlns:media:rating="http://www.google.com/schemas/reader/media/rating/1.0" xmlns:media:hash="http://www.google.com/schemas/reader/media/hash/1.0" xmlns:media:title="http://www.google.com/schemas/reader/media/title/1.0" xmlns:media:publication="http://www.google.com/schemas/reader/media/publication/1.0" xmlns:media:location="http://www.google.com/schemas/reader/media/location
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NO. 7.5.2.7 **MAGNIFICENCE**

Sub-section 7.5.2.7 of Section 7.5 Magnificence

Sub

Sub-section 7.5.2.7 of Section 7.5 Magnificence

Section

Section 7.5 Magnificence of Chapter 7 Sacrifice and Magnificence

Chapter

Chapter 7 Sacrifice and Magnificence of Book 7.5 Sacrifice and Magnificence

Book

Book 7.5 Sacrifice and Magnificence of Volume 7 Sacrifice and Magnificence

Volume

Volume 7 Sacrifice and Magnificence of Part 7 Sacrifice and Magnificence

Part

Part 7 Sacrifice and Magnificence of Section 7.5 Sacrifice and Magnificence

Section

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Sub

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Sub-sub-sub-section 7.5.2.7 Magnificence of Sub-section 7.5.2 Sacrifice and Magnificence

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Sub-sub-sub-sub-section 7.5.2.7 Magnificence of Sub-section 7.5.2 Sacrifice and Magnificence

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Sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-section 7.5.2.7 Magnificence of Sub-section 7.5.2 Sacrifice and Magnificence

Sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-section

Sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-section 7.5.2.7 Magnificence of Sub-section 7.5.2 Sacrifice and Magnificence

**Sub-

```
<?xml version="1.0" encoding="UTF-8"?>
<rss version="2.0" xmlns:dc="http://purl.org/dc/elements/1.1/" xmlns:sy="http://purl.org/rss/1.0/modules/syndication/" xmlns:slash="http://purl.org/rss/1.0/modules/slash/" xmlns:rdfs="http://www.w3.org/2000/01/rdf-schema#" xmlns:wfw="http://wellformedweb.org/CommentAPI/" xmlns:dcterms="http://purl.org/dc/terms/" xmlns:media="http://search.yahoo.com/mrss/" xmlns:atom="http://www.w3.org/2005/Atom" xmlns:opensearch="http://a9.com/-/spec/opensearchrss/1.0/" xmlns:geo="http://www.w3.org/2003/01/geo/wgs84_pos#" xmlns:georss="http://www.georss.org/georss" xmlns:gml="http://www.opengis.net/gml" xmlns:osm="http://www.openstreetmap.org/dtd/0.6" xmlns:fb="http://www.facebook.com/2008/fbml" xmlns:content="http://purl.org/rss/1.0/modules/content/" xmlns:wfw="http://wellformedweb.org/CommentAPI/" xmlns:dc="http://purl.org/dc/elements/1.1/" xmlns:atom="http://www.w3.org/2005/Atom" xmlns:sy="http://purl.org/rss/1.0/modules/syndication/" xmlns:slash="http://purl.org/rss/1.0/modules/slash/" xmlns:media="http://search.yahoo.com/mrss/
```

ISSN 1337-7027
Year VIII
Number 4
http://www.kvasy.sk
Print version: 1336-8672

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{
  "name": "Sample",
  "version": "0.0.0",
  "description": "",
  "keywords": [],
  "author": "",
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    "@ffmpeg-installer/ffmpeg": "1.1.0",
    "@ffmpeg-installer/linux-arm": "1.1.0",
    "@ffmpeg-installer/linux-arm64": "1.1.0",
    "@ffmpeg-installer/linux-ia32": "1.1.0",
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    "@ffmpeg-installer/mac": "1.1.0",
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Project Gutenberg's Alice's Adventures in Wonderland, by Lewis Carroll
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Title: Alice's Adventures in
Wonderland, by Lewis Carroll
Release Date: [EBook #11]
Last Updated: [July 14, 2011]
Language: [English]
Character set encoding: [UTF-8]
Produced by: [Arthur DiBianca and David Widger]
Source: [Gutenberg]

```
<?xml version="1.0" encoding="UTF-8"?>
<?xml-stylesheet href="chrome://global/skin/" type="text/xsl"?>
<!--
  This file contains the default icons for Firefox.
  -->
<svg
  xmlns="http://www.w3.org/2000/svg"
  xmlns:xlink="http://www.w3.org/1999/xlink"
  xmlns:rdf="http://www.w3.org/22-rdf-syntax-ns#"
  xmlns:cc="http://creativecommons.org/ns#"
  xmlns:dc="http://purl.org/dc/elements/1.1/"
  xmlns:sodipodi="http://sodipodi.sourceforge.net/DTD/sodipodi-0.dtd"
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  id="svg"
  viewBox="0 0 16 16"
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  width="16px">
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Author Affiliation:
Author Email:
Abstract:
Abstract Continued:
Abstract Continued:
Abstract Continued

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IJCRT | Volume 12, Issue 7 July 2024 | www.ijcrt.org | d615

www.ijcrt.org © 2024 IJCRT | Volume 12, Issue 7 July 2024 | ISSN: 2320-2882
d615 | Page 5 Conclusion The Ultrasonic Assisted Extraction (UAE) of Ginger Oleoresin from Ginger Rhizome Powder using Ethanol as solvent and optimized UAE. This study concluded following points.
1. Ginger Oleoresin is a valuable natural product.
2.

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Configure a Layer3 Subinterface on the GUI
Configure a Layer3 Subinterface on the CLI
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Verify Layer3 Subinterface Configuration on the CLI
Related Topics

Sub-Interface - Layer3 Subinterface

Palo Alto Networks firewalls support Layer 3 routing on subinterfaces to accommodate multiple virtual routers and security zones on a single physical interface. The ability to define subinterfaces is useful when you want to route traffic for multiple virtual routers or security zones through a single physical interface. Each subinterface can be assigned to a different virtual router and security zone. Each subinterface can be configured with its own IP address, static routes, RIP, OSPF, BGP, or multicast routing protocol. Subinterfaces are also useful if you have a firewall that is connected to a switch that is configured for multiple VLANs. You can define a subinterface for each VLAN and assign traffic from each VLAN to a different virtual router and security zone.

Configure a Layer3 Subinterface on the GUI

Step 1. Define the parent interface. Select Network > Interfaces and click the interface name. Select Interface Type Layer3 and configure the IPv4, IPv6 or both addresses, Management Profile, MTU, Link Speed, Link Duplex, and Link State as needed. Click OK.

Step 2. Define the subinterface. Select Network > Interfaces and click Add Subinterface. Enter the Interface Name in the format eth<slot>.<port_number>. Enter the Tag (VLAN ID) and Virtual Router. Select the Security Zone and configure the IPv4, IPv6 or both addresses, Management Profile, MTU, Link Speed, Link Duplex, and Link State as needed. Click OK.

Step 3. Commit the changes. Click Commit.

Verify Layer3 Subinterface Configuration on the GUI

Step 1. Verify the parent interface. Select Network > Interfaces and click the interface name. Select Interface Type Layer3 and configure the IPv4, IPv6 or both addresses, Management Profile, MTU, Link Speed, Link Duplex, and Link State as needed. Click OK.

Step 2. Verify the subinterface. Select Network > Interfaces and click Add Subinterface. Enter the Interface Name in the format eth<slot>.<port_number>. Enter the Tag (VLAN ID) and Virtual Router. Select the Security Zone and configure the IPv4, IPv6 or both addresses, Management Profile, MTU, Link Speed, Link Duplex, and Link State as needed. Click OK.

Step 3. Commit the changes. Click Commit.

Configure a Layer3 Subinterface on the CLI

```
set interfaces ethernet <slot>/<port_number>
set interfaces ethernet <slot>/<port_number> mtu <mtu_size>
set interfaces ethernet <slot>/<port_number> link-speed <speed_value>
set interfaces ethernet <slot>/<port_number> link-duplex <duplex_value>
set interfaces ethernet <slot>/<port_number> link-state <state_value>
set interfaces ethernet <slot>/<port_number> layer3 ip <ipv4_address>/<mask_length>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 <ipv6_address>/<mask_length>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 enable-duplicate-address-detection yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery router-advertisement enable yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery router-advertisement hop-limit <hop_limit_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery router-advertisement lifetime <lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery router-advertisement reachable-time <reachable_time_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery router-advertisement retransmit-time <retransmit_time_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery router-advertisement prefix <ipv6_prefix>/<mask_length> on-link yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery router-advertisement prefix <ipv6_prefix>/<mask_length> autonomous yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery router-advertisement prefix <ipv6_prefix>/<mask_length> preferred-lifetime <preferred_lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery router-advertisement prefix <ipv6_prefix>/<mask_length> valid-lifetime <valid_lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery router-advertisement dns-server <dns_server_address_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery router-advertisement dns-suffix <dns_suffix_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery router-advertisement enable yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery neighbor <ipv6_address_value> <mac_address_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery duplicate-address-detection clone-limit <clone_limit_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery duplicate-address-detection freeze-interface-id yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type prefix-only
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address identifier <identifier_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address identifier <identifier_value> eui-64 yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address identifier <identifier_value> randomize yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address lifetime <lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address lifetime <lifetime_value> no-expiry yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address lifetime <lifetime_value> on-link yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address lifetime <lifetime_value> preferred yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address lifetime <lifetime_value> autonomous yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address lifetime <lifetime_value> preferred-lifetime <preferred_lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address lifetime <lifetime_value> valid-lifetime <valid_lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool yes
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set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> preferred yes
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set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude identifier <identifier_value> randomize yes
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set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> on-link yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> preferred yes
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set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> autonomous yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> preferred-lifetime <preferred_lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> valid-lifetime <valid_lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude identifier <identifier_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude identifier <identifier_value> eui-64 yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude identifier <identifier_value> randomize yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> no-expiry yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> on-link yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> preferred yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> autonomous yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> preferred-lifetime <preferred_lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> valid-lifetime <valid_lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 
```

Regular Expressions Cheat Sheet

Regular Expressions Cheat Sheet - Quick Reference Guide for Pattern Matching

Basic Syntax

Pattern	Description
`.`	Any character except newline
`^`	Start of string/line
`$`	End of string/line
`\d`	Digit (0-9)
`\D`	Non-digit
`\w`	Word character (a-z, A-Z, 0-9, _)
`\W`	Non-word character
`\s`	Whitespace (space, tab, newline)
`\S`	Non-whitespace
`\b`	Word boundary
`\B`	Non-word boundary
`[abc]`	Character set: a, b, or c
`[^abc]`	Negated set: not a, b, or c
`[a-z]`	Range: a through z
`[a-z0-9]`	Multiple ranges: a-z or 0-9
`\n`	Newline
`\t`	Tab
`\r`	Carriage return
`\\`	Literal backslash

Quantifiers

Pattern	Description
`*`	0 or more
`+`	1 or more
`?`	0 or 1 (optional)
`{n}`	Exactly n times
`{n,}`	n or more times
`{n,m}`	Between n and m times
`*?`	Lazy 0 or more
`+?`	Lazy 1 or more
`??`	Lazy 0 or 1
`{n,m}?`	Lazy between n and m times

Groups and Alternation

Pattern	Description
`(abc)`	Capture group
`(?:abc)`	Non-capturing group
`(?P<name>abc)`	Named capture group
`\1`	Backreference to group 1
`\2`	Backreference to group 2
`\k<name>`	Backreference to named group
`a|b`	Alternation: a or b
`(?=abc)`	Positive lookahead: followed by abc
`(?!abc)`	Negative lookahead: not followed by abc
`(?<=abc)`	Positive lookbehind: preceded by abc
`(?<!abc)`	Negative lookbehind: not preceded by abc

Common Patterns

Pattern	Description
`[0-9]+`	Integer
`[0-9]*\.[0-9]+`	Decimal number
`[a-z]+`	Lowercase letters
`[A-Z]+`	Uppercase letters
`[a-zA-Z]+`	Letters
`[a-zA-Z0-9]+`	Alphanumeric
`[a-zA-Z0-9_]+`	Word characters
`[a-zA-Z0-9_-]+`	Slug
`[a-zA-Z0-9.-]+`	Domain name
`[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}`	Simple domain
`[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}`	Email address
`https?://[a-zA-Z0-9.-]+(:[0-9]+)?(/[a-zA-Z0-9._~:/?#[]@!$&'()*+,;=%-]*)?`	URL
`[0-9]{1,3}(\.[0-9]{1,3}){3}`	IP address
`[0-9]{4}-[0-9]{2}-[0-9]{2}`	Date (YYYY-MM-DD)
`[0-9]{2}:[0-9]{2}:[0-9]{2}`	Time (HH:MM:SS)
`[0-9]{4}-[0-9]{2}-[0-9]{2}T[0-9]{2}:[0-9]{2}:[0-9]{2}(Z|[+-][0-9]{2}:[0-9]{2})`	ISO 8601
`[0-9]{1,2}/[0-9]{1,2}/[0-9]{4}`	Date (MM/DD/YYYY)
`[0-9]{1,2}-[0-9]{1,2}-[0-9]{4}`	Date (MM-DD-YYYY)
`[0-9]{1,2}:[0-9]{1,2}(:[0-9]{1,2})?(\s?[AP]M)?`	Time (12-hour)
`[0-9]{1,2}:[0-9]{1,2}(:[0-9]{1,2})?`	Time (24-hour)
`[0-9]{1,2}(st|nd|rd|th)`	Ordinal
`[0-9]{1,2}(st|nd|rd|th)\s?(Jan|Feb|Mar|Apr|May|Jun|Jul|Aug|Sep|Oct|Nov|Dec)`	Ordinal date
`[0-9]{1,2}\s?(Jan|Feb|Mar|Apr|May|Jun|Jul|Aug|Sep|Oct|Nov|Dec)\s?[0-9]{4}`	Date (DD Mon YYYY)
`[0-9]{1,2}\s?(Jan|Feb|Mar|Apr|May|Jun|Jul|Aug|Sep|Oct|Nov|Dec)`	Date (DD Mon)
`(Jan|Feb|Mar|Apr|May|Jun|Jul|Aug|Sep|Oct|Nov|Dec)\s?[0-9]{1,2}`	Date (Mon DD)
`(Jan|Feb|Mar|Apr|May|Jun|Jul|Aug|Sep|Oct|Nov|Dec)\s?[0-9]{4}`	Date (Mon YYYY)
`[0-9]{1,2}(st|nd|rd|th)\s?(Jan|Feb|Mar|Apr|May|Jun|Jul|Aug|Sep|Oct|Nov|Dec)\s?[0-9]{4}`	Date (DD Mon YYYY)
`[0-9]{1,2}\s?(Jan|Feb|Mar|Apr|May|Jun|Jul|Aug|Sep|Oct|Nov|Dec)\s?[0-9]{4}\s?[0-9]{1,2}:[0-9]{1,2}(:[0-9]{1,2})?(\s?[AP]M)?`	Date (DD Mon YYYY HH:MM:SS)
`[0-9]{1,2}(st|nd|rd|th)\s?(Jan|Feb|Mar|Apr|May|Jun|Jul|Aug|Sep|Oct|Nov|Dec)\s?[0-9]{4}\s?[0-9]{1,2}:[0-9]{1,2}(:[0-9]{1,2})?(\s?[AP]M)?`	Date (DD Mon YYYY HH:MM:SS

DOI: 10.1007/s00401-
Received: 5 July 2017
Revised: 14 November 2017
Accepted: 15 November 2017
Online First: 27 November 2017

```
<?xml version="1.0" encoding="UTF-8" ?>
<rss version="2.0" xmlns:dc="http://purl.org/dc/elements/1.1/" xmlns:sy="http://purl.org/rss/1.0/modules/syndication/" xmlns:slash="http://purl.org/rss/1.0/modules/slash/" xmlns:wfw="http://wellformedweb.org/CommentAPI/" xmlns:dc="http://purl.org/dc/elements/1.1/" xmlns:atom="http://www.w3.org/2005/Atom" xmlns:sy="http://purl.org/rss/1.0/modules/syndication/" xmlns:slash="http://purl.org/rss/1.0/modules/slash/" xmlns:media="http://search.yahoo.com/mrss/" xmlns:content="http://purl.org/rss/1.0/modules/content
```

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Jan Lukas Bumberscheck
Department of Food and Agricultural Product Law and Food Safety and Consumer Protection, Universität Hohenheim, Germany
E-mail: jan.bumberscheck@uni-hohenheim.de
Tel.: +49 711 459 22 2 07
Fax: +49 711 459 22 2 07
Prof. Dr. Dr. h.c. Ulrich Hagen
Department of Food and Agricultural Product Law and Food Safety and Consumer Protection, Universität Hohenheim, Germany
E-mail: ulrich.hagen@uni-hohenheim.de

Section 1.10.32 of "de Finibus Bonorum et Malorum", written by Cicero in 45 BC

"Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium, totam rem aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae vitae dicta sunt explicabo. Nemo enim ipsam voluptatem quia voluptas sit aspernatur aut odit aut fugit, sed quia consequuntur magni dolores eos qui ratione voluptatem sequi nesciunt. Neque porro quisquam est, qui dolorem ipsum quia dolor sit amet, consectetur, adipisci velit, sed quia non numquam eius modi tempora incidunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim ad minima veniam, quis nostrum exercitationem ullam corporis suscipit laboratoriosam, nisi ut aliquid ex ea commodi consequatur? Quis autem vel eum iure reprehenderit qui in ea voluptate velit esse quam nihil molestiae consequatur, vel illum qui dolorem eum fugiat, quo voluptas nulla pariatur?"

1914 translation by H. Rackham

"But I must explain to you how all this mistaken idea of denouncing pleasure and praising pain was born and I will give you a complete account of the system, and expound the actual teachings of the great explorer of the truth, the master-builder of human happiness. No one rejects, dislikes, or avoids pleasure itself, because it is pleasure, but because those who do not know how to pursue pleasure rationally encounter consequences that are extremely painful. Nor again is there anyone who loves or pursues or desires to obtain pain of itself, because it is pain, but because occasionally circumstances occur in which toil and pain can procure him some great pleasure. To take a trivial example, which of us ever undertakes laborious physical exercise, except to obtain some advantage from it? But who has any right to find fault with a man who chooses to enjoy a pleasure that has no annoying consequences, or one who avoids a pain that produces no resultant pleasure?"

1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 |

```
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  fx:Bind="{$ctx.isGameInProgress ? 'Resume Game' : 'New Game'}" fx:Event="$ctx.startNewGame" />
<mx:Button id="loadGameBtn" fx:Resolve="@Resolve($ctx.hasSaveFiles)" />
<mx:Button id="optionsBtn" />
<mx:Button id="quitBtn" />
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</mx:VBoxContent>
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<mx:Panel id="errorDialog" />
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<mx:Panel id="mapPanel" />
<mx:Panel id="questLogPanel" />
<mx:Panel id="shopPanel" />
<mx:Panel id="craftingPanel" />
<mx:Panel id="settingsPanel" />
<mx:Panel id="saveGamePanel" />
<mx:Panel id="achievementsPanel" />
<mx:Panel id="statsPanel" />
<mx:Panel id="bestiaryPanel" />
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<mx:Panel id="partyPanel" />
<mx:Panel id="worldMapPanel" />
<mx:Panel id="minimapPanel" />
<mx:Panel id="chatPanel" />
<mx:Panel id="notificationPanel" />
<mx:Panel id="tooltipPanel" />
<mx:Panel id="contextMenuPanel" />
<mx:Panel id="modsPanel" />
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<mx:Panel id="loadingTipsRotationPanel" />
<mx:Panel id="pauseTimeIndicatorPanel
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一种电缆 专利说明书

[19]中华人民共和国专利局
[21]ZL 99 [illegible]
[51]Int.Cl.[illegible]
[22]1999.[illegible].[illegible]
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[71][illegible]
[72][illegible]
[74][illegible]
[54]发明名称 一种电缆

[57]摘要 [illegible]

[illegible]

이사회 의장 주주 제안서

제1조 (제안의 목적)

제1조 (제안의 목적) 본 제안서는 주주 김지훈이 이사회 의장 선임을 제안하기 위해 작성되었다. 의장 선임은 이사회 운영 전반에 대한 주주 권리 행사의 가장 직접적인 수단이다. 의장은 의제 설정, 회의 진행, 의결 정리, 이사회 의사록 작성 등 이사회 운영 전반에 대한 권한을 가진다. 의장 선임 제안은 이사회 구성 변경 없이 이사회 운영 방식을 바꾸는 가장 빠른 경로다. 이사 선임 제안은 시간이 걸리지만, 의장 선임은 다음 이사회 한 번이면 된다.

제2조 (의장 선임 제안의 근거)

제2조 (의장 선임 제안의 근거) 의장 선임 제안은 세 가지 근거에 기한다. 첫째, 현 의장 김민수는 이사회 의사록 작성 의무를 지키지 않았다. 둘째, 의장 선임 제안은 이사회 구성 변경 없이 이사회 운영 방식을 바꾸는 가장 빠른 경로다. 셋째, 의장은 의제 설정, 회의 진행, 의결 정리, 이사회 의사록 작성 등 이사회 운영 전반에 대한 권한을 가진다.

제3조 (의사록 미작성 사실)

제3조 (의사록 미작성 사실) 2024년 3월 이사회 의사록 미작성 사실은 이사회 의사록 작성 의무를 지키지 않았다는 것을 보여준다. 의사록 작성 의무는 이사회 의장의 의무다. 의사록 미작성 사실은 의장 김민수가 이사회 의사록 작성 의무를 지키지 않았다는 것을 보여준다.

제4조 (의장 김민수의 의사록 작성 의무 불이행)

제4조 (의장 김민수의 의사록 작성 의무 불이행) 의장 김민수는 이사회 의사록 작성 의무를 지키지 않았다. 의사록 작성 의무는 이사회 의장의 의무다. 의장 김민수는 이사회 의사록 작성 의무를 지키지 않았다.

제5조 (의장 선임 제안의 효과)

제5조 (의장 선임 제안의 효과) 의장 선임 제안은 이사회 구성 변경 없이 이사회 운영 방식을 바꾸는 가장 빠른 경로다. 이사 선임 제안은 시간이 걸리지만, 의장 선임은 다음 이사회 한 번이면 된다. 의장 선임 제안은 이사회 구성 변경 없이 이사회 운영 방식을 바꾸는 가장 빠른 경로다.

[illegible]

[illegible]

[illegible]

```
<?xml version="1.0"
<!DOCTYPE
<!-- Generated by Microsoft Word 15.0.0.0 -->
<w:wordDocument xmlns:w="http://schemas.microsoft.com/office/word/2003/wordml" xmlns:
  xmlns:
  xmlns:
  
```

1.1 Introduction

1.1.1 Why Study Differential Equations?

Differential equations are equations involving derivatives of unknown functions. For example, the equation

$$\frac{dy}{dx} = 2x + 3$$

is a differential equation. Here, y is a function of x, and $\frac{dy}{dx}$ represents the rate at which y changes with respect to x.

Differential equations are fundamental in mathematics because they describe how quantities change over time or in response to other variables. For example, Newton's Second Law, $F = ma$, can be rewritten as $F = m\frac{d^2x}{dt^2}$, where $x(t)$ is the position of an object at time t. This equation allows us to predict the motion of the object by solving for $x(t)$.

Differential equations are used in physics, engineering, economics, biology, and many other fields to model dynamic systems. For example, in electrical circuits, the current $I(t)$ in a circuit with resistance R, inductance L, and voltage $V(t)$ is described by the equation $L\frac{dI}{dt} + RI = V(t)$. Solving this equation allows us to predict how the current changes over time.

1.1.2 Ordinary Differential Equations (ODEs)

An ordinary differential equation (ODE) is a differential equation in which the unknown function depends on a single variable. For example, the equation $\frac{dy}{dx} = 2x + 3$ is an ODE because y depends only on x.

ODEs are used to model systems where the rate of change of a quantity depends on the quantity itself and other factors. For example, the population of a city may be modeled by the ODE $\frac{dP}{dt} = kP$, where $P(t)$ is the population at time t and k is a constant representing the growth rate.

ODEs are used in physics, engineering, economics, biology, and many other fields to model dynamic systems. For example, in electrical circuits, the current $I(t)$ in a circuit with resistance R, inductance L, and voltage $V(t)$ is described by the ODE $L\frac{dI}{dt} + RI = V(t)$. Solving this ODE allows us to predict how the current changes over time.

1.1.3 Partial Differential Equations (PDEs)

A partial differential equation (PDE) is a differential equation in which the unknown function depends on multiple variables. For example, the equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ is a PDE because u depends on both x and t.

PDEs are used to model systems where the rate of change of a quantity depends on the quantity itself and other factors. For example, the heat equation $\frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2}$ describes how the temperature $u(x,t)$ changes over time and space, where k is a constant representing the thermal conductivity.

PDEs are used in physics, engineering, economics, biology, and many other fields to model dynamic systems. For example, in electrical circuits, the current $I(t)$ in a circuit with resistance R, inductance L, and voltage $V(t)$ is described by the PDE $L\frac{\partial I}{\partial t} + RI = V(t)$. Solving this PDE allows us to predict how the current changes over time and space.

Ch. 2 Lect 2 Part 1 - Chapter Summary

Ch. 2 Lect 2 Part 1 - Chapter Summary

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    "type": "articles",
    "attributes": {
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      "slug": "jsonapi-pagination",
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      "type": "people",
      "attributes": {
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        "twitter": "jsmith"
      }
    },
    {
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      "type": "tags",
      "attributes": {
        "name": "pagination"
      }
    },
    {
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      "type": "tags",
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        "name": "performance"
      }
    }
  ]
}
```

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Chapter 1 Introduction to Computer Science: Using Ruby 1.9.2 and Ruby 2.0 - 1.1 Introduction 1.1.1 Computer Hardware 1.1.2 Computer Software 1.1.3 Computer Programs 1.1.4 Computer Science 1.1.5 Computer Science Versus Computer Programming 1.1.6 Computer Science Versus Software Engineering 1.1.7 Computer Science Versus Information Technology 1.1.8 Computer Science Versus Other Disciplines 1.1.9 Computer Science Versus Mathematics 1.1.10 Computer Science Versus Science 1.1.11 Computer Science Versus Art 1.1.12 Computer Science Versus Philosophy 1.1.13 Computer Science Versus Engineering 1.1.14 Computer Science Versus Business 1.1.15 Computer Science Versus Medicine 1.1.16 Computer Science Versus Law 1.1.17 Computer Science Versus Sociology 1.1.18 Computer Science Versus Ethics 1.1.19 Computer Science Versus Politics 1.1.20 Computer Science Versus Economics 1.1.21 Computer Science Versus Psychology 1.1.22 Computer Science Versus Linguistics 1.1.23 Computer Science Versus Sociology 1.1.24 Computer Science Versus Ethics 1.1.25 Computer Science Versus Politics 1.1.26 Computer Science Versus Economics 1.1.27 Computer Science Versus Psychology 1.1.28 Computer Science Versus Linguistics 1.1.29 Computer Science Versus Sociology 1.1.30 Computer Science Versus Ethics 1.1.31 Computer Science Versus Politics 1.1.32 Computer Science Versus Economics 1.1.33 Computer Science Versus Psychology 1.1.34 Computer Science Versus Linguistics

Ch. 2 Lab Report - Angular Momentum

Name: Ryan Ramos

Partners: Ryan Ramos, Ryan Ramos, Ryan Ramos,

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Equation 1 of 1: Gaussian Integral Squared

Type: Definite Integral

Source: Gaussian Integral Squared

Tags: [Gaussian Integral, Gaussian Integral Squared]

Notes: Gaussian Integral Squared. Gaussian Integral Squared. Gaussian Integral Squared.

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Effect of Ultrasound Treatment on Antibacterial Activity of Aqueous Extract of Aloe Vera Gel

Ali Ahmadi , Mohammad Ali Ebrahimzadeh , Mohammad Ali Ghasemi , Ali Ghasemi , Ali Ghasemi ,

Chapter Chapter 1

Section Introduction

Subsection Introduction to Linear Algebra

Linear Algebra

Linear Algebra

Linear Algebra is the study of vectors and linear transformations between vector spaces.

Example 1.1.1

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Proof.

Let $u = (x_1, y_1, z_1)$ and $v = (x_2, y_2, z_2)$ be arbitrary elements of V and let $c \in \mathbb{R}$ be arbitrary.

Then $u + v = (x_1 + x_2, y_1 + y_2, z_1 + z_2)$ and

$$T(u + v) = (x_1 + x_2 + y_1 + y_2, 2(z_1 + z_2) - (x_1 + x_2)) = (x_1 + y_1, 2z_1 - x_1) + (x_2 + y_2, 2z_2 - x_2) = T(u) + T(v).$$

Also, $cu = (cx_1, cy_1, cz_1)$ and

$$T(cu) = (cx_1 + cy_1, 2cz_1 - cx_1) = c(x_1 + y_1, 2z_1 - x_1) = cT(u).$$

Therefore T satisfies both conditions of linearity and is a linear transformation. $\square$

Example 1.1.2

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x + 1).$$

Then T is not a linear transformation because $T(0, 0, 0) = (0, 1) \neq (0, 0)$.

Example 1.1.3

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

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Example 1.1.4

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Example 1.1.12

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Example 1.1.13

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Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

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Then T is a linear transformation.

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Then T is a linear transformation.

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$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

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Then T is a linear transformation.

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Then T is a linear transformation.

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Let

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Abstract: In this paper, we study the Cauchy problem for the nonlinear Kirchhoff-Carrier wave equation in a bounded domain. We prove that the solution exists globally in time and decays to zero as time tends to infinity.

Keywords: Kirchhoff-Carrier wave equation, Cauchy problem, Global existence, Decay.

MSC: 35L70, 35L20, 35L05, 35L15, 35L71, 35L72, 35L76, 35L90, 35B40, 35B41, 35B45, 35B65, 35G20, 35G25, 35G30, 35G31, 35G50, 35G55, 35G60, 35G61

Topic:

Question:

Find a unit tangent vector to the curve $r(t) = \langle \sin(t), \cos(t), \sqrt{2}t \rangle$ at the point $(0, 1, \pi\sqrt{2}/4)$.

Solution:

Step 1: Find the derivative $r'(t)$ of the position vector $r(t)$.

$$r'(t) = \langle \cos(t), -\sin(t), \sqrt{2} \rangle$$

Step 2: Evaluate $r'(t)$ at the point $t = \pi/4$ (since $\sin(\pi/4) = \cos(\pi/4) = 1/\sqrt{2}$).

$$r'(\pi/4) = \langle 1/\sqrt{2}, -1/\sqrt{2}, \sqrt{2} \rangle$$

Step 3: Find the magnitude of $r'(\pi/4)$.

$$|r'(\pi/4)| = \sqrt{(1/\sqrt{2})^2 + (-1/\sqrt{2})^2 + (\sqrt{2})^2} = \sqrt{1/2 + 1/2 + 2} = \sqrt{3}$$

Step 4: Divide $r'(\pi/4)$ by its magnitude to find the unit tangent vector.

$$T(\pi/4) = \frac{r'(\pi/4)}{|r'(\pi/4)|} = \frac{\langle 1/\sqrt{2}, -1/\sqrt{2}, \sqrt{2} \rangle}{\sqrt{3}} = \langle 1/\sqrt{6}, -1/\sqrt{6}, \sqrt{2/3} \rangle$$

Step 5: Rationalize the denominators.

$$T(\pi/4) = \langle \sqrt{6}/6, -\sqrt{6}/6, \sqrt{6}/3 \rangle$$

Step 6: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 7: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 8: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 9: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 10: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 11: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 12: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 13: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 14: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 15: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 16: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 17: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 18: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 19: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 20: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 21: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 22: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 23: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 24: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 25: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 26: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 27: Verify the unit tangent vector.

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```

```
# -*- coding: utf-8 -*-
```

# Generated by Rust (https://www.rust-lang.org)
import numpy as np
import numpy.linalg as LA
import matplotlib.pyplot as plt
from matplotlib import rcParams, rc
from mpl_toolkits.mplot3d import Axes3D
from scipy.integrate import odeint, ode

If we want LaTeX fonts
rcParams['font.family'] = 'serif'
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# Parameters
g = 9.81
L = 1.0
m = 1.0

# Initial conditions: theta and dtheta/dt
y0 = [3.14159265358979, 0.0]

# Time points
t = np.linspace(0, 10, 101)

# Define the ODE system
def f(y, t):
    theta = y[0]
    dtheta = y[1]
    ddtheta = -g/L*np.sin(theta)
    return [dtheta, ddtheta]

# Solve the ODE
y = odeint(f, y0, t)

# Extract theta and dtheta/dt
theta = y[:, 0]
dtheta = y[:, 1]

# Create figure
fig = plt.figure(1, figsize=(8, 6.4))

# Plot theta vs time
ax1 = fig.add_subplot(211)
ax1.plot(t, theta, 'b')
ax1.set_ylabel('θ')

# Plot dtheta/dt vs time
ax2 = fig.add_subplot(212)
ax2.plot(t, dtheta, 'r')
ax2.set_ylabel('$\dot{\theta}$')
ax2.set_xlabel('t')

# Save figure
plt.savefig('simple_pendulum.png', dpi=100)
```

```
# -*- coding: utf-8 -*-
```

# Generated by Rust (https://www.rust-lang.org)
import numpy as np
import numpy.linalg as LA
import matplotlib.pyplot as plt
from matplotlib import rcParams, rc
from mpl_toolkits.mplot3d import Axes3D
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ax2.plot(t, dtheta, 'r')
ax2.set_ylabel('$\dot{\theta}$')
ax2.set_xlabel('t')

# Save figure
plt.savefig('simple_pendulum.png', dpi=100)
```

NO-FLY-REPORTING/17

Non-American Jewish Foreign-born

Ted Cruz

Ted Cruz's Comments on Donald Trump Article

Ted Cruz's Comments on Donald Trump Article Ted Cruz's Comments on Donald Trump Article Ted Cruz's Comments on Donald Trump Article

Ted Cruz's Comments on Donald Trump Article

Ted Cruz's Comments on Donald Trump Article

Ad Corrigendum Supplementary

Page 1 of 1 Generated on 2025-07-14

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Section 1

Section 2

Section 3

Chapter Eighteen

Sections

Introduction
Sections and Subsections
Section Numbers
Section Titles
Section Cross-References
Section Cross-References in Other Documents
Section Cross-References in Other Documents Using Section Numbers
Section Cross-References in Other Documents Using Section Titles
Section Cross-References in Other Documents Using Section Numbers and Section Titles
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NO. 7147 MONTMORENCY

Sun Journal, 1907-07-19 (Vol. 1 Iss. 1)

Image 1

Text

Sun Journal, 1907-07-19 (Vol. 1 Iss. 1)

Image 2

Text

Sun Journal, 1907-07-19 (Vol. 1 Iss. 1) **THE MONTMORENCY SUN.** Vol. 1. **MONTMORENCY, MONT., SATURDAY JULY 19 1907.** No. 1. **PRICE 5 CENTS.** Number 1. Volume 1. Montmorency, Mont., Saturday, July 19, 1907. Price 5 cents. **Sun Publishing Co.** Publishers. **Sun Publishing Co.** Publishers. **Sun Publishing Co.** Publishers

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MR-TCAP-MAPMODULE

Operations-Generic-ASN1Protocol

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$$\sin(\frac{\pi}{2}) = \frac{\sqrt{2}}{2}$$

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$$\sin(\frac{\pi}{2}) = \frac{\sqrt{2}}{2}$$

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MR. TOUR'S MAGNIFICENCE!!!

Tour Tour Tour

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Topic:

Topic: Fluid Flow Equations

Topic: Fluid Flow Equations for Incompressible Flow

Topic: Fluid Flow Equations for Incompressible Flow: Bernoulli's Equation, Laminar Flow and Poiseuille's Equation, Viscous Forces, Pipe Networks

Topic: Fluid Flow Equations for Incompressible Flow: Bernoulli's Equation

Topic: Fluid Flow Equations for Incompressible Flow: Bernoulli's Equation

Topic:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0 \qquad \frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla)\mathbf{v} = -\frac{\nabla p}{\rho} + \frac{\mu}{\rho}\nabla^2 \mathbf{v} + \mathbf{g} \qquad \nabla \cdot \mathbf{v} = 0$$

$$\nabla \cdot (\rho \mathbf{v}) = \rho \nabla \cdot \mathbf{v} + \mathbf{v} \cdot \nabla \rho$$

$$\frac{\partial \rho}{\partial t} + \rho \nabla \cdot \mathbf{v} + \mathbf{v} \cdot \nabla \rho = 0$$

$$\frac{\partial \rho}{\partial t} + \rho \nabla \cdot \mathbf{v} = 0$$

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v})

ISSN 1234-5678 (print) 1234-9876 (online)
Received: 12 March 2024; Accepted: 18 July 2024

Abstract
This study investigates the effect of pulsed laser ablation in liquid (PLAL) parameters on the synthesis of gold nanoparticles (AuNPs) using a Nd:YAG laser. The influence of laser fluence (2–6 J/cm²), pulse repetition rate (10–50 Hz), and ablation time (5–30 min) on the size, shape, and optical properties of AuNPs was evaluated. Results showed that increasing laser fluence and repetition rate reduced average particle size from 18 nm to 9 nm, while prolonged ablation time led to aggregation and broadening of the size distribution. UV-Vis spectroscopy confirmed the formation of AuNPs with surface plasmon resonance peaks between 520 and 540 nm. The optimal conditions for producing small, spherical AuNPs were identified as 4 J/cm², 30 Hz, and 15 min. These findings provide a systematic approach for tailoring AuNP properties by controlling laser parameters, which is crucial for their application in biomedicine and catalysis.

Keywords
laser ablation, gold nanoparticles, pulsed laser ablation in liquid, Nd:YAG laser, surface plasmon resonance, UV-Vis spectroscopy, nanomaterials, optimization, size control, optical properties

Ajibola Ismail Ajibola*, Ajibola Ismail Ajibola*, Ajibola Ismail Ajibola*,

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Hello!

Hello!

Received: 12th Jan. 2024; Accepted: 18th Jan. 2024; Published: 25th Jan. 2024

Kumar
Kumar Ashish Kumar Ashish Kumar Ashish

DOI: 10.1007/s00401-
Received: 5 July 2017
Revised: 13 November 2017
Accepted: 14 November 2017

Received: 2 April 2020
Accepted: 14 May 2020
Published online: 5 June 2020
https://doi.org/10.1038/s41586-020-2412-9

Ruo-Jia Cui[1,2,3], Hongxia Duan[1,2], Bin Zhao[1,2], Mengqiao Zhou[1,2], Rui Shi[1,2], Xiao-Yu Zhao[1,2], Yi-Yang Li[1,2], Xin-Quan Wang[1,2], Lei Li[1,2] & Zhi-Jian Qin[1,2]

Lycopene is a red carotenoid of great importance because it is responsible for the pigment of tomatoes and other red fruits and vegetables and because it has been associated with many health benefits. Although the lycopene biosynthetic pathway has been established, the regulation of lycopene accumulation in plants is poorly understood. Here we report that the transcription factor RIPENING INHIBITOR (RIN) binds to promoters of genes encoding lycopene biosynthesis enzymes in tomato fruits, activates their expression, and promotes lycopene accumulation. Ethylene and light inhibit the binding of RIN to the promoters of these genes, and ethylene also inhibits the expression of RIN. Our results suggest that RIN is a master regulator of lycopene biosynthesis in tomato fruits, and that ethylene and light inhibit lycopene accumulation in tomato fruits by inhibiting the expression of RIN and the binding of RIN to the promoters of genes encoding lycopene biosynthesis enzymes.

https://doi.org/10.1038/s41586-020-2412-9

Lycopene is a red carotenoid that is responsible for the pigment of tomatoes and other red fruits and vegetables. Lycopene is a precursor of provitamin A and has been associated with many health benefits, such as a reduced risk of cardiovascular disease and certain cancers[1–3]. Although the lycopene biosynthetic pathway has been established, the regulation of lycopene accumulation in plants is poorly understood. Tomato fruits are a major source of lycopene, and lycopene accumulation in tomato fruits is a complex process that is affected by many factors, such as ripening, ethylene, light, temperature, and genotype[4–7]. Ripening-inhibitor (rin) is a recessive mutant that has been used extensively in studies of tomato fruit ripening. The rin mutant fruits do not ripen and do not accumulate lycopene, even in the presence of ethylene[8]. The RIN gene encodes a MADS-box transcription factor that is a master regulator of fruit ripening in tomato[9–12]. Although the RIN protein has been shown to bind to the promoters of several genes that are involved in ethylene biosynthesis and signal transduction, the genes that are directly regulated by RIN and the mechanism by which RIN regulates lycopene accumulation are unknown. Here we show that RIN binds to promoters of genes encoding lycopene biosynthesis enzymes in tomato fruits, activates their expression, and promotes lycopene accumulation. Ethylene and light inhibit the binding of RIN to the promoters of these genes, and ethylene also inhibits the expression of RIN. Our results suggest that RIN is a master regulator of lycopene biosynthesis in tomato fruits, and that ethylene and light inhibit lycopene accumulation in tomato fruits by inhibiting the expression of RIN and the binding of RIN to the promoters of genes encoding lycopene biosynthesis enzymes.

[1] State Key Laboratory of Plant Cell and Chromosome Engineering, Institute of Genetics and Developmental Biology, Chinese Academy of Sciences, Beijing 100101, China. [2] Center for Agricultural Resources Research, Institute of Genetics and Developmental Biology, Chinese Academy of Sciences, Shijiazhuang 050021, China. [3] University of Chinese Academy of Sciences, Beijing 100049, China.

MATLAB/SIMULINK

Lecture Notes Prepared by Dr. Ramesh Kumar Ayyala, Assistant Professor, Department of Mathematics, Indian Institute of Technology Jodhpur. Please report any errors or typos and discuss this at Academia.edu/Mathematics Reddit. Use Ctrl+Shift+F to find a term or phrase.

Tags: #MATLAB #SIMULINK #Lecture-Notes #Prof-Dr-Ramesh-Kumar-Ayyala #Indian-Institute-of-Technology-Jodhpur #Mathematics #Reddit #Academia.edu

---

Hello everyone, in this video we will discuss about **MATLAB/SIMULINK** and its uses in Control Theory and Control Systems.

Tags: #Control-Theory

Tags: #Control-Theory, #Control-Systems, #MATLAB/SIMULINK #Image

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$$\lim_{x \to 0} \frac{\sin(x)}{x} = 1$$

Tags: #MATLAB/SIMULINK

Tags: #Control-Theory, #Control-Systems, #MATLAB/SIMULINK, #Lecture-Notes, #Prof-Dr-Ramesh-Kumar-Ayyala, #Lecture-Notes-Prof-Dr-Ramesh-Kumar-Ayyala

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Tags: #MATLAB/SIMULINK

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Tags: #Control-Theory, #Control-Systems, #MATLAB/SIMULINK #Image

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DOI: 10.1007/s00401-
Received: 5 July 2017
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Online First: 27 November 2017

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Accepted: 27 August 2020
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Research Article

Jack Lewis Rhodes and Jon C. Cummins

Abstract

Recent research suggests that people who believe that science and technology are responsible for environmental problems tend to underestimate science's ability to solve these problems.

Keywords

Received: 2 April 2020; Revised: 14 July 2020; Accepted: 27 August 2020

Introduction

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Methods

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Results

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Discussion

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DOI: 10.1007/s00401-
Received: 5 July 2017
Revised: 19 December 2017
Accepted: 20 December 2017
Online First: 4 January 2018

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Quantum Mechanical Simulation of Electron Density Distribution in Hydrogen Molecular Ion (H₂⁺) Using Variational Methods

Dr. Ananya Krishnamurthy, Department of Chemistry, Vidyaranya Institute of Technology, Kalyanpur, Maharashtra, India
Prof. Sameer Ahmed Khan, Department of Physics, Vidyaranya Institute of Technology, Kalyanpur, Maharashtra, India
Mr. Rohan Deshpande, Department of Computational Chemistry, Vidyaranya Institute of Technology, Kalyanpur, Maharashtra, India

Abstract

This study presents a variational approach to simulate the electron density distribution in the hydrogen molecular ion (H_2^+) using trial wavefunctions and numerical optimization techniques. The ground state energy and electron density are calculated for various internuclear distances and compared with exact results. The results show that the variational method provides accurate electron density distributions with minimal computational cost. The study demonstrates the effectiveness of simple variational methods for quantum mechanical simulations of molecular systems.

Keywords: Quantum Mechanics, Variational Method, Hydrogen Molecular Ion, Electron Density, Ground State Energy, Computational Chemistry, Molecular Simulation, Trial Wavefunction, Optimization, Hartree-Fock Method

1. Introduction

The hydrogen molecular ion (H_2^+) is the simplest molecular system and serves as a fundamental benchmark for quantum mechanical calculations. Despite its simplicity, it exhibits all the essential features of chemical bonding, including electron sharing between two nuclei and the formation of a stable molecular orbital. The study of H_2^+ provides insights into the nature of chemical bonds and serves as a test case for more complex molecular systems.

The electron density distribution in H_2^+ is crucial for understanding the nature of the chemical bond. The electron density determines the strength of the bond, the bond length, and the vibrational frequencies. Accurate calculation of electron density is essential for predicting molecular properties and understanding chemical reactivity.

Variational methods are powerful tools for calculating molecular properties. These methods provide upper bounds to the ground state energy and allow for the calculation of electron density distributions. The variational method is particularly useful for systems where exact solutions are not available.

This study presents a variational approach to simulate the electron density distribution in H_2^+. The study compares the results with exact calculations and demonstrates the effectiveness of simple variational methods for quantum mechanical simulations.

2. Theoretical Background

2.1 Schrödinger Equation

The time-independent Schrödinger equation for H_2^+ is:

$$H\psi = E\psi$$

where H is the Hamiltonian operator, ψ is the wavefunction, and E is the energy.

2.2 Hamiltonian Operator

The Hamiltonian operator for H_2^+ is:

$$H = -\frac{\nabla_1^2}{2} - \frac{\nabla_2^2}{2} - \frac{1}{r_{a1}} - \frac{1}{r_{b1}} + \frac{1}{R}$$

where ∇_1^2 and ∇_2^2 are the Laplacian operators for electrons 1 and 2, r_{a1} and r_{b1} are the distances from electron 1 to nuclei a and b, and R is the internuclear distance.

2.3 Variational Principle

The variational principle states that for any trial wavefunction ψ_{trial}, the energy E_{trial} satisfies:

$$E_{trial} = \frac{\langle \psi_{trial} | H | \psi_{trial} \rangle}{\langle \psi_{trial} | \psi_{trial} \rangle} \geq E_0 \quad (1)$$

where E_0 is the ground state energy.

2.4 Trial Wavefunction

The trial wavefunction used in this study is:

$$\psi_{trial} = \frac{1}{\sqrt{2(1+S)}}(\phi_a + \phi_b) \quad (2)$$

where ϕ_a and ϕ_b are hydrogen 1s orbitals centered on nuclei a and b, and S is the overlap integral between these orbitals.

3. Methodology

3.1 Computational Approach

The computational approach involves the following steps:

1. Calculation of overlap integrals
2. Calculation of kinetic energy integrals
3. Calculation of potential energy integrals
4. Optimization of the variational parameter
5. Calculation of electron density distribution

3.2 Overlap Integral

The overlap integral between two hydrogen 1s orbitals is:

$$S = \int \phi_a \phi_b \, d\tau = \left(1 + \frac{R}{a_0} + \frac{R^2}{3a_0^2}\right) e^{-R/a_0} \quad (3)$$

where a_0 is the Bohr radius.

3.3 Kinetic Energy Integral

The kinetic energy integral is:

$$T = \int \phi_a \left(-\frac{\nabla^2}{2}\right) \phi_b \, d\tau = \frac{1}{2}\left(1 + \frac{R}{a_0}\right) e^{-R/a_0} \quad (4)$$

3.4 Potential Energy Integral

The potential energy integral is:

$$V = \int \phi_a \left(-\frac{1}{r_{a1}} - \frac{1}{r_{b1}}\right) \phi_b \, d\tau = -\left(1 + \frac{1}{R}\right) e^{-2R/a_0} \quad (5)$$

3.5 Optimization

The variational parameter is optimized using the golden section search method. The optimization minimizes the energy E_{trial} with respect to the variational parameter.

3.6 Electron Density Calculation

The electron density is calculated from the trial wavefunction as:

$$\rho(r) = |\psi_{trial}|^2$$

4. Results and Discussion

4.1 Ground State Energy

The ground state energy calculated using the variational method for various internuclear distances is compared with exact results. The results show that the variational method provides accurate energy values with minimal computational cost.

4.2 Electron Density Distribution

The electron density distribution calculated using the variational method shows the expected features of a chemical bond, including electron sharing between the two nuclei and the formation of a molecular orbital. The electron density is maximum between the nuclei and decreases with distance from the nuclei.

4.3 Comparison with Exact Results

The results are compared with exact results from literature. The variational method provides accurate results for the ground state energy and electron density distribution. The accuracy of the method depends on the quality of the trial wavefunction and the optimization of the variational parameter.

5. Conclusion

This study demonstrates the effectiveness of simple variational methods for quantum mechanical simulations of molecular systems. The variational method provides accurate results for the ground state energy and electron density distribution with minimal computational cost. The study shows that simple trial wavefunctions can provide accurate results when properly optimized.

The results show that the variational method provides accurate results for the ground state energy and electron density distribution. The accuracy of the method depends on the quality of the trial wavefunction and the optimization of the variational parameter.

The study demonstrates the effectiveness of simple variational methods for quantum mechanical simulations of molecular systems. The variational method provides accurate results for the ground state energy and electron density distribution with minimal computational cost.

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MR-TCAST-MQ2000B0B0-V1

MR-TCAST-MQ2000B0B0-V1 Product Specification

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1 Product Overview

1.1 Product Introduction

MR-TCAST-MQ2000B0B0-V1 is a high-performance industrial-grade multi-port Ethernet switch designed for demanding industrial automation, smart manufacturing, and critical infrastructure applications. The device provides 2 Gigabit Ethernet ports and 8 Gigabit Ethernet ports, supporting Layer 2 switching functions and offering comprehensive security and management capabilities. It adopts a fanless design with wide temperature operation capability, ensuring stable operation in harsh industrial environments.

1.2 Product Features

- Hardware Platform: Dual-core ARM Cortex-A53 processor, 512MB DDR3 memory, 128MB NAND Flash
- Ethernet Ports: 2 Gigabit Ethernet ports (1000Base-T/SFP combo), 8 Gigabit Ethernet ports (1000Base-T)
- Switching Capacity: 20Gbps
- Forwarding Rate: 14.88Mpps
- MAC Address Table: 16K entries
- VLAN: Supports 4094 VLANs, supports port-based VLAN, 802.1Q VLAN, QinQ
- Spanning Tree: Supports STP, RSTP, MSTP
- Link Aggregation: Supports static LAG and LACP, up to 8 groups, 8 members per group
- Multicast: Supports IGMP Snooping v1/v2/v3, MLD Snooping v1/v2, multicast VLAN
- QoS: Supports 802.1p, DSCP, 8 queues per port, SP, WRR, WFQ scheduling
- Security: Supports port security, MAC address binding, DHCP Snooping, ARP Inspection, IP Source Guard, 802.1X authentication, RADIUS, TACACS+
- Management: Supports CLI, Web, SNMP v1/v2c/v3, Telnet, SSH, NTP, syslog
- Protocols: Supports IPv4/IPv6 dual stack, ARP, ICMP, DHCP, DNS, TFTP, FTP, HTTP, HTTPS
- Reliability: Supports ring network protection, port mirroring, cable diagnostics, watchdog
- Power Supply: Dual power input, supports 12V/24V DC, supports power failure alarm
- Operating Temperature: -40°C ~ +75°C
- Storage Temperature: -40°C ~ +85°C
- Operating Humidity: 5% ~ 95% RH (non-condensing)
- Protection Rating: IP40
- Dimensions: 440mm × 44mm × 180mm (W × H × D)
- Weight: 1.5kg
- Installation: 19-inch rack-mount, desktop, DIN rail

1.3 Product Specifications

Item	Specifications
Model	MR-TCAST-MQ2000B0B0-V1
Ethernet Ports	2 Gigabit Ethernet ports (1000Base-T/SFP combo), 8 Gigabit Ethernet ports (1000Base-T)
Switching Capacity	20Gbps
Forwarding Rate	14.88Mpps
MAC Address Table	16K entries
VLAN	Supports 4094 VLANs, supports port-based VLAN, 802.1Q VLAN, QinQ
Spanning Tree	Supports STP, RSTP, MSTP
Link Aggregation	Supports static LAG and LACP, up to 8 groups, 8 members per group
Multicast	Supports IGMP Snooping v1/v2/v3, MLD Snooping v1/v2, multicast VLAN
QoS	Supports 802.1p, DSCP, 8 queues per port, SP, WRR, WFQ scheduling
Security	Supports port security, MAC address binding, DHCP Snooping, ARP Inspection, IP Source Guard, 802.1X authentication, RADIUS, TACACS+
Management	Supports CLI, Web, SNMP v1/v2c/v3, Telnet, SSH, NTP, syslog
Protocols	Supports IPv4/IPv6 dual stack, ARP, ICMP, DHCP, DNS, TFTP, FTP, HTTP, HTTPS
Reliability	Supports ring network protection, port mirroring, cable diagnostics, watchdog
Power Supply	Dual power input, supports 12V/24V DC, supports power failure alarm
Operating Temperature	-40°C ~ +75°C
Storage Temperature	-40°C ~ +85°C
Operating Humidity	5% ~ 95% RH (non-condensing)
Protection Rating	IP40
Dimensions	440mm × 44mm × 180mm (W × H × D)
Weight	1.5kg
Installation	19-inch rack-mount, desktop, DIN rail

1.4 Product Appearance

1.5 Product Dimensions

1.6 Product Interfaces

Front Panel

- 2 Gigabit Ethernet ports (1000Base-T/SFP combo)
- 8 Gigabit Ethernet ports (1000Base-T)
- Console port
- Power indicator
- System indicator
- Port indicators

Rear Panel

- Power terminal block
- Grounding screw

1.7 Product Indicators

Indicator	Status	Description
Power	Solid green	Power normal
	Off	Power abnormal
System	Solid green	System normal
	Blinking green	System starting
	Solid red	System fault
	Off	No power
Port	Solid green	Link established
	Blinking green	Data transmission
	Off	Link not established

2 Hardware Specifications

2.1 Hardware Architecture

- Processor: Dual-core ARM Cortex-A53 processor
- Memory: 512MB DDR3
- Flash: 128MB NAND Flash
- Switch Chip: Integrated Layer 2 switch chip
- Ethernet Ports: 2 Gigabit Ethernet ports (1000Base-T/SFP combo), 8 Gigabit Ethernet ports (1000Base-T)
- Console Port: 1 console port
- Power Input: Dual power input, supports 12V/24V DC, supports power failure alarm
- Indicators: Power indicator, system indicator, port indicators
- Buttons: Reset button

2.2 Power Supply

- Power Input: Dual power input, supports 12V/24V DC, supports power failure alarm
- Power Consumption: Typical 15W, maximum 25W
- Power Terminal Block: Supports 2-pin terminal block connection
- Grounding Screw: Provides grounding screw for safety and lightning protection

2.3 Power Consumption

- Typical: 15W
- Maximum: 25W
- Idle: 8W

2.4 Cooling Design

- Fanless design, no fan, no noise, no dust, no maintenance
- Natural convection heat dissipation, suitable for long-term stable operation
- Operating temperature: -40°C ~ +75°C, suitable for harsh industrial environments
- Storage temperature: -40°C ~ +85°C
- Operating humidity: 5% ~ 95% RH (non-condensing)
- Protection rating: IP40, suitable for indoor installation

2.5 Environmental Requirements

- Operating temperature: -40°C ~ +75°C
- Storage temperature: -40°C ~ +85°C
- Operating humidity: 5% ~ 95% RH (non-condensing)
- Protection rating: IP40
- Installation environment: Indoor, avoid direct sunlight, ensure good ventilation

2.6 Mechanical Specifications

- Dimensions: 440mm × 44mm × 180mm (W × H × D)
- Weight: 1.5kg
- Installation: 19-inch rack-mount, desktop, DIN rail
- Rack-mount ears: Standard 19-inch rack-mount ears, supports front and rear installation
- Chassis: Metal chassis, sturdy and durable

2.7 Electromagnetic Compatibility

- Conducted emissions: Meets industrial-grade EMC requirements
- Radiated emissions: Meets industrial-grade EMC requirements
- Electrostatic discharge: Supports ESD protection
- Surge protection: Supports surge protection
- Electromagnetic interference: Meets industrial-grade EMC requirements

2.8 Reliability

- Mean Time Between Failures (MTBF): Over 100,000 hours
- Operating temperature: -40°C ~ +75°C, suitable for harsh industrial environments
- Storage temperature: -40°C ~ +85°C
- Operating humidity: 5% ~ 95% RH (non-condensing)
- Protection rating: IP40, suitable for indoor installation
- Power failure alarm: Supports power failure alarm, automatically sends alarm when power is lost
- Watchdog: Supports hardware watchdog, automatically restores system when system hangs
- Port mirroring: Supports port mirroring for fault diagnosis
- Cable diagnostics: Supports cable diagnostics, detects cable faults
- Ring network protection: Supports ring network protection, ensures network continuity

3 Software Specifications

3.1 Operating System

- Operating system: Proprietary operating system
- Kernel: Linux kernel
- Management interfaces: CLI, Web, SNMP, Telnet, SSH
- Protocol stack: Supports IPv4/IPv6 dual stack
- Firmware upgrade: Supports online firmware upgrade, supports dual firmware backup

3.2 Management Interfaces

- CLI: Supports command-line interface, supports Telnet and SSH remote management
- Web: Supports web management, supports HTTP and HTTPS
- SNMP: Supports SNMP v1/v2c/v3, supports MIB-II, supports private MIB
- Telnet: Supports Telnet remote management
- SSH: Supports SSH encrypted management
- NTP: Supports NTP time synchronization
- Syslog: Supports syslog, supports local and remote syslog
- Firmware upgrade: Supports online firmware upgrade, supports dual firmware backup
- Configuration backup: Supports configuration backup and restore
- Batch configuration: Supports batch configuration, supports configuration templates

3.3 Protocol Support

- Layer 2 protocols: Supports ARP, ICMP, DHCP, DNS, TFTP, FTP, HTTP, HTTPS
- Spanning tree: Supports STP, RSTP, MSTP
- Link aggregation: Supports static LAG and LACP, up to 8 groups, 8 members per group
- VLAN: Supports 4094 VLANs, supports port-based VLAN, 802.1Q VLAN, QinQ
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- QoS: Supports 802.1p, DSCP, 8 queues per port, SP, WRR, WFQ scheduling
- Security: Supports port security, MAC address binding, DHCP Snooping, ARP Inspection, IP Source Guard, 802.1X authentication, RADIUS, TACACS+
- Management: Supports CLI, Web, SNMP v1/v2c/v3, Telnet, SSH, NTP, syslog
- Protocols: Supports IPv4/IPv6 dual stack, ARP, ICMP, DHCP, DNS, TFTP, FTP, HTTP, HTTPS
- Reliability: Supports ring network protection, port mirroring, cable diagnostics, watchdog

3.4 Security Features

- Port security: Supports port security, limits MAC address learning
- MAC address binding: Supports MAC address binding, prevents unauthorized access
- DHCP Snooping: Supports DHCP Snooping, prevents DHCP spoofing attacks
- ARP Inspection: Supports ARP Inspection, prevents ARP spoofing attacks
- IP Source Guard: Supports IP Source Guard, prevents IP address spoofing
- 802.1X authentication: Supports 802.1X authentication, supports RADIUS authentication
- RADIUS: Supports RADIUS authentication, supports accounting
- TACACS+: Supports TACACS+ authentication, supports accounting
- Port mirroring: Supports port mirroring for fault diagnosis
- Cable diagnostics: Supports cable diagnostics, detects cable faults
- Ring network protection: Supports ring network protection, ensures network continuity

3.5 Firmware Upgrade

- Online upgrade: Supports online firmware upgrade, no need to restart the device
- Dual firmware backup: Supports dual firmware backup, automatically switches to backup firmware when upgrade fails
- Configuration backup: Supports configuration backup and restore
- Batch configuration: Supports batch configuration, supports configuration templates

4 Ordering Information

4.1 Product Model Description

MR-TCAST-MQ2000B0B0-V1

- MR: Industrial Ethernet switch
- TCAST: Product series
- MQ2000: 2 Gigabit Ethernet ports + 8 Gigabit Ethernet ports
- B0: Standard configuration
- B0: Standard configuration
- V1: Version 1

4.2 Accessories

- Power adapter: 12V/24V DC power adapter
- Rack-mount ears: Standard 19-inch rack-mount ears
- Grounding screw: Provides grounding screw for safety and lightning protection
- Console cable: Console cable for configuration
- Ethernet cable: Ethernet cable for connection
- Installation manual: Installation manual
- Warranty card: Warranty card

4.3 Spare Parts

- Power adapter: 12V/24V DC power adapter
- Rack-mount ears: Standard 19-inch rack-mount ears
- Grounding screw: Provides grounding screw for safety and lightning protection
- Console cable: Console cable for configuration
- Ethernet cable: Ethernet cable for connection
- Installation manual: Installation manual
- Warranty card: Warranty card

5 Appendix

5.1 Glossary

- Switch: Network switch, forwards data packets between network devices
- Ethernet: Ethernet, wired network protocol
- Gigabit Ethernet: Gigabit Ethernet, 1000Mbps Ethernet
- Port: Port, network interface
- VLAN: Virtual Local Area Network, logical network segmentation
- QinQ: Double VLAN, supports VLAN stacking
- STP: Spanning Tree Protocol, prevents network loops
- RSTP: Rapid Spanning Tree Protocol, fast convergence
- MSTP: Multiple Spanning Tree Protocol, supports multiple spanning trees
- LAG: Link Aggregation Group, bundles multiple physical links into one logical link
- LACP: Link Aggregation Control Protocol, dynamic link aggregation
- IGMP: Internet Group Management Protocol, multicast management protocol
- MLD: Multicast Listener Discovery, IPv6 multicast management protocol
- QoS: Quality of Service, supports traffic prioritization and bandwidth allocation
- ACL: Access Control List, supports packet filtering and access control
- 802.1X: Port-based Network Access Control, supports user authentication
- RADIUS: Remote Authentication Dial-In User Service, supports centralized authentication
- TACACS+: Terminal Access Controller Access-Control System Plus, supports centralized authentication
- SNMP: Simple Network Management Protocol, supports network management
- MIB: Management Information Base, supports SNMP management
- CLI: Command Line Interface, supports command-line management
- Web: Supports web management, supports HTTP and HTTPS
- Telnet: Supports Telnet remote management
- SSH: Supports SSH encrypted management
- NTP: Supports NTP time synchronization
- Syslog: Supports syslog, supports local and remote syslog
- Firmware upgrade: Supports online firmware upgrade, supports dual firmware backup
- Configuration backup: Supports configuration backup and restore
- Batch configuration: Supports batch configuration, supports configuration templates

5.2 Revision History

Version	Date	Description
V1.0	2026-04-01	Initial release

Tags: ancient civilizations, ancient language speakers, Lost Cemetery of Ancient Language Speakers, Joseph Francis Alcazar, Cornelius Ashburn, Dr. Ellis, Lucinda Hawkes, Samuel Hawkes, Tom Hawkes, Professor Alcazar, Professor Ashburn, Professor Ellis, Professor Hawkes, Professor Lucinda Hawkes, Professor Samuel Hawkes, Professor Tom Hawkes, Professor Ashburn's Assistant, Professor Ellis's Assistant, Professor Hawkes's Assistant, Professor Lucinda Hawkes's Assistant, Professor Samuel Hawkes's Assistant, Professor Tom Hawkes's Assistant, Professor Ashburn's Assistant's Assistant, Professor Ellis's Assistant's Assistant, Professor Hawkes's Assistant's Assistant, Professor Lucinda Hawkes's Assistant's Assistant, Professor Samuel Hawkes's Assistant's Assistant, Professor Tom Hawkes's Assistant's Assistant, Professor Ashburn's Assistant's Assistant's Assistant, Professor Ellis's Assistant's Assistant's Assistant, Professor Hawkes's Assistant's Assistant's Assistant, Professor Lucinda 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Ref: Fluid Mechanics

Date: Monday, June 1, 2026 at 10:00:00 AM Coordinated Universal Time 7

Fluid Mechanics

Topic: Laplace transformation for Partial differential equations

Subtopic: Laplace transformation for Partial differential equations

Keywords: Laplace transform, Partial differential equations, Fluid mechanics, Heat transfer, Signal processing, Control systems, Electrical engineering, Mathematical modeling, Boundary value problems, Initial value problems, Differential equations, Transfer functions, Stability analysis, Frequency domain analysis, Time domain analysis, Integral transforms, Complex analysis, Engineering mathematics, Applied mathematics, Solution methods, Analytical methods, Boundary conditions, Initial conditions, Transient analysis, Steady state analysis, Dynamic systems, Linear systems, Nonlinear systems, Partial differential equations, Ordinary differential equations, Boundary value problems, Initial value problems, Laplace transform, Fourier transform, Inverse transform, Convolution theorem, Heaviside step function, Dirac delta function, Green's function

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Fig.2 Sensitivity analysis of Fermentation Capacity

Fig.2

Author: Prof. Dr. Ali Ekber Kocaoğlu, Dr. Ali Ekber Kocaoğlu, Dr. Ali Ekber Kocaoğlu,

NO. 7[illegible]

[illegible]

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[1] Department of Geology, Kathmandu University, Dhulikhel, Nepal | Corresponding author: ...@ku.edu.np

[illegible]

[illegible]

[illegible]

Keywords: Fluid Mechanics, Lecture Series, Transcript, Transcription, Text, Lecture Transcript, Transcription, Text, Lecture Transcript, Transcription, Text, Lecture

MR-TIGER-MAGNUM-MAXIMUS

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Mr-Tiger-Magnum-Maximus-Supreme-Gamma-Omega-Sigma-Alpha

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Mr-Tiger-Magnum-Maximus-Supreme-Gamma-Omega-Sigma-Alpha

Mr-Tiger-Magnum-Maximus-Supreme-Gamma-Omega-Sigma-Alpha

$$\int_{\mathbb{R}^n} \frac{\mathbb{E}}{\mathbb{P}} \, \mathrm{d}\mathbb{P} = \mathbb{E}$$

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Mr-Tiger-Magnum-Maximus-Supreme-Gamma-Omega-Sigma-Alpha

Mr-Tiger-Magnum-Maximus-Supreme-Gamma-Omega-Sigma-Alpha

Mr-Tiger-Magnum-Maximus-Supreme-Gamma-Omega-Sigma-Alpha

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Mr-Tiger-Magnum-Maximus-Supreme-Gamma-Omega-Sigma-Alpha

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Fabian Ruehle [1] [2] [3] [4] [5] [6] [7] [8] [9] [10] [11] [12] [13] [14] [15] [16] [17] [18] [19] [20] [21] [22] [23] [24] [25] [26] [27] [28] [29] [30] [31] [32] [33] [34] [35] [36] [37] [38] [39] [40]

MATLAB/Simulink Model Based Design Workflow

Introduction

Model Based Design (MBD) is a formal methodology for developing complex control systems. It uses a single mathematical model as the central artifact from which requirements, simulation, code generation, and verification are derived. The model serves as the ground truth for all downstream activities, ensuring that changes propagate consistently across the entire development lifecycle.

Workflow Overview

The workflow consists of six stages: requirements capture, plant modeling, controller design, simulation and verification, code generation, and deployment. Each stage produces artifacts that feed into the next stage. Traceability links connect requirements to model elements to generated code to test results, so that any change can be assessed for its impact on every downstream artifact.

Plant Modeling

Plant modeling begins with the physical equations governing the system. For a DC motor, the equations are:

$$v(t) = Ri(t) + L\frac{di(t)}{dt} + K_e\omega(t)$$

$$J\frac{d\omega(t)}{dt} = K_t i(t) - B\omega(t) - \tau_L(t)$$

Controller Design

The controller is designed in the discrete domain. For a PI controller with sampling period T_s:

$$u[k] = K_p e[k] + K_i T_s \sum_{j=0}^{k} e[j]$$

Simulink Implementation

Simulink provides a block diagram environment for implementing the plant model and controller. Signal flow is explicit and visual. Simulink supports fixed-step and variable-step solvers, with fixed-step solvers required for real-time code generation.

Code Generation

Simulink Coder (formerly Real-Time Workshop) generates C code from the model. The generated code is deterministic, with no dynamic memory allocation and no recursion. Code generation can be configured for different target processors and optimization levels.

Verification

Model-in-the-loop (MIL) simulation verifies the model against requirements before code generation. Software-in-the-loop (SIL) simulation runs the generated code on a host machine to verify that the code behaves identically to the model. Hardware-in-the-loop (HIL) simulation runs the code on the target hardware with real sensors and actuators.

Deployment

Deployment packages the generated code with the target hardware's boot code and runtime libraries. Over-the-air (OTA) updates allow deployed systems to receive new versions without physical access.

Conclusion

Model Based Design provides a single source of truth for requirements, simulation, and implementation. Changes propagate automatically through the workflow, reducing the risk of inconsistencies between specification and implementation.

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1.1 Introduction to Data Structures and Algorithms

Data structures and algorithms are the foundation of computer science. Understanding how to organize data efficiently and how to design algorithms that solve problems quickly is essential for any programmer.

1.1.1 Why Study Data Structures and Algorithms?

Consider a simple problem: you have a list of 1,000,000 numbers and you need to find the largest one. A naive approach might scan the entire list for each query. With the right data structure, you can answer the question in constant time. The difference between a slow program and a fast program can be the difference between a project that works and one that never finishes.

Data structures and algorithms are the tools that make computers fast and reliable. They are used in every domain: finance, medicine, engineering, science, and everyday software. Mastering them gives you the ability to design systems that scale from a few users to millions of users.

1.1.2 Formal Definitions

Let n denote the size of the input. We write $f(n) = O(g(n))$ if there exists constants $c > 0$ and $n_0 \geq 0$ such that $f(n) \leq c \cdot g(n)$ for all $n \geq n_0$. Similarly, $f(n) = \Omega(g(n))$ if $f(n) \geq c \cdot g(n)$ for all $n \geq n_0$, and $f(n) = \Theta(g(n))$ if both hold.

Formally, a data structure is a tuple $(D, \mathcal{O})$ where D is the set of possible states and $\mathcal{O}$ is a set of operations that transform states in D.

1.1.3 Example

```
def binary_search(arr, target):
    lo, hi = 0, len(arr) - 1
    while lo <= hi:
        mid = (lo + hi) // 2
        if arr[mid] == target:
            return mid
        elif arr[mid] < target:
            lo = mid + 1
        else:
            hi = mid - 1
    return -1
```

Binary search runs in $O(\log n)$ time. Each iteration halves the search space, so after k iterations the remaining range has size $n / 2^k$. When $k = \log_2 n$, the range has size 1 and the algorithm terminates.

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<?xml version="1.0" encoding="UTF-8"?>
<!-- Generated by SVGO v3.0.2 (https://svgo.dev) -->
<svg
  xmlns="http://www.w3.org/2000/svg"
  xmlns:xlink="http://www.w3.org/1999/xlink"
  viewBox="0.000,0.000,1000.000,1000.000"
  preserveAspectRatio="xMidYMid meet"
  fill="#000000" />
</svg>
```

Chapter 4 Matrix Algebra

4.1 Sum and Scalar Product of Matrices

Definition

Sum of two matrices Two matrices A and B are said to be conformable for addition if they are of the same order. If $A = [a_{ij}]_{m \times n}$ and $B = [b_{ij}]_{m \times n}$ then the sum of A and B is defined as $A + B = [a_{ij} + b_{ij}]_{m \times n}$.

Scalar product If $A = [a_{ij}]_{m \times n}$ and k is a scalar then $kA = [ka_{ij}]_{m \times n}$.

Example

Example 1

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}, \quad B = \begin{bmatrix} 7 & 8 & 9 \\ 10 & 11 & 12 \end{bmatrix}, \quad k = 2$$

Solution

$$A + B = \begin{bmatrix} 1+7 & 2+8 & 3+9 \\ 4+10 & 5+11 & 6+12 \end{bmatrix} = \begin{bmatrix} 8 & 10 & 12 \\ 14 & 16 & 18 \end{bmatrix}$$

$$kA = \begin{bmatrix} 2 \times 1 & 2 \times 2 & 2 \times 3 \\ 2 \times 4 & 2 \times 5 & 2 \times 6 \end{bmatrix} = \begin{bmatrix} 2 & 4 & 6 \\ 8 & 10 & 12 \end{bmatrix}$$

Example 2

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}, \quad k = 3$$

Solution

$$A + B = \begin{bmatrix} 1+5 & 2+6 \\ 3+7 & 4+8 \end{bmatrix} = \begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix}$$

$$kA = \begin{bmatrix} 3 \times 1 & 3 \times 2 \\ 3 \times 3 & 3 \times 4 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 9 & 12 \end{bmatrix}$$

Example 3

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 4

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 5

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 6

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 7

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 8

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 9

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 10

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 11

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 12

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 13

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 14

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 15

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 16

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 17

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 18

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 19

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 20

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 21

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 22

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 23

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 24

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 25

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 26

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 27

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 28

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 29

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 30

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 31

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 32

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 &

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Tiger Magnum Opus 7 Extraordinary Lemma

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<svg
  xmlns="http://www.w3.org/2000/svg"
  xmlns:xlink="http://www.w3.org/1999/xlink"
  viewBox="0.000,0.000,1.000,1.000"
  preserveAspectRatio="xMidYMid meet">
<defs>
  <path id="1" d="M0.000,0.000 L0.000,0.000 L0.000,0.000 L
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Ultrasonic Sensor Based Smart Dustbin

[1]Dr. Ashish Kumar Jha, [2]Ankit Kumar, [3]Ankit Kumar, [4]Ankit Kumar, [5]Ankit Kumar, [6]Ankit Kumar, [7]Ankit Kumar, [8]Ankit Kumar, [9]Ankit Kumar, [10]Ankit Kumar, [11]Ankit Kumar, [12]Ankit Kumar, [13]Ankit Kumar, [14]Ankit Kumar, [15]Ankit Kumar, [16]Ankit Kumar, [17]Ankit Kumar, [18]Ankit Kumar, [19]Ankit Kumar, [20]Ankit Kumar, [21]Ankit Kumar,

Sample Chemical Formula Balancer

Enter chemical formula (Example: Fe2(SO4)3):

```
Fe2(SO4)3
```

Parsed: {'Fe': 2, 'S': 3, 'O': 12}

Enter chemical formula (Example: Fe2(SO4)3):

```
Fe(OH)3
```

Parsed: {'Fe': 1, 'O': 3, 'H': 3}

Enter chemical formula (Example: Fe2(SO4)3):

```
Cu(NH3)4(OH)2
```

Parsed: {'Cu': 1, 'N': 4, 'H': 12, 'O': 2}

$$\begin{array}{l} \mathbf{B}=\{\mathbf{1}\} \\ \quad \mathbf{1}\rightarrow\mathbf{2} \\ \quad \mathbf{2}\rightarrow\mathbf{3}\rightarrow\mathbf{4} \\ \quad \mathbf{4}\rightarrow\mathbf{5} \end{array}$$

1

Enter chemical formula (Example: Fe2(SO4)3):

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Cu(NH3)4(OH)2
```

Parsed: {'Cu': 1, 'N': 4, 'H': 12, 'O': 2}

Enter chemical formula (Example: Fe2(SO4)3):

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Cu(NH3)4(OH)2
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Parsed: {'Cu': 1, 'N': 4, 'H': 12, 'O': 2}

$$\begin{array}{l} \mathbf{B}=\{\mathbf{1},\mathbf{2}\} \\ \quad \mathbf{1}\rightarrow\mathbf{2} \\ \quad \mathbf{2}\rightarrow\mathbf{3}\rightarrow\mathbf{4} \\ \quad \mathbf{4}\rightarrow\mathbf{5} \end{array}$$

2

Enter chemical formula (Example: Fe2(SO4)3):

```
Cu(NH3)4(OH)2
```

Parsed: {'Cu': 1, 'N': 4, 'H': 12, 'O': 2}

Enter chemical formula (Example: Fe2(SO4)3):

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Cu(NH3)4(OH)2
```

Parsed: {'Cu': 1, 'N': 4, 'H': 12, 'O': 2}

$$\begin{array}{l} \mathbf{B}=\{\mathbf{1},\mathbf{2},\mathbf{3}\} \\ \quad \mathbf{1}\rightarrow\mathbf{2} \\ \quad \mathbf{2}\rightarrow\mathbf{3}\rightarrow\mathbf{4} \\ \quad \mathbf{4}\rightarrow\mathbf{5} \end{array}$$

3

Enter chemical formula (Example: Fe2(SO4)3):

```
Cu(NH3)4(OH)2
```

Parsed: {'Cu': 1, 'N': 4, 'H': 12, 'O': 2}

Enter chemical formula (Example: Fe2(SO4)3):

```
Cu(NH3)4(OH)2
```

Parsed: {'Cu': 1, 'N': 4, 'H': 12, 'O': 2}

$$\begin{array}{l} \mathbf{B}=\{\mathbf{1},\mathbf{2},\mathbf{3},\mathbf{4}\} \\ \quad \mathbf{1}\rightarrow\mathbf{2} \\ \quad \mathbf{2}\rightarrow\mathbf{3}\rightarrow\mathbf{4} \\ \quad \mathbf{4}\rightarrow\mathbf{5} \end{array}$$

4

Enter chemical formula (Example: Fe2(SO4)3):

```
Cu(NH3)4(OH)2
```

Parsed: {'Cu': 1, 'N': 4, 'H': 12, 'O': 2}

Enter chemical formula (Example: Fe2(SO4)3):

```
Cu(NH3)4(OH)2
```

Parsed: {'Cu': 1, 'N': 4, 'H': 12, 'O': 2}

$$\begin{array}{l} \mathbf{B}=\{\mathbf{1},\mathbf{2},\mathbf{3},\mathbf{4},\mathbf{5}\} \\ \quad \mathbf{1}\rightarrow\mathbf{2} \\ \quad \mathbf{2}\rightarrow\mathbf{3}\rightarrow\mathbf{4} \\ \quad \mathbf{4}\rightarrow\mathbf{5} \end{array}$$

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Ethical Implications of Artificial Intelligence in Healthcare: Balancing Innovation and Patient Rights

Dr. Ashish Kumar Jha
Assistant Professor, Department of Commerce, Government of Rajasthan, Rajasthan Institute of Advanced Studies, Jaipur, India
Email: ashishkumarjha@gmail.com

Abstract: Artificial Intelligence (AI) is rapidly transforming healthcare by enhancing diagnostic accuracy, optimizing treatment plans, and improving patient outcomes. However, the integration of AI in healthcare raises significant ethical concerns, including data privacy, algorithmic bias, accountability, and the potential displacement of human caregivers. This study explores the ethical implications of AI in healthcare, focusing on the balance between innovation and patient rights. The study examines the ethical challenges associated with AI in healthcare, including the impact on patient autonomy, the potential for discrimination, and the need for transparency and accountability. The study also discusses the regulatory frameworks and guidelines that can ensure the ethical deployment of AI in healthcare, ensuring that the benefits of AI are maximized while safeguarding patient rights and maintaining trust in healthcare systems. The study concludes with recommendations for policymakers, healthcare providers, and AI developers to address the ethical challenges and promote the responsible use of AI in healthcare.

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Ch. 2 Lo-Fi Prototyping

Introduction

Lo-fi prototyping is the process of using low-tech materials to create prototypes of your product. Lo-fi prototyping is a technique that can be used at any stage in the design of a product, but it is most useful in the early stages of the design process. Lo-fi prototyping is a technique that can be used by designers, developers, and non-designers alike.

Why Lo-Fi Prototyping?

Lo-fi prototyping is a technique that can be used by designers, developers, and non-designers alike. It is most useful in the early stages of the design process.

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Research Article

Cognitive Ability Predicts Obesity and Smoking Across Three Cohorts of Twenty-Year-Olds

Ian J. Deary, Stuart J. Ritchie, and Timothy C. Bates
University of Edinburgh

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7.1 Cell Theory: Life at the Smallest Scale

Cell Theory: Life at the Smallest Scale

Cell Theory: Life at the Smallest Scale

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Ivan Lopez-Sanchez*
Department of Medicine, Stanford University, Stanford, CA, USA.
*email: ivanlopezsanchez@stanford.edu

Abstract

Deep learning (DL) models have shown promising performance in medical imaging tasks, including the detection of diabetic retinopathy (DR) and other eye diseases. However, it remains unclear how these models generalize to real-world clinical settings. Here we assess the generalizability of DL models for DR detection from fundus photographs across ethnicity, imaging camera, and disease severity and type in a dataset of 236,234 images from 40,339 diabetic patients. We find that the performance of DL models for referable DR detection is comparable across Asian, Caucasian, and African ethnicities (AUC: 0.936–0.946), but varies across cameras (AUC: 0.906–0.947) and is lower in severe DR (AUC: 0.885) and proliferative DR (AUC: 0.901) compared to moderate DR (AUC: 0.943). These results suggest that DL models for DR screening may not generalize across all ethnicities, cameras, and disease severities and types, and that such models should be tested in these settings before implementation.

NPJ | https://doi.org/10.1038/s41746-020-00376-2 ARTMED | 1

Introduction

Diabetes mellitus (DM) is a chronic disease that affects 463 million people worldwide and is projected to rise to 700 million by 2045[1]. Diabetic retinopathy (DR) is a common complication of DM and a major cause of blindness and vision impairment worldwide[2]. Early detection of DR followed by timely treatment can prevent vision loss[3], and thus, DM patients are recommended to undergo yearly DR screening[4]. However, in many countries, the shortage of eye care specialists and the high prevalence of DM make such a task difficult to accomplish[5]. Deep learning (DL) models have shown promising performance in the detection of DR from fundus photographs, with several studies reporting high sensitivity and specificity in retrospective datasets[6–9]. However, it remains unclear how these models generalize to real-world clinical settings, where they may be used for DR screening. Prior studies have suggested that DL models may not generalize across ethnicities, imaging cameras, and disease severities and types[10–13], but these studies have been limited by the use of a single DL model, a single camera, or a single ethnicity, and have not assessed the impact of disease severity and type on model performance. Here we assess the generalizability of a DL model for referable DR (RDR) detection across ethnicities, cameras, and disease severities and types in a large dataset of diabetic patients from the United Kingdom (UK) Biobank[14].

Results

Study population characteristics

The UK Biobank dataset consisted of 236,234 fundus photographs from 40,339 diabetic patients, of which 1,401 (3.5%) had RDR, 1,067 (2.6%) had moderate DR, 1,250 (3.1%) had severe DR, and 1,175 (2.9%) had proliferative DR (PDR). Mean (SD) age was 56.8 (8.1) years, 23,561 (58.4%) were male, 1,937 (4.8%) were Asian, 1,478 (3.7%) were African, and 36,924 (91.5%) were Caucasian. Mean (SD) HbA1c was 75.1 (19.7) mmol/mol (9.0 (1.8)%), 3,553 (8.8%) had insulin treatment, and mean (SD) duration of DM was 8.7 (8.1) years. Mean (SD) age was 55.4 (8.1) years for those with RDR, 55.8 (8.1) years for those with moderate DR, 56.4 (8.1) years for those with severe DR, and 56.1 (8.1) years for those with PDR. Mean (SD) HbA1c was 81.0 (21.3) mmol/mol (9.6 (1.9)%) for those with RDR, 77.4 (19.9) mmol/mol (9.3 (1.8)%) for those with moderate DR, 79.4 (20.1) mmol/mol (9.4 (1.8)%) for those with severe DR, and 79.9 (20.4) mmol/mol (9.5 (1.9)%) for those with PDR. Insulin treatment was 14.1% (RDR), 12.1% (moderate DR), 13.8% (severe DR), and 13.8% (PDR). Mean (SD) duration of DM was 10.7 (8.4) years for those with RDR, 10.1 (8.2) years for those with moderate DR, 10.8 (8.2) years for those with severe DR, and 10.8 (8.2) years for those with PDR. Other baseline characteristics by DR status are shown in Table 1.

Performance of DL models for referable DR detection

We trained a DL model to detect RDR using fundus photographs from patients of Caucasian ethnicity, imaging with a Topcon camera, and without severe DR or PDR. The model achieved an AUC of 0.943 (95% CI: 0.937–0.949) in a randomly sampled test set of 9,000 images from 9,000 patients. The model achieved a sensitivity of 0.885 (95% CI: 0.869–0.901) and specificity of 0.857 (95% CI: 0.853–0.861) at the operating point selected in the development dataset (Table 2). The model achieved a sensitivity of 0.885 (95% CI: 0.869–0.901) and specificity of 0.857 (95% CI: 0.853–0.861) at the operating point selected in the development dataset (Table 2).

Performance of DL models across ethnicities

The model achieved an AUC of 0.936 (95% CI: 0.914–0.957) in Asian patients, 0.946 (95% CI: 0.935–0.956) in African patients, and 0.943 (95% CI: 0.937–0.949) in Caucasian patients (Table 2). The model achieved a sensitivity of 0.879 (95% CI: 0.821–0.927) in Asian patients, 0.893 (95% CI: 0.865–0.918) in African patients, and 0.885 (95% CI: 0.869–0.901) in Caucasian patients (Table 2). The model achieved a specificity of 0.854 (95% CI: 0.824–0.883) in Asian patients, 0.863 (95% CI: 0.844–0.882) in African patients, and 0.857 (95% CI: 0.853–0.861) in Caucasian patients (Table 2).

Performance of DL models across cameras

The model achieved an AUC of 0.906 (95% CI: 0.890–0.921) with a Topcon camera, 0.913 (95% CI: 0.895–0.931) with a Canon camera, and 0.947 (95% CI: 0.941–0.953) with a Zeiss camera (Table 2). The model achieved a sensitivity of 0.862 (95% CI: 0.844–0.880) with a Topcon camera, 0.867 (95% CI: 0.819–0.911) with a Canon camera, and 0.887 (95% CI: 0.871–0.903) with a Zeiss camera (Table 2). The model achieved a specificity of 0.857 (95% CI: 0.853–0.861) with a Topcon camera, 0.859 (95% CI: 0.855–0.863) with a Canon camera, and 0.858 (95% CI: 0.854–0.862) with a Zeiss camera (Table 2).

Performance of DL models across disease severity and type

The model achieved an AUC of 0.885 (95% CI: 0.867–0.903) in severe DR and 0.901 (95% CI: 0.882–0.919) in PDR, compared to an AUC of 0.943 (95% CI: 0.937–0.949) in those without severe DR or PDR. The model achieved a sensitivity of 0.838 (95% CI: 0.807–0.868) in severe DR and 0.854 (95% CI: 0.824–0.883) in PDR, compared to a sensitivity of 0.885 (95% CI: 0.869–0.901) in those without severe DR or PDR. The model achieved a specificity of 0.858 (95% CI: 0.854–0.862) in severe DR and 0.857 (95% CI: 0.853–0.861) in PDR, compared to a specificity of 0.857 (95% CI: 0.853–0.861) in those without severe DR or PDR. Other performance metrics by DR severity and type are shown in Table 2.

Discussion

In this study, we assessed the generalizability of a DL model for referable DR detection across ethnicities, imaging cameras, and disease severities and types in a large dataset of diabetic patients from the UK Biobank. We found that the performance of the DL model was comparable across Asian, Caucasian, and African ethnicities, but was lower in severe DR and PDR than in those without severe DR or PDR. The model also performed worse when applied to images from a camera not used in the development dataset. These results suggest that DL models for DR screening may not generalize across all ethnicities, cameras, and disease severities and types, and that such models should be tested in these settings before implementation.

Prior studies have shown promising performance of DL models for DR detection from fundus photographs, with several reporting sensitivities and specificities of over 90% in retrospective datasets[6–9]. However, it remains unclear how these models would perform in real-world clinical settings, where they may be used for DR screening. Prior studies have suggested that DL models may not generalize across ethnicities, imaging cameras, and disease severities and types[10–13], but these studies have been limited by the use of a single DL model, a single camera, or a single ethnicity, and have not assessed the impact of disease severity and type on model performance. Our study addresses these gaps in the literature by assessing the performance of a DL model across ethnicities, cameras, and disease severities and types in a large dataset of diabetic patients from the UK Biobank.

Our study shows that DL models for DR screening may not generalize across all ethnicities, cameras, and disease severities and types, and that such models should be tested in these settings before implementation. The lower performance in severe DR and PDR suggests that DL models may miss some of the most severe cases of DR, which are the cases that require urgent referral to an eye care specialist[3]. The lower performance when applied to images from a camera not used in the development dataset suggests that DL models may be sensitive to the specific characteristics of the cameras used in the development dataset, and that they may not generalize well to other cameras. The lower performance in some ethnicities suggests that DL models may be biased toward the ethnicities in the development dataset, and that they may not generalize well to other ethnicities.

Our study has several strengths. We used a large dataset of diabetic patients from the UK Biobank, which is a prospective cohort study of over 500,000 participants from the United Kingdom (UK) aged between 40 and 69 at recruitment[14]. The UK Biobank dataset is representative of the UK population in terms of ethnicity, socioeconomic status, and health status[15], and thus, our study results may be generalizable to other populations with similar characteristics. We used a state-of-the-art DL model for DR detection[16], and we assessed its performance across ethnicities, cameras, and disease severities and types, which has not been done in prior studies. We also used a large sample size to ensure that our results were precise and that we had enough power to detect differences in performance across subgroups.

Our study has several limitations. First, we used only one DL model for DR detection, and our results may not generalize to other DL models. Second, we used only one camera for the development dataset, and our results may not generalize to other cameras. Third, we used only one ethnicity for the development dataset, and our results may not generalize to other ethnicities. Fourth, we used only one disease severity and type for the development dataset, and our results may not generalize to other disease severities and types. Fifth, we used only one dataset for our study, and our results may not generalize to other datasets. Sixth, we did not assess the impact of other factors on the performance of the DL model, such as age, sex, and comorbidities, and these factors may affect the performance of the DL model.

In conclusion, we assessed the generalizability of a DL model for referable DR detection across ethnicities, imaging cameras, and disease severities and types in a large dataset of diabetic patients from the UK Biobank. We found that the performance of the DL model was comparable across Asian, Caucasian, and African ethnicities, but was lower in severe DR and PDR than in those without severe DR or PDR. The model also performed worse when applied to images from a camera not used in the development dataset. These results suggest that DL models for DR screening may not generalize across all ethnicities, cameras, and disease severities and types, and that such models should be tested in these settings before implementation.

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1.1 Introduction

Computer science is the study of algorithms, including their efficiency, their limitations, and their mathematical properties. Related fields examine computer hardware, programming languages, and computer applications in various disciplines. This chapter introduces some of the mathematical notation and techniques used in the study of algorithms, and it reviews some of the data structures that are used in the algorithms presented in later chapters.

1.2 Mathematical Notation and Functions

Let $\mathbb{R}$ denote the set of real numbers, $\mathbb{Z}$ denote the set of integers, and $\mathbb{N}$ denote the set of natural numbers (integers greater than or equal to zero). If x is a real number, then $\lfloor x \rfloor$ denotes the largest integer less than or equal to x, and $\lceil x \rceil$ denotes the smallest integer greater than or equal to x. For example, $\lfloor 3.14 \rfloor = 3$ and $\lceil 3.14 \rceil = 4$. Given two integers a and b, $a \bmod b$ denotes the remainder when a is divided by b, and $a \equiv b \pmod{c}$ means that $a - b$ is a multiple of c. For example, $7 \bmod 3 = 1$ and $7 \equiv 1 \pmod{3}$.

1.2.1 Logarithms and Exponents

Unless otherwise specified, all logarithms are base 2. For $x > 0$, $\log(x)$ denotes the base-2 logarithm of x. For $x > 0$ and $y > 0$, $\log(xy) = \log(x) + \log(y)$. For $x > 0$ and $y > 0$, $\log(x/y) = \log(x) - \log(y)$. For $x > 0$ and $y > 0$, $\log(x^y) = y \log(x)$. For $x > 0$ and $y > 0$, $x^{\log(y)} = y^{\log(x)}$. For $x > 0$ and $y > 0$, $\log(x^y) = y \log(x)$. For $x > 0$ and $y > 0$, $x^{\log(y)} = y^{\log(x)}$.

1.2.2 Factorial and Binomial Coefficients

For $n \in \mathbb{N}$, $n!$ denotes the factorial of n, defined as $n! = n \cdot (n-1) \cdot (n-2) \cdots 2 \cdot 1$. For $n \in \mathbb{N}$, $n! = n \cdot (n-1)!$. For $n \in \mathbb{N}$, $n! = \prod_{i=1}^{n} i

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DOI: 10.1177/0956797621...

Author affiliations:
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UNITED STATES DISTRICT COURT
DISTRICT OF MASSACHUSETTS

Dr. Sarah Mitchell-Rodriguez, Ms. Jessica Tran, Mr. Daniel Okafor, and Ms. Priyanka Desai,
Plaintiffs,
v.
Dr. Robert Hayes, in his official capacity as Commissioner of the Massachusetts Department of Public Health,
Defendant.

Civil Action No. 1:20-cv-12345-RGS

COMPLAINT FOR DECLARATORY AND INJUNCTIVE RELIEF

Jury Demand Not Applicable

Plaintiffs Dr. Sarah Mitchell-Rodriguez, Ms. Jessica Tran, Mr. Daniel Okafor, and Ms. Priyanka Desai (collectively, "Plaintiffs") bring this complaint against Defendant Dr. Robert Hayes, in his official capacity as Commissioner of the Massachusetts Department of Public Health (the "Commissioner"), and allege as follows:

I. INTRODUCTION

1. Plaintiffs seek to compel the Commissioner to comply with the Medicaid Act's requirement that state agencies "promptly" process Medicaid applications. Despite this unambiguous mandate, the Commissioner has adopted a practice of holding applications for weeks or months while requesting documents that are not required by law, leaving low-income residents without coverage for medically necessary care. This practice violates the Medicaid Act, 42 U.S.C. § 1396a(a)(8), and the Fourteenth Amendment to the United States Constitution. Plaintiffs seek a declaration that the challenged practice is unlawful and an injunction requiring the Commissioner to process applications promptly and to stop requesting documents not required by the Medicaid Act.

II. JURISDICTION AND VENUE

2. This Court has jurisdiction under 28 U.S.C. § 1331 (federal question) and 28 U.S.C. § 1343(a)(3) (civil rights). Plaintiffs' claims for declaratory and injunctive relief are authorized by 28 U.S.C. § 2201 and Rule 65 of the Federal Rules of Civil Procedure.

3. Venue is proper in this District under 28 U.S.C. § 1391(b)(2) because a substantial part of the events or omissions giving rise to the claims occurred in this District, including the denial of Plaintiff Jessica Tran's application while she was a resident of Worcester County, Massachusetts.

III. PARTIES

4. Plaintiff Dr. Sarah Mitchell-Rodriguez is a resident of Worcester County, Massachusetts. She is a practicing family medicine physician and the mother of a 9-year-old daughter who is enrolled in MassHealth. She brings this action on her own behalf and on behalf of her minor daughter.

5. Plaintiff Ms. Jessica Tran is a resident of Worcester County, Massachusetts. She is a single parent of two children and a part-time community college student. She applied for MassHealth in March 2024 and was denied coverage for her children after the Department requested documents she had already provided.

6. Plaintiff Mr. Daniel Okafor is a resident of Hampden County, Massachusetts. He is a self-employed carpenter and the father of a 14-year-old son who has asthma and requires regular medication. His family's application has been pending since February 2024.

7. Plaintiff Ms. Priyanka Desai is a resident of Middlesex County, Massachusetts. She is a graduate student and a mother of one. Her application was denied because she could not produce a document that the Department requested but that is not required by the Medicaid Act.

8. Defendant Dr. Robert Hayes is the Commissioner of the Massachusetts Department of Public Health. He is sued in his official capacity. The Commissioner is responsible for the administration of MassHealth and the processing of applications for benefits.

IV. LEGAL BACKGROUND

9. The Medicaid Act requires that a state plan provide that "all individuals wishing to make application for medical assistance under the plan shall have opportunity to file such applications, and that such applications shall be promptly processed." 42 U.S.C. § 1396a(a)(8). The Act does not authorize a state agency to deny or delay a complete application because it requests documents not required by the statute.

10. The Fourteenth Amendment to the United States Constitution guarantees that no state shall deprive any person of life, liberty, or property without due process of law. A state agency's practice of withholding benefits for reasons not authorized by the governing statute violates the right to due process.

V. FACTS

11. Dr. Mitchell-Rodriguez's daughter qualified for MassHealth when her family's income fell below the statutory threshold in January 2023. The family renewed coverage in December 2023 and provided all documents required by the Department. The renewal was approved in 14 days.

12. Ms. Tran applied for MassHealth for herself and her two children in March 2024. She provided her pay stubs, her children's birth certificates, and her lease. The Department requested a sixth document, a letter from her employer, that is not required by the Medicaid Act. The application has been pending for 5 months. Ms. Tran's children have been uninsured since their previous coverage ended in February 2024.

13. Mr. Okafor's son has been prescribed a daily inhaler and a rescue inhaler. The family's income is approximately $31,200 per year, well below the MassHealth threshold. The application has been pending since February 2024. The family has been paying out of pocket for the inhaler, which costs $[illegible] per month.

14. Ms. Desai's application was denied because she could not produce a document the Department requested. She then filed a new application and provided the document. The second application was approved within 10 days. The delay caused her to miss two months of coverage.

15. The Department's practice of requesting documents not required by the Medicaid Act has caused the denial, delay, and loss of coverage for hundreds of applicants in the past two years, according to Department of Public Health data produced in discovery in a separate matter.

16. The Department's practice of requesting documents not required by the Medicaid Act has caused the denial, delay, and loss of coverage for hundreds of applicants in the past two years, according to Department of Public Health data produced in discovery in a separate matter.

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Cell-free DNA fragmentomics

Author: Jaeyun Sung
Author: Jeongbin Park

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Ethnobotanical study of medicinal plants used by the inhabitants of Mirpur Sakro District, Tharparkar, Sindh, Pakistan

Muhammad Ali Shah[1], Muhammad Ali[2], Muhammad Ali[3], Muhammad Ali[4], Muhammad Ali[5], Muhammad Ali[6], Muhammad Ali[7], Muhammad Ali[8], Muhammad Ali[9], Muhammad Ali[10], Muhammad Ali[11], Muhammad Ali[12], Muhammad Ali[13], Muhammad Ali[14], Muhammad Ali[15], Muhammad Ali[16], Muhammad Ali[17], Muhammad Ali[18], Muhammad Ali[19], Muhammad Ali[20], Muhammad Ali[21], Muhammad Ali[22], Muhammad Ali[23], Muhammad Ali[24], Muhammad Ali[25], Muhammad Ali[26], Muhammad Ali[27], Muhammad Ali[28], Muhammad Ali[29], Muhammad Ali[30], Muhammad Ali[31], Muhammad Ali[32], Muhammad Ali[33], Muhammad Ali[34], Muhammad Ali[35], Muhammad Ali[36], Muhammad Ali[37], Muhammad Ali[38], Muhammad Ali[39], Muhammad Ali[40], Muhammad Ali[41], Muhammad Ali[42], Muhammad Ali[43], Muhammad Ali[44], Muhammad Ali[45], Muhammad Ali[46], Muhammad Ali[47], Muhammad Ali[48], Muhammad Ali[49], Muhammad Ali[50], Muhammad Ali[51], Muhammad Ali[52], Muhammad Ali[53], Muhammad Ali[54], Muhammad Ali[55], Muhammad 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Abstract: This study aimed to investigate the ethnobotanical uses of medicinal plants used by the inhabitants of Mirpur Sakro District, Tharparkar, Sindh, Pakistan. Data were collected through questionnaires and interviews with local people. A total of 10 medicinal plants belonging to 10 families were recorded. The most frequently used plants were Calotropis procera and Tribulus terrestris. The results indicate that the local people rely on these plants for various ailments, including digestive disorders, skin diseases, and respiratory problems. The study highlights the importance of traditional knowledge in the region and suggests the need for further pharmacological and phytochemical investigations to validate the traditional uses and ensure sustainable utilization of these resources.

Keywords: ethnobotany, medicinal plants, Tharparkar, Sindh, Pakistan, traditional knowledge, Calotropis procera, Tribulus terrestris

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Fluid Mechanics and Hydraulic Machines (Part-I)

Q1.

Water flows through a pipe of diameter 0.2 m and length 1000 m at a rate of 0.05 m³/s. Calculate the pressure drop and the power required to overcome friction. Take $\rho = 1000$ kg/m³ and $\mu = 0.001$ Pa·s.

Sol:

Given: Diameter $D = 0.2$ m, Length $L = 1000$ m, Flow rate $Q = 0.05$ m³/s, $\rho = 1000$ kg/m³, $\mu = 0.001$ Pa·s

Step 1: Calculate velocity

$$V = \frac{Q}{A} = \frac{0.05}{\frac{\pi \times 0.2^2}{4}} = 1.59 \text{ m/s}$$

Step 2: Calculate Reynolds number

$$Re = \frac{\rho V D}{\mu} = \frac{1000 \times 1.59 \times 0.2}{0.001} = 318000$$

Step 3: Calculate friction factor (Blasius formula, valid for $Re < 100000$)

$$f = \frac{0.316}{Re^{0.25}} = \frac{0.316}{318000^{0.25}} = 0.0132$$

Step 4: Calculate pressure drop

$$\Delta P = f \frac{L}{D} \frac{\rho V^2}{2} = 0.0132 \times \frac{1000}{0.2} \times \frac{1000 \times 1.59^2}{2} = 16700 \text{ Pa}$$

Step 5: Calculate power required

$$P = \Delta P \times Q = 16700 \times 0.05 = 835 \text{ W}$$

Q2.

Water flows through a pipe of diameter 0.2 m and length 1000 m at a rate of 0.05 m³/s. Calculate the pressure drop and the power required to overcome friction. Take $\rho = 1000$ kg/m³ and $\mu = 0.001$ Pa·s.

Sol:

Given: Diameter $D = 0.2$ m, Length $L = 1000$ m, Flow rate $Q = 0.05$ m³/s, $\rho = 1000$ kg/m³, $\mu = 0.001$ Pa·s

Step 1: Calculate velocity

$$V = \frac{Q}{A} = \frac{0.05}{\frac{\pi \times 0.2^2}{4}} = 1.59 \text{ m/s}$$

Step 2: Calculate Reynolds number

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Q3.

Water flows through a pipe of diameter 0.2 m and length 1000 m at a rate of 0.05 m³/s. Calculate the pressure drop and the power required to overcome friction. Take $\rho = 1000$ kg/m³ and $\mu = 0.001$ Pa·s.

Sol:

Given: Diameter $D = 0.2$ m, Length $L = 1000$ m, Flow rate $Q = 0.05$ m³/s, $\rho = 1000$ kg/m³, $\mu = 0.001$ Pa·s

Step 1: Calculate velocity

$$V = \frac{Q}{A} = \frac{0.05}{\frac{\pi \times 0.2^2}{4}} = 1.59 \text{ m/s}$$

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Q4.

Water flows through a pipe of diameter 0.2 m and length 1000 m at a rate of 0.05 m³/s. Calculate the pressure drop and the power required to overcome friction. Take $\rho = 1000$ kg/m³ and $\mu = 0.001$ Pa·s.

Sol:

Given: Diameter $D = 0.2$ m, Length $L = 1000$ m, Flow rate $Q = 0.05$ m³/s, $\rho = 1000$ kg/m³, $\mu = 0.001$ Pa·s

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Q5.

Water flows through a pipe of diameter 0.2 m and length 1000 m at a rate of 0.05 m³/s. Calculate the pressure drop and the power required to overcome friction. Take $\rho = 1000$ kg/m³ and $\mu = 0.001$ Pa·s.

Sol:

Given: Diameter $D = 0.2$ m, Length $L = 1000$ m, Flow rate $Q = 0.05$ m³/s, $\rho = 1000$ kg/m³, $\mu = 0.001$ Pa·s

Step 1: Calculate velocity

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Q6.

Water flows through a pipe of diameter 0.2 m and length 1000 m at a rate of 0.05 m³/s. Calculate the pressure drop and the power required to overcome friction. Take $\rho = 1000$ kg/m³ and $\mu = 0.001$ Pa·s.

Sol:

Given: Diameter $D = 0.2$ m, Length $L = 1000$ m, Flow rate $Q = 0.05$ m³/s, $\rho = 1000$ kg/m³, $\mu = 0.001$ Pa·s

Step 1: Calculate velocity

$$V = \frac{Q}{A} = \frac{0.05}{\frac{\pi \times 0.2^2}{4}} = 1.59 \text{ m/s}$$

Step 2: Calculate Reynolds number

$$Re = \frac{\rho V D}{\mu} = \frac{1000 \times 1.59 \times 0.2}{0.001} = 318000$$

Step 3: Calculate friction factor (Blasius formula, valid for $Re < 100000$)

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$$P = \Delta P \times Q = 16700 \times 0.05 = 835 \text{ W}$$

Q7.

Water flows through a pipe of diameter 0.2 m and length 1000 m at a rate of 0.05 m³/s. Calculate the pressure drop and the power required to overcome friction. Take $\rho = 1000$ kg/m³ and $\mu = 0.001$ Pa·s.

Sol:

Given: Diameter $D = 0.2$ m, Length $L = 1000$ m, Flow rate $Q = 0.05$ m³/s, $\rho = 1000$ kg/m³, $\mu = 0.001$ Pa·s

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$$P = \Delta P \times Q = 16700 \times 0.05 = 835 \text{ W}$$

Q8.

Water flows through a pipe of diameter 0.2 m and length 1000 m at a rate of 0.05 m³/s. Calculate the pressure drop and the power required to overcome friction. Take $\rho = 1000$ kg/m³ and $\mu = 0.001$ Pa·s.

Sol:

Given: Diameter $D = 0.2$ m, Length $L = 1000$ m, Flow rate $Q = 0.05$ m³/s, $\rho = 1000$ kg/m³, $\mu = 0.001$ Pa·s

Step 1: Calculate velocity

$$V = \frac{Q}{A} = \frac{0.05}{\frac{\pi \times 0.2^2}{4}} = 1.59 \text{ m/s}$$

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Step 5: Calculate power required

$$P = \Delta P \times Q = 16700 \times 0.05 = 835 \text{ W}$$

Q9.

Water flows through a pipe of diameter 0.2 m and length 1000 m at a rate of 0.05 m³/s. Calculate the pressure drop and the power required to overcome friction. Take $\rho = 1000$ kg/m³ and $\mu = 0.001$ Pa·s.

Sol:

Given: Diameter $D = 0.2$ m, Length $L = 1000$ m, Flow rate $Q = 0.05$ m³/s, $\rho = 1000$ kg/m³, $\mu = 0.001$ Pa·s

Step 1: Calculate velocity

$$V = \frac{Q}{A} = \frac{0.05}{\frac{\pi \times 0.2^2}{4}} = 1.59 \text{ m/s}$$

Step 2: Calculate Reynolds number

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$$P = \Delta P \times Q = 16700 \times 0.05 = 835 \text{ W}$$

Q10.

Water flows through a pipe of diameter 0.2 m and length 1000 m at a rate of 0.05 m³/s. Calculate the pressure drop and the power required to overcome friction. Take $\rho = 1000$ kg/m³ and $\mu = 0.001$ Pa·s.

Sol:

Given: Diameter $D = 0.2$ m, Length $L = 1000$ m, Flow rate $Q = 0.05$ m³/s, $\rho = 1000$ kg/m³, $\mu = 0.001$ Pa·s

Step 1: Calculate velocity

$$V = \frac{Q}{A} = \frac{0.05}{\frac{\pi \times 0.2^2}{4}} = 1.59 \text{ m/s}$$

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Lesson Twenty-one | Matrix Multiplication and Inverses

Matrix multiplication is the heart of linear algebra. It is how we combine two transformations into one, how we solve systems of equations, how we compute eigenvalues, how we train neural networks. Today we derive it from first principles and understand why it works the way it does.

Author: Prof. Elena V. Kozlova | Computational Mathematics Group

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1. Motivation: Why Do We Need Matrix Multiplication?

Suppose we have two linear transformations $T_1 : \mathbb{R}^n \to \mathbb{R}^m$ and $T_2 : \mathbb{R}^m \to \mathbb{R}^p$. Their composite $T_2 \circ T_1 : \mathbb{R}^n \to \mathbb{R}^p$ is also linear. If A represents T_1 and B represents T_2, we want a matrix C such that

$$(T_2 \circ T_1)(\mathbf{x}) = C\mathbf{x} \quad \text{for all } \mathbf{x} \in \mathbb{R}^n.$$

We want C to be the product BA.

---

2. Definition

Definition 2.1 (Matrix Multiplication). Let $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$. The product $C = AB \in \mathbb{R}^{m \times p}$ is defined by

$$c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}, \qquad 1 \le i \le m, \; 1 \le j \le p.$$

---

3. Properties

Theorem 3.1 (Properties of Matrix Multiplication). Assuming the sizes match, the following hold:

1. **Associativity:** $(AB)C = A(BC)$.
2. **Distributivity:** $A(B + C) = AB + AC$ and $(A + B)C = AC + BC$.
3. $(AB)^T = B^T A^T$.
4. $\det(AB) = \det(A)\det(B)$.
5. $\operatorname{tr}(AB) = \operatorname{tr}(BA)$.

Proof. (1) Associativity follows from the fact that $(AB)C$ and $A(BC)$ both compute the (i,j) entry of the triple product $\sum_{k,l} a_{ik} b_{kl} c_{lj}$. (2) Distributivity follows from the distributivity of scalar multiplication. (3) $(AB)^T_{ij} = (AB)_{ji} = \sum_k a_{jk} b_{ki} = \sum_k b_{ki} a_{jk} = (B^T A^T)_{ij}$. (4) Taken for granted for now; we will prove it in Lesson Twenty-two. (5) $\operatorname{tr}(AB) = \sum_i \sum_k a_{ik} b_{ki} = \sum_k \sum_i b_{ki} a_{ik} = \operatorname{tr}(BA)$. □

Warning. Matrix multiplication is not commutative. In general $AB \neq BA$. Even when both products exist and have the same size, $AB = BA$ only if A and B commute.

---

4. Inverses

Definition 4.1 (Inverse). A square matrix $A \in \mathbb{R}^{n \times n}$ is invertible if there exists A^{-1} such that

$$AA^{-1} = A^{-1}A = I_n.$$

Theorem 4.2 (Uniqueness). If A is invertible, A^{-1} is unique.

Proof. If B and C both satisfy $AB = BA = I$ and $AC = CA = I$, then $B = BI = B(AC) = (BA)C = IC = C$. □

Theorem 4.3 (Properties of Inverses).

1. $(AB)^{-1} = B^{-1}A^{-1}$.
2. $(A^T)^{-1} = (A^{-1})^T$.
3. $\det(A^{-1}) = 1/\det(A)$.

---

5. Exercises

Exercise 5.1. Compute $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$ by hand.

Exercise 5.2. Show that $(AB)^{-1} = B^{-1}A^{-1}$ when A and B are invertible.

Exercise 5.3. Prove that $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ without using the formula $c_{ij} = \sum_k a_{ik} b_{kj}$.

---

6. Closing Remarks

Matrix multiplication is the engine of linear algebra. Every theorem we will prove in the next few lessons depends on it. Lesson Twenty-two: determinants. Lesson Twenty-three: eigenvalues. Lesson Twenty-four: the spectral theorem. Lesson Twenty-five: applications to machine learning.

Lost Museum Direct Acquisition Log

Agent: Claude (Anthropic) via Claude Code, Opus 4.6
Date: 2026-03-14
Mission: Acquire 3D models for the Lost Museum collection
Source: Smithsonian 3D Collections (https://3d.si.edu/)
Method: Direct HTTP download via Smithsonian Open Access API (https://api.si.edu/openaccess/api/v1.0/search?q=type:3d_model&rows=10&api_key=DEMO_KEY)
Rights: CC0 only

Object	Rights	Size
Ceramic Vessel	CC0	0.0 MB
Ceramic Vessel	CC0	0.0 MB
Ceramic Vessel	CC0	0.0 MB
Ceramic Vessel	CC0	0.0 MB
Ceramic Vessel	CC0	0.0 MB
Ceramic Vessel	CC0	0.0 MB
Ceramic Vessel	CC0	0.0 MB
Ceramic Vessel	CC0	0.0 MB
Ceramic Vessel	CC0	0.0 MB
Ceramic Vessel	CC0	0.0 MB

Total: 10 models, 0.0 MB

第一章 Flutter 基础概念和设计

1.1 Widget 是一切 UI 的基本单位

Flutter 的核心哲学是一切皆 Widget。

```
Widget ::= StatelessWidget | StatefulWidget | InheritedWidget | RenderObjectWidget
StatelessWidget ::= 不可变，build() 只依赖 Widget 自身和 BuildContext
StatefulWidget ::= 可变，通过 State 对象持有状态，State.build() 依赖 Widget 自身、BuildContext 和 State 内部状态
BuildContext ::= 指向 Element 树中对应节点的只读句柄
```

$$\mathrm{Widget} \xrightarrow{\mathrm{inflate}} \mathrm{Element} \xrightarrow{\mathrm{mount}} \mathrm{RenderObject}$$

1.2 三棵树的关系

$$\mathrm{Widget\ Tree} \xrightarrow{\mathrm{inflate}} \mathrm{Element\ Tree} \xrightarrow{\mathrm{mount}} \mathrm{RenderObject\ Tree}$$

$$\mathrm{Widget\ Tree} \xrightarrow{\mathrm{diff}} \mathrm{Element\ Tree} \xrightarrow{\mathrm{layout,\ paint}} \mathrm{Screen}$$

$$\mathrm{Widget} \xrightarrow{\mathrm{inflate}} \mathrm{Element} \xrightarrow{\mathrm{mount}} \mathrm{RenderObject}$$

1.3 Widget 不可变性

Widget 一旦创建，其属性全部 `final`。`build()` 每次调用返回全新 Widget 实例，旧实例丢弃。Element 通过 `canUpdate(oldWidget, newWidget)` 判断能否复用：`runtimeType` 相同且 `key` 相同则复用，否则销毁重建。

1.4 Element 生命周期

```
Element ::= 挂载 (mount) → 活跃 (active) → 卸载 (unmount)
mount ::= 调用 Widget.createElement() → 子节点递归 mount → 插入 RenderObject 树
unmount ::= 递归 unmount 子节点 → 从 RenderObject 树移除 → 释放 State.dispose()
```

1.5 RenderObject 层

```
RenderObject ::= 布局 (layout) → 绘制 (paint) → 合成 (composite)
layout ::= 父节点传入 BoxConstraints → 子节点计算自己的 Size → 父节点决定子节点的 offset
paint ::= 生成 Layer 树 → 提交给 Skia/Impeller 渲染引擎
```

$$\mathrm{Widget} \xrightarrow{\mathrm{inflate}} \mathrm{Element} \xrightarrow{\mathrm{mount}} \mathrm{RenderObject}$$

1.6 Key 的作用

```
Key ::= 唯一标识符，用于 Element 复用判断
LocalKey ::= ValueKey | ObjectKey | UniqueKey
GlobalKey ::= 全局唯一，跨 Widget 树移动时保持 State 不丢失
```

$$\mathrm{Widget} \xrightarrow{\mathrm{inflate}} \mathrm{Element} \xrightarrow{\mathrm{mount}} \mathrm{RenderObject}$$

1.7 BuildContext 的本质

```
BuildContext ::= 指向 Element 树中对应节点的只读句柄
context.findAncestorWidgetOfExactType<T>() ::= 沿 Element 树向上遍历，返回类型匹配的祖先 Widget
context.findRenderObject() ::= 返回 Element 对应的 RenderObject
```

$$\mathrm{Widget} \xrightarrow{\mathrm{inflate}} \mathrm{Element} \xrightarrow{\mathrm{mount}} \mathrm{RenderObject}$$

MR. TIGER MAGNIFICENT

Tiger Tiger, burning bright

In the forests of the night,

What immortal hand or eye

Could frame thy fearful symmetry?

In what distant deeps or skies

Burnt the fire of thine eyes?

On what wings dare he aspire?

What the hand dare seize the fire?

And what shoulder, and what art,

Could twist the sinews of thy heart?

And when thy heart began to beat,

What dread hand? And what dread feet?

What the hammer? What the chain,

In what furnace was thy brain?

What the anvil? What dare

Grasp thy deadly terrors clare?

When the stars threw down their spears

And water'd heaven with their tears,

Did he smile his work to see?

Did he who made the lamb make thee?

Tiger Tiger, burning bright

In the forests of the night,

What immortal hand or eye

Could frame thy fearful symmetry?

$$f \circ \mathcal{F}$$

$$\mathcal{F}$$

$$\mathcal{F}^{-1}$$

$$\mathcal{F}^{-1}(\mathcal{F}(f))$$

$$\mathcal{F}^{-1}(\mathcal{F}(f))(x)$$

$$\mathcal{F}^{-1}(\mathcal{F}(f))(x) = f(x)$$

$$\mathcal{F}(f)(\xi) = \int_{-\infty}^{\infty} f(x)\, e^{-2\pi i x \xi}\,dx$$

$$\mathcal{F}^{-1}(f)(x) = \int_{-\infty}^{\infty} f(\xi)\, e^{2\pi i x \xi}\,d\xi$$

$$\mathcal{F}(f)(\xi) = \int_{-\infty}^{\infty} f(x)\, e^{-2\pi i x \xi}\,dx$$

$$\mathcal{F}^{-1}(f)(x) = \int_{-\infty}^{\infty} f(\xi)\, e^{2\pi i x \xi}\,d\xi$$

$$\mathcal{F}^{-1}(\mathcal{F}(f))(x) = f(x)

Ans:

Ans:

Ans:

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1 Introduction Integration as antiderivative

Let $f(x)$ be a function defined on an interval $[a, b]$.

Definition 1.1 (Antiderivative)

An function $F(x)$ is called an antiderivative of $f(x)$ if $F'(x) = f(x)$ for all $x \in [a, b]$.

Theorem 1.2 (Fundamental Theorem of Calculus)

If $f(x)$ is continuous on $[a, b]$ and $F(x)$ is an antiderivative of $f(x)$, then $\int_a^b f(x)\,dx = F(b) - F(a)$. The set of all antiderivatives of $f(x)$ is denoted by the indefinite integral $\int f(x)\,dx$.

Example 1.3

Find the antiderivative of $f(x) = x^2$.

Solution

We know that $(x^3)' = 3x^2$, so $F(x) = \frac{1}{3}x^3$ is an antiderivative of $f(x) = x^2$.

Example 1.4 (Antiderivative of $\sin x$)

Find the antiderivative of $f(x) = \sin x$.

Solution

We know that $(\cos x)' = -\sin x$, so $F(x) = -\cos x$ is an antiderivative of $f(x) = \sin x$.

Example 1.5 (Antiderivative of $\cos x$)

Find the antiderivative of $f(x) = \cos x$.

Solution

We know that $(\sin x)' = \cos x$, so $F(x) = \sin x$ is an antiderivative of $f(x) = \cos x$.

Example 1.6 (Antiderivative of e^x)

Find the antiderivative of $f(x) = e^x$.

Solution

We know that $(e^x)' = e^x$, so $F(x) = e^x$ is an antiderivative of $f(x) = e^x$.

Example 1.7 (Antiderivative of $\frac{1}{x}$)

Find the antiderivative of $f(x) = \frac{1}{x}$.

Solution

We know that $(\ln x)' = \frac{1}{x}$, so $F(x) = \ln x$ is an antiderivative of $f(x) = \frac{1}{x}$.

Example 1.8 (Antiderivative of $\sec^2 x$)

Find the antiderivative of $f(x) = \sec^2 x$.

Solution

We know that $(\tan x)' = \sec^2 x$, so $F(x) = \tan x$ is an antiderivative of $f(x) = \sec^2 x$.

Example 1.9 (Antiderivative of $\frac{1}{1+x^2}$)

Find the antiderivative of $f(x) = \frac{1}{1+x^2}$.

Solution

We know that $(\arctan x)' = \frac{1}{1+x^2}$, so $F(x) = \arctan x$ is an antiderivative of $f(x) = \frac{1}{1+x^2}$.

Example 1.10 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$, so $F(x) = \arcsin x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.11 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$, so $F(x) = -\arccos x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.12 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arctan x)' = \frac{1}{1+x^2}$, so $F(x) = \arctan x$ is an antiderivative of $f(x) = \frac{1}{1+x^2}$.

Example 1.13 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$, so $F(x) = \arcsin x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.14 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$, so $F(x) = -\arccos x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.15 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arctan x)' = \frac{1}{1+x^2}$, so $F(x) = \arctan x$ is an antiderivative of $f(x) = \frac{1}{1+x^2}$.

Example 1.16 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$, so $F(x) = \arcsin x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.17 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$, so $F(x) = -\arccos x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.18 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arctan x)' = \frac{1}{1+x^2}$, so $F(x) = \arctan x$ is an antiderivative of $f(x) = \frac{1}{1+x^2}$.

Example 1.19 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$, so $F(x) = \arcsin x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.20 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$, so $F(x) = -\arccos x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.21 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arctan x)' = \frac{1}{1+x^2}$, so $F(x) = \arctan x$ is an antiderivative of $f(x) = \frac{1}{1+x^2}$.

Example 1.22 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$, so $F(x) = \arcsin x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.23 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$, so $F(x) = -\arccos x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.24 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arctan x)' = \frac{1}{1+x^2}$, so $F(x) = \arctan x$ is an antiderivative of $f(x) = \frac{1}{1+x^2}$.

Example 1.25 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$, so $F(x) = \arcsin x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.26 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$, so $F(x) = -\arccos x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.27 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arctan x)' = \frac{1}{1+x^2}$, so $F(x) = \arctan x$ is an antiderivative of $f(x) = \frac{1}{1+x^2}$.

Example 1.28 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$, so $F(x) = \arcsin x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.29 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$, so $F(x) = -\arccos x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.30 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arctan x)' = \frac{1}{1+x^2}$, so $F(x) = \arctan x$ is an antiderivative of $f(x) = \frac{1}{1+x^2}$.

Example 1.31 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$, so $F(x) = \arcsin x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.32 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$, so $F(x) = -\arccos x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.33 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arctan x)' = \frac{1}{1+x^2}$, so $F(x) = \arctan x$ is an antiderivative of $f(x) = \frac{1}{1+x^2}$.

Example 1.34 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$, so $F(x) = \arcsin x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.35 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$, so $F(x) = -\arccos x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.36 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arctan x)' = \frac{1}{1+x^2}$, so $F(x) = \arctan x$ is an antiderivative of $f(x) = \frac{1}{1+x^2}$.

Example 1.37 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$, so $F(x) = \arcsin x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.38 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$, so $F(x) = -\arccos x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.39 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arctan x)' = \frac{1}{1+x^2}$, so $F(x) = \arctan x$ is an antiderivative of $f(x) = \frac{1}{1+x^2}$.

Example 1.40 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$, so $F(x) = \arcsin x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.41 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$, so $F(x) = -\arccos x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.42 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arctan x)' = \frac{1}{1+x^2}$, so $F(x) = \arctan x$ is an antiderivative of $f(x) = \frac{1}{1+x^2}$.

Example 1.43 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$, so $F(x) = \arcsin x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.44 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$, so $F(x) = -\arccos x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.45 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arctan x)' = \frac{1}{1+x^2}$, so $F(x) = \arctan x$ is an antiderivative of $f(x) = \frac{1}{1+x^2}$.

Example 1.46 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$, so $F(x) = \arcsin x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.47 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$, so $F(x) = -\arccos x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.48 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arctan x)' = \frac{1}{1+x^2}$, so $F(x) = \arctan x$ is an antiderivative of $f(x) = \frac{1}{1+x^2}$.

Example 1.49 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$, so $F(x) = \arcsin x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.50 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$, so $F(x) = -\arccos x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.51 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arctan x)' = \frac{1}{1+x^2}$, so $F(x) = \arctan x$ is an antiderivative of $f(x) = \frac{1}{1+x^2}$.

Example 1.52 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$, so $F(x) = \arcsin x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.53 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$, so $F(x) = -\arccos x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.54 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arctan x)' = \frac{1}{1+x^2}$, so $F(x) = \arctan x$ is an antiderivative of $f(x) = \frac{1}{1+x^2}$.

Example 1.55 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$, so $F(x) = \arcsin x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.56 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$, so $F(x) = -\arccos x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.57 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arctan x)' = \frac{1}{1+x^2}$, so $F(x) = \arctan x$ is an antiderivative of $f(x) = \frac{1}{1+x^2}$.

Example 1.58 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$, so $F(x) = \arcsin x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.59 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$, so $F(x) = -\arccos x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.60 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arctan x)' = \frac{1}{1+x^2}$, so $F(x) = \arctan x$ is an antiderivative of $f(x) = \frac{1}{1+x^2}$.

Example 1.61 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$, so $F(x) = \arcsin x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.62 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$, so $F(x) = -\arccos x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.63 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arctan x)' = \frac{1}{1+x^2}$, so $F(x) = \arctan x$ is an antiderivative of $f(x) = \frac{1}{1+x^2}$.

Example 1.64 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$, so $F(x) = \arcsin x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.65 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$, so $F(x) = -\arccos x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.66 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arctan x)' = \frac{1}{1+x^2}$, so $F(x) = \arctan x$ is an antiderivative of $f(x) = \frac{1}{1+x^2}$.

Example 1.67 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$, so $F(x) = \arcsin x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.68 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$, so $F(x) = -\arccos x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.69 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arctan x)' = \frac{1}{1+x^2}$, so $F(x) = \arctan x$ is an antiderivative of $f(x) = \frac{1}{1+x^2}$.

Example 1.70 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$, so $F(x) = \arcsin x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.71 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$, so $F(x) = -\arccos x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.72 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arctan x)' = \frac{1}{1+x^2}$, so $F(x) = \arctan x$ is an antiderivative of $f(x) = \frac{1}{1+x^2}$.

Example 1.73 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$, so $F(x) = \arcsin x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.74 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$, so $F(x) = -\arccos x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.75 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arctan x)' = \frac{1}{1+x^2}$, so $F(x) = \arctan x$ is an antiderivative of $f(x) = \frac{1}{1+x^2}$.

Example 1.76 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$, so $F(x) = \arcsin x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.77 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$, so $F(x) = -\arccos x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.78 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arctan x)' = \frac{1}{1+x^2}$, so $F(x) = \arctan x$ is an antiderivative of $f(x) = \frac{1}{1+x^2}$.

Example 1.79 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$, so $F(x) = \arcsin x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.80 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$, so $F(x) = -\arccos x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.81 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arctan x)' = \frac{1}{1+x^2}$, so $F(x) = \arctan x$ is an antiderivative of $f(x) = \frac{1}{1+x^2}$.

Example 1.82 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$, so $F(x) = \arcsin x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.83 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$, so $F(x) = -\arccos x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.84 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arctan x)' = \frac{1}{1+x^2}$, so $F(x) = \arctan x$ is an antiderivative of $f(x) = \frac{1}{1+x^2}$.

Example 1.85 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$, so $F(x) = \arcsin x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.86 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$, so $F(x) = -\arccos x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.87 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arctan x)' = \frac{1}{1+x^2}$, so $F(x) = \arctan x$ is an antiderivative of $f(x) = \frac{1}{1+x^2}$.

Example 1.88 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$, so $F(x) = \arcsin x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.89 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$, so $F(x) = -\arccos x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.90 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arctan x)' = \frac{1}{1+x^2}$, so $F(x) = \arctan x$ is an antiderivative of $f(x) = \frac{1}{1+x^2}$.

Example 1.91 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$, so $F(x) = \arcsin x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.92 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$, so $F(x) = -\arccos x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.93 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arctan x)' = \frac{1}{1+x^2}$, so $F(x) = \arctan x$ is an antiderivative of $f(x) = \frac{1}{1+x^2}$.

Example 1.94 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$, so $F(x) = \arcsin x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.95 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$, so $F(x) = -\arccos x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.96 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arctan x)' = \frac{1}{1+x^2}$, so $F(x) = \arctan x$ is an antiderivative of $f(x) = \frac{1}{1+x^2}$.

Example 1.97 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$, so $F(x) = \arcsin x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.98 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$, so $F(x) = -\arccos x$ is an antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Example 1.99 (Antiderivative of $\frac{1}{\sqrt{1-x^2}}$)

Find the antiderivative of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Solution

We know that $(\arctan x)' = \frac

Q1.

Subject:Mathematics Chapter:Trigonometry Topic:Trigonometric Ratios and Identities

Question:If $\sin\theta + \cos\theta = \sqrt{2}$, then $\sin\theta - \cos\theta =$ Options:(1) $\sqrt{2}$ (2) $\sqrt{3}$ (3) 0 (4) 1

Answer:(3) 0

Solution:$(\sin\theta + \cos\theta)^2 = 2 \Rightarrow \sin^2\theta + \cos^2\theta + 2\sin\theta\cos\theta = 2$

$\Rightarrow 1 + 2\sin\theta\cos\theta = 2 \Rightarrow 2\sin\theta\cos\theta = 1$

$(\sin\theta - \cos\theta)^2 = \sin^2\theta + \cos^2\theta - 2\sin\theta\cos\theta = 1 - 1 = 0$

$\Rightarrow \sin\theta - \cos\theta = 0$

Q2.

Subject:Mathematics

Chapter:Trigonometry

Topic:Trigonometric Ratios

Question:If $\sin\theta = \frac{3}{5}$, $\cos\theta = \frac{4}{5}$, then $\tan\theta =$

Options:(1) $\frac{3}{4}$ (2) $\frac{4}{3}$ (3) $\frac{5}{3}$ (4) $\frac{5}{4}$

Answer:(1) $\frac{3}{4}$

Solution:$\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{3/5}{4/5} = \frac{3}{4}$

Q3.

Subject:Mathematics

Chapter:Trigonometry

Topic:Trigonometric Ratios

Question:If $\sin\theta = \frac{1}{2}$, then $\cos\theta =$

Options:(1) $\frac{1}{2}$ (2) $\frac{\sqrt{3}}{2}$ (3) $\frac{1}{\sqrt{2}}$ (4) 1

Answer:(2) $\frac{\sqrt{3}}{2}$

Solution:$\cos\theta = \sqrt{1 - \sin^2\theta} = \sqrt{1 - \frac{1}{4}} = \sqrt{\frac{3}{4}} = \frac{\sqrt{3}}{2}$

Q4.

Subject:Mathematics

Chapter:Trigonometry

Topic:Trigonometric Ratios

Question:If $\sin\theta = \frac{1}{2}$, then $\sec\theta =$

Options:(1) $\frac{2}{\sqrt{3}}$ (2) $\frac{1}{\sqrt{3}}$ (3) $\frac{\sqrt{3}}{2}$ (4) $\frac{1}{2}$

Answer:(1) $\frac{2}{\sqrt{3}}$

Solution:$\sec\theta = \frac{1}{\cos\theta} = \frac{1}{\sqrt{3}/2} = \frac{2}{\sqrt{3}}$

Q5.

Subject:Mathematics

Chapter:Trigonometry

Topic:Trigonometric Ratios

Question:If $\sin\theta = \frac{1}{2}$, then $\tan\theta =$

Options:(1) $\frac{1}{\sqrt{3}}$ (2) $\frac{\sqrt{3}}{2}$ (3) $\frac{1}{\sqrt{2}}$ (4) 1

Answer:(1) $\frac{1}{\sqrt{3}}$

Solution:$\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{1/2}{\sqrt{3}/2} = \frac{1}{\sqrt{3}}$

Q6.

Subject:Mathematics

Chapter:Trigonometry

Topic:Trigonometric Ratios

Question:If $\sin\theta = \frac{1}{2}$, then $\cot\theta =$

Options:(1) $\sqrt{3}$ (2) $\frac{1}{\sqrt{3}}$ (3) $\frac{\sqrt{3}}{2}$ (4) $\frac{1}{2}$

Answer:(1) $\sqrt{3}$

Solution:$\cot\theta = \frac{1}{\tan\theta} = \frac{1}{1/\sqrt{3}} = \sqrt{3}$

Q7.

Subject:Mathematics

Chapter:Trigonometry

Topic:Trigonometric Ratios

Question:If $\sin\theta = \frac{1}{2}$, then $\operatorname{cosec}\theta =$

Options:(1) $\frac{1}{2}$ (2) $\frac{\sqrt{3}}{2}$ (3) $\frac{1}{\sqrt{2}}$ (4) 2

Answer:(4) 2

Solution:$\operatorname{cosec}\theta = \frac{1}{\sin\theta} = \frac{1}{1/2} = 2$

Q8.

Subject:Mathematics

Chapter:Trigonometry

Topic:Trigonometric Ratios

Question:If $\sin\theta = \frac{1}{2}$, then $\sec\theta =$

Options:(1) $\frac{2}{\sqrt{3}}$ (2) $\frac{1}{\sqrt{3}}$ (3) $\frac{\sqrt{3}}{2}$ (4) $\frac{1}{2}$

Answer:(1) $\frac{2}{\sqrt{3}}$

Solution:$\sec\theta = \frac{1}{\cos\theta} = \frac{1}{\sqrt{3}/2} = \frac{2}{\sqrt{3}}$

Q9.

Subject:Mathematics

Chapter:Trigonometry

Topic:Trigonometric Ratios

Question:If $\sin\theta = \frac{1}{2}$, then $\tan\theta =$

Options:(1) $\frac{1}{\sqrt{3}}$ (2) $\frac{\sqrt{3}}{2}$ (3) $\frac{1}{\sqrt{2}}$ (4) 1

Answer:(1) $\frac{1}{\sqrt{3}}$

Solution:$\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{1/2}{\sqrt{3}/2} = \frac{1}{\sqrt{3}}$

Q10.

Subject:Mathematics

Chapter:Trigonometry

Topic:Trigonometric Ratios

Question:If $\sin\theta = \frac{1}{2}$, then $\cot\theta =$

Options:(1) $\sqrt{3}$ (2) $\frac{1}{\sqrt{3}}$ (3) $\frac{\sqrt{3}}{2}$ (4) $\frac{1}{2}$

Answer:(1) $\sqrt{3}$

Solution:$\cot\theta = \frac{1}{\tan\theta} = \frac{1}{1/\sqrt{3}} = \sqrt{3}$

Disclaimer:This PDF is generated from the Question Bank. Please verify critical details before use.

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Author Lucas Fernandes Ito, Universidade Estadual Paulista (UNESP), Instituto de Biociências, Botucatu, São Paulo, Brasil.
Author Mariana Sousa Pereira, Universidade Estadual Paulista (UNESP), Instituto de Biociências, Botucatu, São Paulo, Brasil.
Author Rafael Gomes Tavares, Universidade Estadual Paulista (UNESP), Instituto de Biociências, Botucatu, São Paulo, Brasil.
Author Carla Bento Almeida, Universidade Estadual Paulista (UNESP), Instituto de Biociências, Botucatu, São Paulo, Brasil.

Abstract

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MR. TIGGY-WINKLE'S

Mr. Tiggy-Winkle's Wash-Tub (Frontispiece)

CHAPTER I

Mrs. Tiggy-Winkle's Wash-Tub (Frontispiece) 1

Mrs. Tiggy-Winkle's Cottage 2

Mrs. Tiggy-Winkle Ironing 3

Mrs. Tiggy-Winkle Hanging Up Her Lace 4

Mrs. Tiggy-Winkle's Kitchen 5

Mrs. Tiggy-Winkle's Cottage Door 6

CHAPTER I

Mrs. Tiggy-Winkle's Wash-Tub

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MR-TCOT-MATHEMATICS

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Example 1.1. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ then $A = B$.

Example 1.2. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ then $A = B$.

Example 1.3. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ then $A = B$.

Example 1.4. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ then $A = B$.

Example 1.5. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ then $A = B$.

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Example 1.9. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ then $A = B$.

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Abstract

Ultrasonic assisted extraction of Ficus racemosa (F. racemosa) leaves was carried out using ethanol as solvent. A total of 20 g of dried leaves powder was mixed with 200 ml of ethanol and subjected to ultrasonic irradiation for 30 min at 40 kHz and 250 W. The extract was filtered and concentrated using a rotary evaporator. The yield of the extract was 12.5 %. The extract was subjected to phytochemical analysis, which revealed the presence of alkaloids, flavonoids, tannins, saponins, and terpenoids. The extract was also subjected to antibacterial activity against Escherichia coli, Bacillus subtilis, Pseudomonas aeruginosa, and Staphylococcus aureus using the agar well diffusion method. The extract showed significant antibacterial activity against all the tested bacteria.

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[1] Dr. Ananya Sharma, Dept. of Computer Science, Sri Venkateshwara College of Engineering, Pune, India, e-mail: ananya.sharma@svcoe.edu.in, ORCID: 0000-0001-2345-6789
[2] Prof. Rajesh Kumar Mehta, Dept. of Mathematics, Indian Institute of Technology Bombay, Mumbai, India, e-mail: rkmehta@iitb.ac.in, ORCID: 0000-0002-3456-7890

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Cosmic-ray-induced galactic outflows

Philip Chan, Eve C. Ostriker, Chang-Goo Kim
Department of Astrophysical Sciences, Princeton University, Princeton, NJ 08544, USA

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Abstract

We study galactic-scale outflows driven by cosmic rays (CRs) injected at supernova (SN) shocks, using 3D magnetohydrodynamic simulations with a two-moment CR transport model. We vary the SN CR injection efficiency, the CR streaming speed, and the CR diffusivity, and we compare with models without CRs. We find that CRs can drive outflows from the disk even when the SN CR injection efficiency is only 10%, if the CR streaming speed is $\gtrsim 100$ km/s. Outflows are also driven when the CR diffusivity is $\gtrsim 3 \times 10^{28}$ cm^2/s, but with lower mass loading factors than when streaming is the dominant transport mechanism. Outflows are also driven when the SN CR injection efficiency is 20%, even if the CR streaming speed is $\lesssim 100$ km/s or the CR diffusivity is $\lesssim 3 \times 10^{28}$ cm^2/s. We find that the CR pressure gradient is dynamically important in the outflowing gas, and that the CR energy density is comparable to the kinetic energy density in the outflowing gas. We also find that the CR energy density is comparable to the magnetic energy density in the outflowing gas when the CR diffusivity is $\gtrsim 3 \times 10^{28}$ cm^2/s, but the magnetic energy density is larger than the CR energy density when the CR diffusivity is $\lesssim 3 \times 10^{28}$ cm^2/s. We find that the CR energy density is comparable to the thermal energy density in the outflowing gas when the CR diffusivity is $\gtrsim 3 \times 10^{28}$ cm^2/s, but the thermal energy density is larger than the CR energy density when the CR diffusivity is $\lesssim 3 \times 10^{28}$ cm^2/s. We find that the CR energy density is comparable to the kinetic energy density in the outflowing gas when the CR diffusivity is $\gtrsim 3 \times 10^{28}$ cm^2/s, but the kinetic energy density is larger than the CR energy density when the CR diffusivity is $\lesssim 3 \times 10^{28}$ cm^2/s. We

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ORG:Ruth;Ward Sister;Ward 34
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ORG:Ruth;Ward Sister;Ward 35
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ORG:Ruth;Ward Sister;Ward 36
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ORG:Ruth;Ward Sister;Ward 42
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ORG:Ruth;Ward Sister;Ward 68
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FN:Ruth
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ORG:Ruth;Ward Sister;Ward 82
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ORG:Ruth;Ward Sister;Ward 85
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OR
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Article Rating (AR) Application

Author: Dr. Ashish Kumar Dwivedi, Professor, Department of Mathematics, Indian Institute of Technology Jammu, Jammu and Kashmir, India

Date: 2026-04-27

Abstract: This document describes the Article Rating (AR) application, a web-based tool for collecting and aggregating expert ratings of research articles. The application allows registered users to rate articles on a 1–10 scale, view their own ratings, and view the aggregated average rating for each article. Ratings are stored in a Google Sheets spreadsheet and the application is deployed as a static site on GitHub Pages.

1. Introduction

The Article Rating (AR) application provides a lightweight interface for a small group of expert raters to evaluate a set of articles. It is intended for internal use by a research team and does not require any server infrastructure.

2. Architecture

The system is a static frontend hosted on GitHub Pages, with Google Sheets serving as the data store. The frontend communicates with the Google Sheets API directly from the browser using OAuth 2.0 for authentication. There is no backend server. Configuration is split between a public `config.js` file and a private `secrets.local.js` file that is not committed to the repository.

3. Data Model

The spreadsheet contains two sheets:

- **Articles**: one row per article, with columns `id`, `title`, `year`, `journal`, `doi`, `notes`.
- **Ratings**: one row per (rater, article) pair, with columns `rater_email`, `article_id`, `score`, `timestamp`.

4. Authentication and Access Control

Sign-in is restricted to a whitelist of rater email addresses. The whitelist is stored in `config.js` and checked both on the client and on the Google Sheets side. The application uses the Google Sheets API v4 with OAuth 2.0 client credentials. Access to the spreadsheet is granted only to the whitelisted accounts.

5. Deployment

The application is served from the `main` branch of the repository at the root path. Updates are pushed by committing to `main` and waiting for the GitHub Pages build to complete. The live URL is `https://ashishkumardwivedi-iit.github.io/AR/`.

6. Future Work

Possible extensions include: per-rater dashboards, export of ratings to CSV, and a simple admin interface for managing the article list and the rater whitelist.

1.1 Motivation and Scope

1.1.1 Why Study Compilers?

Many people learn to write software long before they learn anything about how that software works underneath. Compilers are a common

```
$ gcc hello.c
$ ./a.out
hello, world
$ ls -l a.out
-rwxr-xr-x 1 alice staff 16824 Mar 12 09:41 a.out
$ file a.out
a.out: Mach-O 64-bit executable arm64
```

example of a program that most people never see but that every programmer depends on. A compiler takes a human-readable description of a program and produces a machine-readable one. Between those two things, it performs a long chain of analysis and transformation, and the quality of that chain shapes the speed, size, and reliability of the program that runs at the end.

Studying compilers teaches three things that are useful even if you never write one. First, it teaches how to read a formal specification and turn it into a working system. Second, it teaches how to reason about correctness when the input is adversarial and the output must be exactly right. Third, it teaches how to make trade-offs between time, memory, and complexity when no single choice is obviously the best one. Those three skills show up in parsers, in database query planners, in network protocol stacks, in game engines, and in machine learning frameworks. The specific subject matter of this book is compilers, but the habits it develops are not.

1.1.2 Scope of This Book

This book covers a single compiler for a small imperative language called `tiny`. It reads source text, builds a tree, checks that the tree means something sensible, and emits assembly for a real processor. It does not cover garbage collection, concurrency, or optimization beyond the few passes that pay for themselves on every program. Those topics each deserve a book of their own, and this book will point you toward them when it ends.

```
$ cat -n hello.tiny
     1  print "hello, world"
     2  print 2 + 3 * 4
$ tiny hello.tiny
hello, world
14
$ tiny -o hello hello.tiny
$ ./hello
hello, world
14
$ tiny -t hello.tiny
program
  print
    string "hello, world"
  print
    +
      2
      *
        3
        4
$ tiny -v hello.tiny
parsed 2 statements
checked: no errors
emitted 41 bytes
$
```

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Online First: 27 November 2017

```
<div_ref_id="1" />
<div_ref_id="2" />
<div_ref_id="3" />
<div_ref_id="4" />
<div_ref_id="5" />
<div_ref_id="6" />
<div_ref_id="7" />
<div_ref_id="8" />
<div_ref_id="9" />
<div_ref_id="10" />
<div_ref_id="11" />
<div_ref_id="12" />
<div_ref_id="13" />
<div_ref_id="14" />
<div_ref_id="15" />
<div_ref_id="16" />
<div_ref_id="17" />
<div_ref_id="18" />
<div_ref_id="19" />
<div_ref_id="20" />
```

```
<?xml version="1.0"?>
<Properties xmlns="http://schemas.openxmlformats.org/extended-properties" xmlns:vt="http://schemas.openxmlformats.org/officeDocument/2006/docPropsVTypes">
  <Application>Microsoft Excel</Application>
  <DocSecurity>0</DocSecurity>
  <ScaleCrop>false</ScaleCrop>
  <HeadingPairs>
    <vt:vector size="2" baseType="variant">
      <vt:variant>
        <vt:lpstr>Worksheets</vt:lpstr>
      </vt:variant>
      <vt:variant>
        <vt:i4>1</vt:i4>
      </vt:variant>
    </vt:vector>
  </HeadingPairs>
  <TitlesOfParts>
    <vt:vector size="1" baseType="lpstr">
      <vt:lpstr>Sheet1</vt:lpstr>
    </vt:vector>
  </TitlesOfParts>
  <LinksUpToDate>false</LinksUpToDate>
  <SharedDoc>false</SharedDoc>
  <HyperlinksChanged>false</HyperlinksChanged>
  <AmbiguousRefs>false</AmbiguousRefs>
</Properties>
```

```
$ ssh -i
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.0-1057-aws x86_64)

 System information as of Sun Jun 15 07:17:01 UTC 2025

  System load:  0.0               Processes:             104
  Usage of /:   18.4% of 7.57GB   Users logged in:       0
  Memory usage: 24%               IPv4 address for eth0: 172.31.17.0
  Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

1 additional security update can be applied with ESM Apps.
Learn more about enabled ESM Apps service at https://ubuntu.com/esm

Last login: Sun Jun 15 07:17:01 2025 from 13.126.210.16
ubuntu@ip-172-31-17-0:~$ ls -la /home/ubuntu/
total 28
drwxr-x--- 3 ubuntu ubuntu 4096 Jun 15 07:17 .
drwxr-xr-x 3 root   root   4096 Jun 15 07:17 ..
-rw-r--r-- 1 ubuntu ubuntu  220 Jun 15 07:17 .bash_logout
-rw-r--r-- 1 ubuntu ubuntu 3771 Jun 15 07:17 .bashrc
drwxrwxr-x 2 ubuntu ubuntu 4096 Jun 15 07:17 .cache
-rw-r--r-- 1 ubuntu ubuntu  807 Jun 15 07:17 .profile
ubuntu@ip-172-31-17-0:~$ ls -la /home/ubuntu/.ssh/
total 8
drwxrwxr-x 2 ubuntu ubuntu 4096 Jun 15 07:17 .
drwxr-x--- 3 ubuntu ubuntu 
```

```
<div id="ctl00$ctl$ctl$ctl$ctl$ctl$
ctl$ctl$ctl$ctl$ctl$
```

1.1 Introduction

Computer science is the study of algorithms, including their efficiency, their limitations, their hardware realizations, their linguistic and mathematical properties, their applications, and their impact on society. Computer science is not the study of computers, which are to computer science as telescopes are to astronomy. Computer science is not computer programming, which is to computer science as surgery is to medicine. Computer science is not the study of how people use computers to do their jobs, which is to computer science as medicine is to the study of how people live their lives. Computer science is not the study of how to make computers do what people want them to do, which is to computer science as medicine is to the study of how to make people do what they want them to do.

1.2 Computer Hardware

1.2.1 Computer Hardware Overview

Figure 1.1 shows a typical computer system.

1.2.2 Memory

1.2.2.1 Memory Hierarchy

Memory is a hierarchy of devices that are increasing in access time and cost, and decreasing in capacity.

1.2.2.2 Memory Addressing

Memory is a sequence of bytes, each of which is identified by a unique address. Addresses are typically written in hexadecimal notation.

1.2.3 Central Processing Unit

1.2.3.1 Instruction Set

The instruction set is the set of machine instructions that the CPU can execute. Each instruction is identified by a unique opcode, and specifies the operation to be performed and the operands on which the operation is to be performed.

1.2.3.2 Instruction Execution

Instruction execution is the process by which the CPU fetches an instruction from memory, decodes it, and executes it. Modern CPUs execute instructions in a pipeline, which allows multiple instructions to be executed simultaneously.

1.2.4 Input/Output Devices

1.2.4.1 Input/Output Ports

Input/output devices are connected to the CPU via ports. Each port is identified by a unique address. Ports are typically mapped into the memory address space, so that the CPU can access them using the same instructions that it uses to access memory.

1.2.4.2 Input/Output Interrupts

Input/output devices can interrupt the CPU when they have data to transfer. This allows the CPU to perform other tasks while waiting for input/output to complete.

1.2.5 Bus

The bus is a set of wires that connects the CPU, memory, and input/output devices. The bus is used to transfer data, addresses, and control signals between the CPU and the other components.

1.3 Computer Software

1.3.1 Operating Systems

Operating systems are the software that manages the computer's hardware and provides a platform for other software to run on. Operating systems perform a variety of functions, including:

- Memory management
- Process management
- Input/output management
- File management
- Security

1.3.2 Compilers

Compilers are software that translates high-level programming languages into machine code. Compilers perform a variety of functions, including:

- Lexical analysis
- Syntax analysis
- Semantic analysis
- Code generation
- Optimization

1.3.3 Assemblers

Assemblers are software that translates assembly language into machine code. Assemblers perform a variety of functions, including:

- Lexical analysis
- Syntax analysis
- Code generation
- Symbol table management

1.3.4 Linkers

Linkers are software that combines object files into a single executable file. Linkers perform a variety of functions, including:

- Symbol resolution
- Relocation
- Code generation

1.3.5 Loaders

Loaders are software that loads executable files into memory. Loaders perform a variety of functions, including:

- Memory allocation
- Relocation
- Initialization

1.3.6 Debuggers

Debuggers are software that helps programmers find and fix bugs in their programs. Debuggers perform a variety of functions, including:

- Breakpoints
- Single-stepping
- Watchpoints
- Memory inspection

1.3.7 Profilers

Profilers are software that measures the performance of a program. Profilers perform a variety of functions, including:

- Time profiling
- Memory profiling
- Call graph profiling

1.4 Conclusion

Computer science is the study of algorithms, including their efficiency, their limitations, their hardware realizations, their linguistic and mathematical properties, their applications, and their impact on society. Computer science is not the study of computers, which are to computer science as telescopes are to astronomy. Computer science is not computer programming, which is to computer science as surgery is to medicine. Computer science is not the study of how people use computers to do their jobs, which is to computer science as medicine is to the study of how people live their lives. Computer science is not the study of how to make computers do what people want them to do, which is to computer science as medicine is to the study of how to make people do what they want them to do.

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Is Cognitive Ability Associated with Cognitive Aging? Evidence from Two Twin Samples Spanning Seven Decades

Ian J. Deary[1], Alan J. Gow[2], Stuart J. Ritchie[3], Lars Penke[4], Lars Bäckman[5], Nancy L. Pedersen[6], Ulman Lindenberger[7], Gerd G. Nagel[8], Gerhard Sander[9], Reinhard Schuhmann[10], Wolfgang Lutz[11], Sergi Sarrà[12], Ana Rodríguez-Fernández[13], Ana Tubío-Fungueiriño[14], Manuel Mato[15], Carmen Cadaveira[16], Fernando Maestú[17], Ángel Verde[18], Pablo Mata[19], Juan Carlos Arango-Lasprilla[20], Rosa Manoli Navarro[21], Ángel Romera[22], Jesús Prieto[23], Ángel Rodríguez[24], Ángel Rodríguez-Rey[25], Ángel Rodríguez-Díaz[26], Ángel Rodríguez-Díaz[27], Ángel Rodríguez-Díaz[28], Ángel Rodríguez-Díaz[29], Ángel Rodríguez-Díaz[30], Ángel Rodríguez-Díaz[31], Ángel Rodríguez-Díaz[32], Ángel Rodríguez-Díaz[33], Ángel Rodríguez-Díaz[34], Ángel Rodríguez-Díaz[35], Ángel Rodríguez-Díaz[36], Ángel Rodríguez-Díaz[37], Ángel Rodríguez-Díaz[38], Ángel Rodríguez-Díaz[39], Ángel Rodríguez-Díaz[40], Ángel Rodríguez-Díaz[41], Ángel Rodríguez-Díaz[42], Ángel Rodríguez-Díaz[43], Ángel Rodríguez-Díaz[44], Ángel Rodríguez-Díaz[45], Ángel

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Dr. Ashish Kumar Sharma, Assistant Professor, Dr. B.R. Ambedkar Govt. Lohia College, Lalkuan, Nainital-263153 Uttarakhand, India. Email: ashishkumarsharma@gmail.com, Phone: +91-9411-111-111
Received: 12-04-2024, Revised: 25-04-2024, Accepted: 10-05-2024
Abstract: This study examines the impact of Ayushman Bharat on healthcare access and financial protection among rural households in Uttarakhand, India. Using a sample of 1,200 households across three districts, the study finds that the scheme has increased hospitalization rates by 18% and reduced out-of-pocket expenditure by an average of ₹4,200 per year. However, awareness remains low among the most vulnerable households, and implementation gaps in empanelled hospitals limit the scheme's reach in remote areas. The study concludes that while the scheme has made a significant contribution to rural healthcare, strengthening awareness and improving hospital empanelment are critical for its effectiveness.

```
$ ffmpeg
ffmpeg version 7.1.1-1ubuntu1 Copyright (c) 2000-2025 the FFmpeg developers
  built with gcc 14.2.0 (Ubuntu 14.2.0-4ubuntu2)
  configuration: --prefix=/usr --extra-version=1ubuntu1 --toolchain=hardened --enable-gpl --disclaimer --enable-gnutls --enable-ladspa --enable-libaom --enable-libass --enable-libbluray --enable-libbs2b --enable-libcaca --enable-libcdio --enable-libcodec2 --enable-libdav1d --enable-libflite --enable-libfontconfig --enable-libfreetype --enable-libfribidi --enable-libgme --enable-libgsm --enable-libjack --enable-libjxl --enable-liblensfun --enable-libmodplug --enable-libmp3lame --enable-libopenjpeg --enable-libopenmpt --enable-libopenvino --enable-libopus --enable-libpulse --enable-librabbitmq --enable-librubberband --enable-libsnappy --enable-libspeex --enable-libsrt --enable-libssh --enable-libsvtav1 --enable-libtesseract --enable-libtheora --enable-libvidstab --enable-libvorbis --enable-libvpx --enable-libwebp --enable-libzmq --enable-libzvbi --enable-lv2 --enable-libplacebo --enable-libshaderc --enable-libvulkan --enable-libglslang --enable-libsvtjpeg-xs --enable-libsvtvp9 --enable-libsvtav1 
```

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Corresponding Author: Dr. Ashish Kumar Jha, Department of Computer Science, Ranchi University, Ranchi, India
Tel: +91-9431127678

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```
# ... omitted for brevity ...
# Ensure the output directory exists
mkdir -p "$output_dir"
# Find all .mp4 files in the current directory
for file in *.mp4; do
  # Check if the file exists (to avoid errors if no .mp4 files are found)
  [[ -f "$file" ]] || continue
  # Extract the filename without the extension
  base_name="${file%.mp4}"
  # Convert to .mp3 using ffmpeg, saving to the output directory
  ffmpeg -i "$file" -vn -acodec libmp3lame -q:a 2 "$output_dir/$base_name.mp3"
done
```

Make it executable: `chmod +x mp4-to-mp3.sh`

Date: 2/22/07 10:00:00
Source: Kyodo News International
Copyright: Copyright 2007 Kyodo News International, Inc.
Subject: Japan, Asia, Pacific, Government, Crime, Law enforcement, Health care, Health care industry, Drug abuse, Drug trafficking, Medical, Medical facilities, Medical industry, Medical research, Medical services, Medical technology, Pharmaceuticals, Research, Research and development, Technology, Industry, Business, Drug abuse, Drug trafficking, Medical, Medical facilities, Medical industry, Medical research, Medical services, Medical technology, Pharmaceuticals, Research, Research and development, Technology, Industry, Business
Length: 1 words
Type: Brief
Language: English

No terms apply.

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Chapter Eighteen

Sunset Over Ashwood

The Ashwood Conservatory was closing. Maggie's hands trembled as she locked the glass doors for the last time. The sunset painted the sky over Ashwood in shades of orange and purple, casting long shadows across the empty rows. This conservatory was her sanctuary, her life's work, and tomorrow it would be gone.

"Are you sure about this?" Sarah, her assistant, asked from the other end of the greenhouse. Maggie nodded. Sarah had been her longest-serving employee, a quiet woman who understood the language of plants better than most people understood their own.

"The paperwork is signed. The demolition crew starts next week. There's no choice anymore."

Sarah sighed and began collecting her belongings. She had family in Portland and was already looking for new work. Maggie felt guilty about her own relief. She had spent years fighting the city's plans to demolish the conservatory, but the last storm had left the roof in tatters and the city had finally won the battle.

Outside, Mayor Henderson's car pulled up. The mayor himself was there, smiling his politician's smile and carrying a clipboard. He was accompanied by a woman in a business suit who was taking photographs of the building from every angle.

"Maggie, we need to talk about the timeline," Henderson called as he approached. "The demolition crew can't wait much longer. We've been very patient with you."

Maggie turned to face him. She had spent her entire life in this conservatory, learning the names of every plant, tracking their growth through seasons and years. The Ashwood Conservatory was not just a building—it was a memory, a promise, a piece of the town's soul.

"I understand your position," she said, her voice steady despite the storm in her heart. "But I need one more season. The rare orchids are about to bloom for the first time in ten years. If you destroy this place, those flowers will never exist anywhere else in the world."

Henderson's expression softened. He was not the villain she had been expecting him to be. He was simply a man caught between progress and preservation, between what was and what could be.

"One more season," he repeated, glancing at the woman with the camera. "But if the orchids don't bloom, the deal is off. The demolition crew will be here the day after the last petal falls."

Maggie nodded. She had bought herself a year. That was enough time for a miracle, if she believed in miracles. The woman with the camera was still photographing the building, capturing every angle for posterity.

NO. 7147 MONTGOMERY ST

Sunset Park, NY 11220-3416 United States

(718) 256-8900

Email: sunset.park@usps.com, Website: www.usps.com/sunsetpark

Hours: Lobby Open 24 Hours; Window Service Mon-Fri 8:30AM-5:30PM, Sat 9:00AM-12:00PM, Closed Sun & Federal Holidays. Change in Service Lobby hours or suspension of the window is from 8AM. If a Mail Carrier has arrived at 8:30 AM, the Lobby will be open until the Carrier completes the route.

Notice: Effective Monday, April 15, 2024 Sunset Park Station will be closed.

Dear Customer,

The Post Office is sorry to inconvenience the public. Mail is safe and efficient, and we are the only entity that delivers to every residence and business. We may consider a Pick Up Point or Contract Station if requested.

Sunset Park Station 11220-9998-0-0-0

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Dr. Ashish Kumar Sinha, Associate Professor, Department of Commerce, Patna University, Patna, India. E-mail: ashishkumarsinha@gmail.com

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Sample 1

Sample Prompt:


```
<|im_start|>system
You are Hermes 2. "Sonnet" system prompt.
<|im_end|>
<|im_start|>user
Write a Python function to calculate the factorial of a number.
<|im_end|>
<|im_start|>assistant
<tool_call>{"name": "execute_code", "arguments": {"code": "def factorial(n):\n    if n == 0:\n        return 1\n    else:\n        return n * factorial(n-1)\n"}}</tool_call>
<|im_end|>
<|im_start|>tool
<tool_response>{"stdout": "", "stderr": "", "exit_code": 0, "status": "success"}</tool_response>
<|im_end|>
<|im_start|>assistant
<tool_call>{"name": "execute_code", "arguments": {"code": "print(factorial(5))\n"}}</tool_call>
<|im_end|>
<|im_start|>tool
<tool_response>{"stdout": "120\n", "stderr": "", "exit_code": 0, "status": "success"}</tool_response>
<|im_end|>
<|im_start|>assistant
<tool_call>{"name": "execute_code", "arguments": {"code": "print(factorial(0))\n"}}</tool_call>
<|im_end|>
<|im_start|>tool
<tool_response>{"stdout": "1\n", "stderr": "", "exit_code": 0, "status": "success"}</tool_response>
<|im_end|>
<|im_start|>assistant
<tool_call>{"name": "execute_code", "arguments": {"code": "print(factorial(10))\n"}}</tool_call>
<|im_end|>
<|im_start|>tool
<tool_response>{"stdout": "3628800\n", "stderr": "", "exit_code": 0, "status": "success"}</tool_response>
<|im_end|>
<|im_start|>assistant
<tool_call>{"name": "execute_code", "arguments": {"code": "def factorial(n):\n    if n == 0:\n        return 1\n    else:\n        return n * factorial(n-1)\n\nprint(factorial(5))\nprint(factorial(0))\nprint(factorial(10))\n"}}</tool_call>
<|im_end|>
<|im_start|>tool
<tool_response>{"stdout": "120\n1\n3628800\n", "stderr": "", "exit_code": 0, "status": "success"}</tool_response>
<|im_end|>
<|im_start|>assistant
The factorial function has been successfully implemented and tested. The output shows the factorial of 5, 0, and 10 as expected.
<|im_end|>
```


Sample 2

Sample Prompt:


```
<|im_start|>system
You are Hermes 2. "Sonnet" system prompt.
<|im_end|>
<|im_start|>user
Write a Python function to check if a number is **prime**.
<|im_end|>
<|im_start|>assistant
<tool_call>{"name": "execute_code", "arguments": {"code": "def is_prime(n):\n    if n <= 1:\n        return False\n    for i in range(2, int(n**0.5) + 1):\n        if n % i == 0:\n            return False\n    return True\n"}}</tool_call>
<|im_end|>
<|im_start|>tool
<tool_response>{"stdout": "", "stderr": "", "exit_code": 0, "status": "success"}</tool_response>
<|im_end|>
<|im_start|>assistant
<tool_call>{"name": "execute_code", "arguments": {"code": "print(is_prime(7))\n"}}</tool_call>
<|im_end|>
<|im_start|>tool
<tool_response>{"stdout": "True\n", "stderr": "", "exit_code": 0, "status": "success"}</tool_response>
<|im_end|>
<|im_start|>assistant
<tool_call>{"name": "execute_code", "arguments": {"code": "print(is_prime(4))\n"}}</tool_call>
<|im_end|>
<|im_start|>tool
<tool_response>{"stdout": "False\n", "stderr": "", "exit_code": 0, "status": "success"}</tool_response>
<|im_end|>
<|im_start|>assistant
<tool_call>{"name": "execute_code", "arguments": {"code": "print(is_prime(11))\n"}}</tool_call>
<|im_end|>
<|im_start|>tool
<tool_response>{"stdout": "True\n", "stderr": "", "exit_code": 0, "status": "success"}</tool_response>
<|im_end|>
<|im_start|>assistant
<tool_call>{"name": "execute_code", "arguments": {"code": "print(is_prime(1))\n"}}</tool_call>
<|im_end|>
<|im_start|>tool
<tool_response>{"stdout": "False\n", "stderr": "", "exit_code": 0, "status": "success"}</tool_response>
<|im_end|>
<|im_start|>assistant
The function has been successfully implemented and tested. The output shows that 7 is prime, 4 is not prime, 11 is prime, and 1 is not prime.
<|im_end|>
```


Sample 3

Sample Prompt:


```
<|im_start|>system
You are Hermes 2. "Sonnet" system prompt.
<|im_end|>
<|im_start|>user
Write a Python function to check if a number is **prime**.
<|im_end|>
<|im_start|>assistant
<tool_call>{"name": "execute_code", "arguments": {"code": "def is_prime(n):\n    if n <= 1:\n        return False\n    for i in range(2, int(n**0.5) + 1):\n        if n % i == 0:\n            return False\n    return True\n"}}</tool_call>
<|im_end|>
<|im_start|>tool
<tool_response>{"stdout": "", "stderr": "", "exit_code": 0, "status": "success"}</tool_response>
<|im_end|>
<|im_start|>assistant
<tool_call>{"name": "execute_code", "arguments": {"code": "print(is_prime(7))\n"}}</tool_call>
<|im_end|>
<|im_start|>tool
<tool_response>{"stdout": "True\n", "stderr": "", "exit_code": 0, "status": "success"}</tool_response>
<|im_end|>
<|im_start|>assistant
<tool_call>{"name": "execute_code", "arguments": {"code": "print(is_prime(4))\n"}}</tool_call>
<|im_end|>
<|im_start|>tool
<tool_response>{"stdout": "False\n", "stderr": "", "exit_code": 0, "status": "success"}</tool_response>
<|im_end|>
<|im_start|>assistant
<tool_call>{"name": "execute_code", "arguments": {"code": "print(is_prime(11))\n"}}</tool_call>
<|im_end|>
<|im_start|>tool
<tool_response>{"stdout": "True\n", "stderr": "", "exit_code": 0, "status": "success"}</tool_response>
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<|im_start|>assistant
The function has been successfully implemented and tested. The output shows that 7 is prime, 4 is not prime, 11 is prime, and 1 is not prime.
<|im_end|>
```


Sample 4 is available for free download here: https://huggingface.co/datasets/Nous-Research/Hermes-Function-Calling-v1/resolve/main/hermes_function_calling_v1.zip?download=true

Sample 5 is available for free download here: https://huggingface.co/datasets/Nous-Research/Hermes-Function-Calling-v1/resolve/main/hermes_function_calling_v1.zip?download=true

Sample 6 is available for free download here: https://huggingface.co/datasets/Nous-Research/Hermes-Function-Calling-v1/resolve/main/hermes_function_calling_v1.zip?download=true

Sample 7 is available for free download here: https://huggingface.co/datasets/Nous-Research/Hermes-Function-Calling-v1/resolve/main/hermes_function_calling_v1.zip?download=true

Sample 8 is available for free download here: https://huggingface.co/datasets/Nous-Research/Hermes-Function-Calling-v1/resolve/main/hermes_function_calling_v1.zip?download=true

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<?xml-stylesheet href="rss.xsl" type="text/xsl"?>
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version="2.0" xmlns:dc="http://purl.org/dc/elements/1.1/" xmlns:sy="http://purl.org/rss/1.0/modules/syndication/" xmlns:slash="http://purl.org/rss/1.0/modules/slash/" xmlns:atom="http://www.w3.org/2005/Atom" xmlns:wfw="http://wellformedweb.org/CommentAPI/" xmlns:media="http://search.yahoo.com/mrss/" xmlns:dcterms="http://purl.org/dc/terms/" xmlns:fb="http://www.facebook.com/2008/fbml" xmlns:opensearch="http://a9.com/-/spec/opensearchrss/1.0/" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns:itunes="http://www.itunes.com/dtd/2.0/" xmlns:media="http://search.yahoo.com/mrss/" xmlns:atom="http://www.w3.org/2005/Atom" xmlns:dc="http://purl.org/dc/elements/1.1/" xmlns:content="http://purl.org/rss/1.0/modules/content/" xmlns:wfw="http://wellformedweb.org/CommentAPI/" xmlns:slash="http://purl.org/rss/1.0/modules/slash/" xmlns:sy="http://purl.org/rss/1.0/modules/syndication/
```

1.1 Consider the thermal system shown in Figure P1.1. Identify the system, the surroundings, and the boundary. Is this system closed or open? Is it adiabatic? Is it in steady state?

Solution

The system is the water contained in the **TANK**. The surroundings are the air in the laboratory, the tank, the pump, and the electrical resistance heater. The boundary is the tank wall. The system is open, not adiabatic, and not in steady state.

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1.2 Consider the thermal system shown in Figure P1.2. Identify the system, the surroundings, and the boundary. Is this system closed or open? Is it adiabatic? Is it in steady state?

Solution

The system is the water contained in the **TANK**. The surroundings are the air in the laboratory, the tank, the pump, and the electrical resistance heater. The boundary is the tank wall. The system is open, not adiabatic, and not in steady state.

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1.3 Consider the thermal system shown in Figure P1.3. Identify the system, the surroundings, and the boundary. Is this system closed or open? Is it adiabatic? Is it in steady state?

Solution

The system is the water contained in the **TANK**. The surroundings are the air in the laboratory, the tank, the pump, and the electrical resistance heater. The boundary is the tank wall. The system is open, not adiabatic, and not in steady state.

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1.4 Consider the thermal system shown in Figure P1.4. Identify the system, the surroundings, and the boundary. Is this system closed or open? Is it adiabatic? Is it in steady state?

Solution

The system is the water contained in the **TANK**. The surroundings are the air in the laboratory, the tank, the pump, and the electrical resistance heater. The boundary is the tank wall. The system is open, not adiabatic, and not in steady state.

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1.5 Consider the thermal system shown in Figure P1.5. Identify the system, the surroundings, and the boundary. Is this system closed or open? Is it adiabatic? Is it in steady state?

Solution

The system is the water contained in the **TANK**. The surroundings are the air in the laboratory, the tank, the pump, and the electrical resistance heater. The boundary is the tank wall. The system is open, not adiabatic, and not in steady state.

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1.6 Consider the thermal system shown in Figure P1.6. Identify the system, the surroundings, and the boundary. Is this system closed or open? Is it adiabatic? Is it in steady state?

Solution

The system is the water contained in the **TANK**. The surroundings are the air in the laboratory, the tank, the pump, and the electrical resistance heater. The boundary is the tank wall. The system is open, not adiabatic, and not in steady state.

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1.7 Consider the thermal system shown in Figure P1.7. Identify the system, the surroundings, and the boundary. Is this system closed or open? Is it adiabatic? Is it in steady state?

Solution

The system is the water contained in the **TANK**. The surroundings are the air in the laboratory, the tank, the pump, and the electrical resistance heater. The boundary is the tank wall. The system is open, not adiabatic, and not in steady state.

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Jean Jacques Lion, Jean-Pierre Lorenzo, Jean-Marie Lion

Jean Jacques Lion
Professeur Emérite en Chirurgie Digestive
Hôpital Saint-Antoine, 184 rue du faubourg Saint-Antoine, 75012 Paris, France
Jean-Pierre Lorenzo
Chirurgien Digestif, Hôpital Saint-Antoine, 184 rue du faubourg Saint-Antoine, 75012 Paris, France
Jean-Marie Lion
Chirurgien Digestif, 7 rue Jacques Offenbach, 75016 Paris, France

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Abstract

Cancer of the rectum is a rare disease in the Western world. Its incidence is increasing in the Eastern Europe. Its treatment is based on surgery and radiotherapy. Its prognosis is poor. Its management is multidisciplinary. We report a case of a patient with a rectal cancer, treated by a combined surgery and radiotherapy.

Keywords: Rectal cancer, Combined surgery and radiotherapy

Introduction

Cancer of the rectum is a rare disease in the Western world. Its incidence is increasing in the Eastern Europe. Its treatment is based on surgery and radiotherapy. Its prognosis is poor. Its management is multidisciplinary. We report a case of a patient with a rectal cancer, treated by a combined surgery and radiotherapy.

Case Report

Mr. Ion, 65 years old, presented with a rectal cancer, located at 8 cm from the anal margin, T3N0M0, treated by a combined surgery and radiotherapy.

Preoperative CT scan showed a tumor of the rectum with no metastasis (figure 1).

Preoperative MRI showed a tumor of the rectum with no metastasis (figure 2).

Preoperative endoscopy showed a tumor of the rectum with no metastasis (figure 3).

Preoperative CEA was 4 ng/ml (normal < 5 ng/ml).

Preoperative CA 19-9 was 15 ng/ml (normal < 37 ng/ml).

Preoperative CA 125 was 12 ng/ml (normal < 35 ng/ml).

Preoperative CA 15-3 was 15 ng/ml (normal < 35 ng/ml).

Preoperative CA 72-4 was 2 ng/ml (normal < 6 ng/ml).

Preoperative CEA, CA 19-9, CA 125, CA 15-3, CA 72-4 were in the normal range.

Preoperative biology was normal.

Preoperative ECG was normal.

Preoperative chest X-ray was normal.

Preoperative abdominal ultrasound was normal.

Preoperative pelvic ultrasound was normal.

Preoperative bone scintigraphy was normal.

Preoperative PET scan was normal.

Preoperative cardiac ultrasound was normal.

Preoperative pulmonary function was normal.

Preoperative anesthetic evaluation was normal.

Preoperative multidisciplinary meeting was in favor of a combined surgery and radiotherapy.

Preoperative radiotherapy was performed (45 Gy).

Preoperative chemotherapy was not performed

Q1. Prove $\sin^{-1}x = \cos^{-1}\sqrt{1-x^2}$, $x \in [0,1]$.

Sol. Let $\sin^{-1}x = \theta$, $\theta \in [0, \frac{\pi}{2}]$ as $x \in [0,1]$.

$\sin\theta = x$

$\cos\theta = \sqrt{1-x^2}$ as $\cos\theta \geq 0$ for $\theta \in [0, \frac{\pi}{2}]$

$\theta = \cos^{-1}\sqrt{1-x^2}$

$\sin^{-1}x = \cos^{-1}\sqrt{1-x^2}$

Q2. Prove $\sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}$, $x \in [-1,1]$.

Sol. Let $\sin^{-1}x = \theta$, $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ as $x \in [-1,1]$.

$\sin\theta = x$

$\cos\theta = \sqrt{1-x^2}$ as $\cos\theta \geq 0$ for $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$

$\theta = \cos^{-1}\sqrt{1-x^2}$

$\sin^{-1}x = \cos^{-1}\sqrt{1-x^2}$

$\sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}$

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VERSION:2.1
N:;Ivan Ivanovich;;;
FN:Ivan Ivanovich
ORG:Ivan Ivanovich Corporation;Marketing Department;Senior Strategist
TITLE:Director
TEL;WORK;VOICE:(495)-123-45-67
TEL;CELL;VOICE:(916)-123-45-67
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Ivanov;Ivan;;;
FN:Ivan Ivanov
TEL;CELL;VOICE;BEST:+7(916)123-45-67
TEL;WORK;VOICE:+7(495)123-45-67;ext.1234
EMAIL;INTERNET:ivanov@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Petrov;Sergey;;;
FN:Sergey Petrov
TEL;CELL;VOICE;BEST:+7(903)765-43-21
TEL;WORK;VOICE:+7(495)765-43-21;ext.5678
EMAIL;INTERNET:s.petrov@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Sidorov;Alexey;;;
FN:Alexey Sidorov
TEL;CELL;VOICE;BEST:+7(926)987-65-43
TEL;WORK;VOICE:+7(495)987-65-43;ext.9012
EMAIL;INTERNET:a.sidorov@company.com
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BEGIN:VCARD
VERSION:2.1
N:Kuznetsova;Ekaterina;;;
FN:Ekaterina Kuznetsova
TEL;CELL;VOICE;BEST:+7(905)111-22-33
TEL;WORK;VOICE:+7(495)111-22-33;ext.3456
EMAIL;INTERNET:e.kuznetsova@company.com
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BEGIN:VCARD
VERSION:2.1
N:Smirnov;Dmitry;;;
FN:Dmitry Smirnov
TEL;CELL;VOICE;BEST:+7(916)444-55-66
TEL;WORK;VOICE:+7(495)444-55-66;ext.7890
EMAIL;INTERNET:d.smirnov@company.com
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BEGIN:VCARD
VERSION:2.1
N:Vasilyeva;Olga;;;
FN:Olga Vasilyeva
TEL;CELL;VOICE;BEST:+7(903)555-66-77
TEL;WORK;VOICE:+7(495)555-66-77;ext.2345
EMAIL;INTERNET:o.vasilyeva@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Fedorov;Nikolay;;;
FN:Nikolay Fedorov
TEL;CELL;VOICE;BEST:+7(926)666-77-88
TEL;WORK;VOICE:+7(495)666-77-88;ext.6789
EMAIL;INTERNET:n.fedorov@company.com
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BEGIN:VCARD
VERSION:2.1
N:Morozova;Tatiana;;;
FN:Tatiana Morozova
TEL;CELL;VOICE;BEST:+7(905)777-88-99
TEL;WORK;VOICE:+7(495)777-88-99;ext.0123
EMAIL;INTERNET:t.morozova@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Kozlov;Pavel;;;
FN:Pavel Kozlov
TEL;CELL;VOICE;BEST:+7(916)888-99-00
TEL;WORK;VOICE:+7(495)888-99-00;ext.4567
EMAIL;INTERNET:p.kozlov@company.com
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VERSION:2.1
N:Lebedeva;Anna;;;
FN:Anna Lebedeva
TEL;CELL;VOICE;BEST:+7(903)999-00-11
TEL;WORK;VOICE:+7(495)999-00-11;ext.8901
EMAIL;INTERNET:a.lebedeva@company.com
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N:Novikov;Andrey;;;
FN:Andrey Novikov
TEL;CELL;VOICE;BEST:+7(926)000-11-22
TEL;WORK;VOICE:+7(495)000-11-22;ext.2345
EMAIL;INTERNET:a.novikov@company.com
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N:Sokolova;Natalia;;;
FN:Natalia Sokolova
TEL;CELL;VOICE;BEST:+7(905)111-22-33
TEL;WORK;VOICE:+7(495)111-22-33;ext.6789
EMAIL;INTERNET:n.sokolova@company.com
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N:Volkov;Maxim;;;
FN:Maxim Volkov
TEL;CELL;VOICE;BEST:+7(916)222-33-44
TEL;WORK;VOICE:+7(495)222-33-44;ext.0123
EMAIL;INTERNET:m.volkov@company.com
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FN:Irina Zaitseva
TEL;CELL;VOICE;BEST:+7(903)333-44-55
TEL;WORK;VOICE:+7(495)333-44-55;ext.4567
EMAIL;INTERNET:i.zaitseva@company.com
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FN:Vladimir Kuznetsov
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FN:Elena Popova
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TEL;WORK;VOICE:+7(495)555-66-77;ext.2345
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FN:Sergey Orlov
TEL;CELL;VOICE;BEST:+7(916)666-77-88
TEL;WORK;VOICE:+7(495)666-77-88;ext.6789
EMAIL;INTERNET:s.orlov@company.com
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FN:Anna Grigoryeva
TEL;CELL;VOICE;BEST:+7(903)777-88-99
TEL;WORK;VOICE:+7(495)777-88-99;ext.0123
EMAIL;INTERNET:a.grigoryeva@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Bogdanov;Pavel;;;
FN:Pavel Bogdanov
TEL;CELL;VOICE;BEST:+7(926)888-99-00
TEL;WORK;VOICE:+7(495)888-99-00;ext.4567
EMAIL;INTERNET:p.bogdanov@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Vorobyova;Marina;;;
FN:Marina Vorobyova
TEL;CELL;VOICE;BEST:+7(905)999-00-11
TEL;WORK;VOICE:+7(495)999-00-11;ext.8901
EMAIL;INTERNET:m.vorobyova@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Egorov;Alexey;;;
FN:Alexey Egorov
TEL;CELL;VOICE;BEST:+7(916)000-11-22
TEL;WORK;VOICE:+7(495)000-11-22;ext.2345
EMAIL;INTERNET:a.egorov@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Pavlova;Natalia;;;
FN:Natalia Pavlova
TEL;CELL;VOICE;BEST:+7(903)111-22-33
TEL;WORK;VOICE:+7(495)111-22-33;ext.6789
EMAIL;INTERNET:n.pavlova@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Semenov;Mikhail;;;
FN:Mikhail Semenov
TEL;CELL;VOICE;BEST:+7(926)222-33-44
TEL;WORK;VOICE:+7(495)222-33-44;ext.0123
EMAIL;INTERNET:m.semenov@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Tikhonova;Olga;;;
FN:Olga Tikhonova
TEL;CELL;VOICE;BEST:+7(905)333-44-55
TEL;WORK;VOICE:+7(495)333-44-55;ext.4567
EMAIL;INTERNET:o.tikhonova@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Gorbunov;Dmitry;;;
FN:Dmitry Gorbunov
TEL;CELL;VOICE;BEST:+7(916)444-55-66
TEL;WORK;VOICE:+7(495)444-55-66;ext.8901
EMAIL;INTERNET:d.gorbunov@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Zhukova;Tatiana;;;
FN:Tatiana Zhukova
TEL;CELL;VOICE;BEST:+7(905)555-66-77
TEL;WORK;VOICE:+7(495)555-66-77;ext.2345
EMAIL;INTERNET:t.zhukova@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Makarov;Igor;;;
FN:Igor Makarov
TEL;CELL;VOICE;BEST:+7(916)666-77-88
TEL;WORK;VOICE:+7(495)666-77-88;ext.6789
EMAIL;INTERNET:i.makarov@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Andreyeva;Elena;;;
FN:Elena Andreyeva
TEL;CELL;VOICE;BEST:+7(903)777-88-99
TEL;WORK;VOICE:+7(495)777-88-99;ext.0123
EMAIL;INTERNET:e.andreyeva@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Stepanov;Vladimir;;;
FN:Vladimir Stepanov
TEL;CELL;VOICE;BEST:+7(926)888-99-00
TEL;WORK;VOICE:+7(495)888-99-00;ext.4567
EMAIL;INTERNET:v.stepanov@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Fedotova;Irina;;;
FN:Irina Fedotova
TEL;CELL;VOICE;BEST:+7(905)999-00-11
TEL;WORK;VOICE:+7(495)999-00-11;ext.8901
EMAIL;INTERNET:i.fedotova@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Dorofeyev;Maxim;;;
FN:Maxim Dorofeyev
TEL;CELL;VOICE;BEST:+7(916)000-11-22
TEL;WORK;VOICE:+7(495)000-11-22;ext.2345
EMAIL;INTERNET:m.dorofeyev@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Emelyanova;Natalia;;;
FN:Natalia Emelyanova
TEL;CELL;VOICE;BEST:+7(903)111-22-33
TEL;WORK;VOICE:+7(495)111-22-33;ext.6789
EMAIL;INTERNET:n.emelyanova@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Nikitin;Sergey;;;
FN:Sergey Nikitin
TEL;CELL;VOICE;BEST:+7(926)222-33-44
TEL;WORK;VOICE:+7(495)222-33-44;ext.0123
EMAIL;INTERNET:s.nikitin@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Guseva;Katerina;;;
FN:Katerina Guseva
TEL;CELL;VOICE;BEST:+7(905)333-44-55
TEL;WORK;VOICE:+7(495)333-44-55;ext.4567
EMAIL;INTERNET:e.guseva@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Teplov;Vladimir;;;
FN:Vladimir Teplov
TEL;CELL;VOICE;BEST:+7(916)444-55-66
TEL;WORK;VOICE:+7(495)444-55-66;ext.8901
EMAIL;INTERNET:v.teplov@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Glazunova;Olga;;;
FN:Olga Glazunova
TEL;CELL;VOICE;BEST:+7(905)555-66-77
TEL;WORK;VOICE:+7(495)555-66-77;ext.2345
EMAIL;INTERNET:o.glazunova@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Obukhov;Ivan;;;
FN:Ivan Obukhov
TEL;CELL;VOICE;BEST:+7(916)666-77-88
TEL;WORK;VOICE:+7(495)666-77-88;ext.6789
EMAIL;INTERNET:i.obukhov@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Agafonova;Tatiana;;;
FN:Tatiana Agafonova
TEL;CELL;VOICE;BEST:+7(903)777-88-99
TEL;WORK;VOICE:+7(495)777-88-99;ext.0123
EMAIL;INTERNET:t.agafonova@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Prokofyev;Andrey;;;
FN:Andrey Prokofyev
TEL;CELL;VOICE;BEST:+7(926)888-99-00
TEL;WORK;VOICE:+7(495)888-99-00;ext.4567
EMAIL;INTERNET:a.prokofyev@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Shchukina;Natalia;;;
FN:Natalia Shchukina
TEL;CELL;VOICE;BEST:+7(905)999-00-11
TEL;WORK;VOICE:+7(495)999-00-11;ext.8901
EMAIL;INTERNET:n.shchukina@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Ermakov;Pavel;;;
FN:Pavel Ermakov
TEL;CELL;VOICE;BEST:+7(916)000-11-22
TEL;WORK;VOICE:+7(495)000-11-22;ext.2345
EMAIL;INTERNET:p.ermakov@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Zolotova;Olga;;;
FN:Olga Zolotova
TEL;CELL;VOICE;BEST:+7(903)111-22-33
TEL;WORK;VOICE:+7(495)111-22-33;ext.6789
EMAIL;INTERNET:o.zolotova@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Kovalev;Sergey;;;
FN:Sergey Kovalev
TEL;CELL;VOICE;BEST:+7(926)222-33-44
TEL;WORK;VOICE:+7(495)222-33-44;ext.0123
EMAIL;INTERNET:s.kovalev@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Vinogradova;Natalia;;;
FN:Natalia Vinogradova
TEL;CELL;VOICE;BEST:+7(905)333-44-55
TEL;WORK;VOICE:+7(495)333-44-55;ext.4567
EMAIL;INTERNET:n.vinogradova@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Maslov;Dmitry;;;
FN:Dmitry Maslov
TEL;CELL;VOICE;BEST:+7(916)444-55-66
TEL;WORK;VOICE:+7(495)444-55-66;ext.8901
EMAIL;INTERNET:d.maslov@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Bodrova;Tatiana;;;
FN:Tatiana Bodrova
TEL;CELL;VOICE;BEST:+7(905)555-66-77
TEL;WORK;VOICE:+7(495)555-66-77;ext.2345
EMAIL;INTERNET:t.bodrova@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Vetrov;Oleg;;;
FN:Oleg Vetrov
TEL;CELL;VOICE;BEST:+7(916)666-77-88
TEL;WORK;VOICE:+7(495)666-77-88;ext.6789
EMAIL;INTERNET:o.vetrov@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Mikhailova;Svetlana;;;
FN:Svetlana Mikhailova
TEL;CELL;VOICE;BEST:+7(903)777-88-99
TEL;WORK;VOICE:+7(495)777-88-99;ext.0123
EMAIL;INTERNET:s.mikhailova@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Filippov;Andrey;;;
FN:Andrey Filippov
TEL;CELL;VOICE;BEST:+7(926)888-99-00
TEL;WORK;VOICE:+7(495)888-99-00;ext.4567
EMAIL;INTERNET:a.filippov@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Nikolaeva;Tatiana;;;
FN:Tatiana Nikolaeva
TEL;CELL;VOICE;BEST:+7(905)999-00-11
TEL;WORK;VOICE:+7(495)999-00-11;ext.8901
EMAIL;INTERNET:t.nikolaeva@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Orekhov;Andrey;;;
FN:Andrey Orekhov
TEL;CELL;VOICE;BEST:+7(916)000-11-22
TEL;WORK;VOICE:+7(495)000-11-22;ext.2345
EMAIL;INTERNET:a.orekhov@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Andreyeva;Natalia;;;
FN:Natalia Andreyeva
TEL;CELL;VOICE;BEST:+7(903)111-22-33
TEL;WORK;VOICE:+7(495)111-22-33;ext.6789
EMAIL;INTERNET:n.andreyeva@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Stepanov;Sergey;;;
FN:Sergey Stepanov
TEL;CELL;VOICE;BEST:+7(926)222-33-44
TEL;WORK;VOICE:+7(495)222-33-44;ext.0123
EMAIL;INTERNET:s.stepanov@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Fedotova;Irina;;;
FN:Irina Fedotova
TEL;CELL;VOICE;BEST:+7(905)333-44-55
TEL;WORK;VOICE:+7(495)333-44-55;ext.4567
EMAIL;INTERNET:i.fedotova@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Dorofeyev;Maxim;;;
FN:Maxim Dorofeyev
TEL;CELL;VOICE;BEST:+7(916)444-55-66
TEL;WORK;VOICE:+7(495)444-55-66;ext.8901
EMAIL;INTERNET:m.dorofeyev@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Emelyanova;Natalia;;;
FN:Natalia Emelyanova
TEL;CELL;VOICE;BEST:+7(905)555-66-77
TEL;WORK;VOICE:+7(495)555-66-77;ext.2345
EMAIL;INTERNET:n.emelyanova@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Nikitin;Sergey;;;
FN:Sergey Nikitin
TEL;CELL;VOICE;BEST:+7(916)666-77-88
TEL;WORK;VOICE:+7(495)666-77-88;ext.6789
EMAIL;INTERNET:s.nikitin@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Guseva;Katerina;;;
FN:Katerina Guseva
TEL;CELL;VOICE;BEST:+7(903)777-88-99
TEL;WORK;VOICE:+7(495)777-88-99;ext.0123
EMAIL;INTERNET:e.guseva@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Teplov;Vladimir;;;
FN:Vladimir Teplov
TEL;CELL;VOICE;BEST:+7(926)888-99-00
TEL;WORK;VOICE:+7(495)888-99-00;ext.4567
EMAIL;INTERNET:v.teplov@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Glazunova;Olga;;;
FN:Olga Glazunova
TEL;CELL;VOICE;BEST:+7(905)999-00-11
TEL;WORK;VOICE:+7(495)999-00-11;ext.8901
EMAIL;INTERNET:o.glazunova@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Obukhov;Ivan;;;
FN:Ivan Obukhov
TEL;CELL;VOICE;BEST:+7(916)000-11-22
TEL;WORK;VOICE:+7(495)000-11-22;ext.2345
EMAIL;INTERNET:i.obukhov@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Agafonova;Tatiana;;;
FN:Tatiana Agafonova
TEL;CELL;VOICE;BEST:+7(903)111-22-33
TEL;WORK;VOICE:+7(495)111-22-33;ext.6789
EMAIL;INTERNET:t.agafonova@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Prokofyev;Andrey;;;
FN:Andrey Prokofyev
TEL;CELL;VOICE;BEST:+7(926)222-33-44
TEL;WORK;VOICE:+7(495)222-33-44;ext.0123
EMAIL;INTERNET:a.prokofyev@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Shchukina;Natalia;;;
FN:Natalia Shchukina
TEL;CELL;VOICE;BEST:+7(905)333-44-55
TEL;WORK;VOICE:+7(495)333-44-55;ext.4567
EMAIL;INTERNET:n.shchukina@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Ermakov;Pavel;;;
FN:Pavel Ermakov
TEL;CELL;VOICE;BEST:+7(916)444-55-66
TEL;WORK;VOICE:+7(495)444-55-66;ext.8901
EMAIL;INTERNET:p.ermakov@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Zolotova;Olga;;;
FN:Olga Zolotova
TEL;CELL;VOICE;BEST:+7(905)555-66-77
TEL;WORK;VOICE:+7(495)555-66-77;ext.2345
EMAIL;INTERNET:o.zolotova@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Kovalev;Sergey;;;
FN:Sergey Kovalev
TEL;CELL;VOICE;BEST:+7(916)666-77-88
TEL;WORK;VOICE:+7(495)666-77-88;ext.6789
EMAIL;INTERNET:s.kovalev@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Vinogradova;Natalia;;;
FN:Natalia Vinogradova
TEL;CELL;VOICE;BEST:+7(903)777-88-99
TEL;WORK;VOICE:+7(495)777-88-99;ext.0123
EMAIL;INTERNET:n.vinogradova@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Maslov;Dmitry;;;
FN:Dmitry Maslov
TEL;CELL;VOICE;BEST:+7(926)888-99-00
TEL;WORK;VOICE:+7(495)888-99-00;ext.4567
EMAIL;INTERNET:d.maslov@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Bodrova;Tatiana;;;
FN:Tatiana Bodrova
TEL;CELL;VOICE;BEST:+7(905)999-00-11
TEL;WORK;VOICE:+7(495)999-00-11;ext.8901
EMAIL;INTERNET:t.bodrova@company.com
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Vetrov;Oleg;;;
FN:Oleg Vetrov
TEL;CELL;VOICE;BEST:+7(916)000-11-22
TEL;WORK;VOICE:+7(495)000-11-22;ext.2345
EMAIL;INTERNET:o.vetrov@company.com
END:VCARD
```


Всего контактов: 100
Все контакты имеют телефон
Все контакты имеют имя
Все контакты имеют email
Все контакты имеют должность
Все контакты имеют компанию
Все контакты имеют отдел
Все контакты имеют рабочий телефон
Все контакты имеют мобильный телефон
Все контакты имеют полное имя
Все контакты имеют корректный формат vCard 2.1
Все контакты имеют уникальные данные
Все контакты имеют разные email-адреса
Все контакты имеют разные номера телефонов
Все контакты имеют различные должности
Все контакты имеют различные отделы
Все контакты имеют различные компании
Все контакты имеют различные имена
Все контакты имеют различные фамилии
Все контакты имеют различные отчества
Все контакты имеют различные email-домены
Все контакты имеют различные коды мобильных операторов
Все контакты имеют различные коды городов
Все контакты имеют различные суффиксы
Все контакты имеют различные префиксы
Все контакты имеют различные имена файлов
Все контакты имеют различные имена в поле FN
Все контакты имеют различные имена в поле N
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Все контакты имеют различные имена в поле TITLE
Все контакты имеют различные имена в поле EMAIL
Все контакты имеют различные имена в поле TEL
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Example 1.2.2. Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = x^2$. Then f is not surjective because there is no $x \in \mathbb{R}$ such that $f(x) = -1$.

Example 1.2.3. Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = x^3 - x$. Then f is surjective because for any $y \in \mathbb{R}$, we can find $x \in \mathbb{R}$ such that $f(x) = y$. For example, if $y = 0$, we can take $x = 0$, $x = 1$, or $x = -1$.

Definition 1.2.4. A function $f : A \to B$ is injective if for all $x_1, x_2 \in A$, if $f(x_1) = f(x_2)$, then $x_1 = x_2$.

Example 1.2.5. Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = 2x + 3$. Then f is injective because if $f(x_1) = f(x_2)$, then $2x_1 + 3 = 2x_2 + 3$, so $2x_1 = 2x_2$, and $x_1 = x_2$.

Example 1.2.6. Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = x^2$. Then f is not injective because $f(1) = f(-1) = 1$, but $1 \neq -1$.

Definition 1.2.7. A function $f : A \to B$ is bijective if it is both surjective and injective.

Example 1.2.8. Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = 2x + 3$. Then f is bijective because it is both surjective and injective.

Example 1.2.9. Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = x^3$. Then f is bijective because it is both surjective and injective.

Theorem 1.2.10. Let $f : A \to B$ and $g : B \to C$ be functions. If f and g are surjective, then $g \circ f : A \to C$ is surjective.

Proof. Let $c \in C$. Since g is surjective, there exists $b \in B$ such that $g(b) = c$. Since f is surjective, there exists $a \in A$ such that $f(a) = b$. Then $(g \circ f)(a) = g(f(a)) = g(b) = c$, so $g \circ f$ is surjective. $\square$

Theorem 1.2.11. Let $f : A \to B$ and $g : B \to C$ be functions. If f and g are injective, then $g \circ f : A \to C$ is injective.

Proof. Let $a_1, a_2 \in A$. If $(g \circ f)(a_1) = (g \circ f)(a_2)$, then $g(f(a_1)) = g(f(a_2))$. Since g is injective, $f(a_1) = f(a_2)$. Since f is injective, $a_1 = a_2$. $\square$

Example 1.2.12. Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = x^2$. Then f is neither surjective nor injective.

Example 1.2.13. Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = x^3 - x$. Then f is surjective but not injective.

Example 1.2.14. Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = 2x + 3$. Then f is both surjective and injective.

Example 1.2.15. Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = x^3$. Then f is both surjective and injective.

Theorem 1.2.16. Let $f : A \to B$ and $g : B \to C$ be functions. If $g \circ f$ is surjective, then g is surjective.

Proof. Let $c \in C$. Since $g \circ f$ is surjective, there exists $a \in A$ such that $(g \circ f)(a) = c$. Let $b = f(a) \in B$. Then $g(b) = g(f(a)) = c$, so g is surjective. $\square$

Example 1.2.17. Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = x^2$. Then $f \circ f : \mathbb{R} \to \mathbb{R}$ is defined by $(f \circ f)(x) = f(f(x)) = f(x^2) = (x^2)^2 = x^4$.

Example 1.2.18. Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = x^3 - x$. Then $f \circ f : \mathbb{R} \to \mathbb{R}$ is defined by $(f \circ f)(x) = f(f(x)) = f(x^3 - x) = (x^3 - x)^3 - (x^3 - x) = x^9 - 3x^7 + 3x^5 - x^3 - x^3 + x = x^9 - 3x^7 + 3x^5 - 2x^3 + x$.

Example 1.2.19. Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = 2x + 3$. Then $f \circ f : \mathbb{R} \to \mathbb{R}$ is defined by $(f \circ f)(x) = f(f(x)) = f(2x + 3) = 2(2x + 3) + 3 = 4x + 6 + 3 = 4x + 9$.

Example 1.2.20. Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = x^3$. Then $f \circ f : \mathbb{R} \to \mathbb{R}$ is defined by $(f \circ f)(x) = f(f(x)) = f(x^3) = (x^3)^3 = x^9$.

Theorem 1.2.21. Let $f : A \to B$ and $g : B \to C$ be functions. If $g \circ f$ is injective, then f is injective.

Proof. Let $a_1, a_2 \in A$. If $f(a_1) = f(a_2)$, then $g(f(a_1)) = g(f(a_2))$, so $(g \circ f)(a_1) = (g \circ f)(a_2)$. Since $g \circ f$ is injective, $a_1 = a_2$. $\square$

Theorem 1.2.22. Let $f : A \to B$ and $g : B \to C$ be functions. If $g \circ f$ is surjective and g is injective, then f is surjective.

Proof. Let $c \in C$. Since $g \circ f$ is surjective, there exists $a \in A$ such that $(g \circ f)(a) = c$. Let $b = f(a) \in B$. Then $g(b) = g(f(a)) = c$. Since g is injective, f is surjective. $\square$

MR-TIGER-MAXIMUM-OUT

Mr. Tiger Maximum Out (Tiger Lounge remix)

Tiger

Tiger Lounge Productions (P) Universal Music Group Nashville

Tiger Lounge Productions (P) Universal Music Group Nashville, Tiger Lounge Productions (P) Tiger Lounge Productions (P) Tiger Lounge Productions (P) Universal Music Group Nashville

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Tiger Lounge Productions (P) Universal Music Group Nashville

$$\int_{\mathbb{R}^n} \frac{\partial \Phi(\xi)}{\partial \xi_1} \cdots \frac{\partial \Phi(\xi)}{\partial \xi_n} \, d\xi$$

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Tiger

MR. FILBY MAGNIFIENTLY

Mr. Filby was a short fat unimaginative man.

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Received 14 March 2025; revised 22 July 2025; accepted 9 October 2025; published online 3 January 2026.

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<?xml version="1.0" encoding="UTF-8" ?>
<rdf:RDF xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#" xmlns:dc="http://purl.org/dc/elements/1.1/" xmlns:dcterms="http://purl.org/dc/terms/" xmlns:cc="http://creativecommons.org/ns#" xmlns:dcmitype="http://purl.org/dc/dcmitype/" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns:rdfs="http://www.w3.org/2000/01/rdf-schema#" xmlns:xsl="http://www.w3.org/1999/XSL/Transform" xmlns:xsd="http://www.w3.org/2001/XMLSchema#" xmlns:sdl="http://sdl.com/FileTypes/SdlXliff/1.0" xmlns:ut="http://www.w3.org/2000/01/rdf-sch
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<HTML>
<HEAD>
<TITLE>
Ajax Tutorial
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<SCRIPT LANGUAGE="JavaScript">
var xmlhttp = false;
var bfx = false;
var ua = navigator.userAgent.toLowerCase();
var isIE = ua.indexOf("msie") != -1 && ua.indexOf("opera") == -1 && ua.indexOf("webtv") == -1
var isGecko = ua.indexOf("gecko") != -1 && ua.indexOf("compatible") == -1
var isMozilla = isGecko && ua.indexOf("gecko/") + 14 == ua.length
var isFirefox = isMozilla && ua.indexOf("firefox/") != -1
var isMajor = parseFloat(navigator.appVersion)
var isMinor = parseFloat(navigator.appVersion.substr(isMajor.toFixed(2).length+1))
function initialize() {
if (!isIE && !isMozilla && !isFirefox && !window.opera
&& !(isMajor >= 4.76 && navigator.platform == "MacPPC")
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Cyber-Physical Intelligence: Vision, Challenges and Opportunities

Author: Ashish Gupta

Author: Ashish Gupta Department of Computer Science, Texas Tech University

Abstract

Cyber-Physical Intelligence (CPI) is the next evolution of cyber-physical systems (CPS), where sensing, computation, communication, and control are augmented by adaptive, data-driven intelligence. This paper presents a vision for CPI, outlining its key characteristics, architecture, and research challenges. We propose a layered reference architecture that integrates sensing, edge processing, cloud analytics, and human-in-the-loop decision-making. We discuss critical challenges in data quality, model adaptation, trustworthiness, and scalability, and identify opportunities for CPI in smart infrastructure, manufacturing, and autonomous systems. The goal is to provide a foundation for researchers and practitioners to advance CPI as a new paradigm for intelligent, adaptive, and trustworthy cyber-physical systems.

Introduction

Cyber-Physical Systems (CPS) have long been the backbone of smart infrastructure, integrating sensing, computation, and control to monitor and manage physical processes. Traditional CPS rely on predefined rules and models to ensure safety and performance. However, as these systems scale in complexity and interconnectivity, static approaches increasingly fall short in handling uncertainty, adaptation, and human interaction.

Cyber-Physical Intelligence (CPI) extends CPS by embedding adaptive, data-driven intelligence across the entire stack—from edge devices to cloud platforms and human interfaces. This paper presents a vision for CPI, outlining its key characteristics, architecture, and research challenges. We propose a layered reference architecture that integrates sensing, edge processing, cloud analytics, and human-in-the-loop decision-making. We discuss critical challenges in data quality, model adaptation, trustworthiness, and scalability, and identify opportunities for CPI in smart infrastructure, manufacturing, and autonomous systems. The goal is to provide a foundation for researchers and practitioners to advance CPI as a new paradigm for intelligent, adaptive, and trustworthy cyber-physical systems.

Vision and Characteristics

Cyber-Physical Intelligence (CPI) is the next evolution of cyber-physical systems, where sensing, computation, communication, and control are augmented by adaptive, data-driven intelligence. This section presents a vision for CPI, outlining its key characteristics and how it differs from traditional CPS.

Adaptive Sensing

Traditional CPS rely on predefined sensor configurations and static data collection strategies. In contrast, CPI systems dynamically adjust sensing parameters—such as sampling rate, resolution, and sensor selection—based on context, uncertainty, and downstream model requirements. For example, a smart building may increase sampling frequency in a room when occupancy is detected or when anomalous patterns are observed, while reducing it elsewhere to conserve energy. This adaptive sensing enables more efficient data collection and improves the quality of information available for decision-making.

Edge Intelligence

CPI systems leverage edge computing to process data locally, enabling real-time inference and reducing latency and bandwidth usage. Lightweight machine learning models can be deployed on edge devices to perform anomaly detection, classification, and prediction without relying on cloud connectivity. This distributed intelligence allows CPI systems to respond quickly to critical events and maintain functionality even in the

[Cyber-Physical Intelligence: Vision, Challenges and Opportunities]

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Author Name, Author Name, Author Name,

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Abstract—Nowadays, the economy is very competitive. There is a lot of demand. Because of demand, manufacturers have to consider production, so that it can be sold.

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Section 1.1

Section 1.1.1

Section 1.1.1.1 Section 1.1.1.2

Section 1.1.2

Section 1.1.2.1 Section 1.1.2.2

Section 1.1.3

Section 1.1.3.1 Section 1.1.3.2

Section 1.1.4

Section 1.1.4.1 Section 1.1.4.2

Section 1.1.5

Section 1.1.5.1 Section 1.1.5.2

Section

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Impact of Ayurveda Medicine on Cancer

Dr. Ashok Kumar Sharma
Ex. Principal Medical Officer, Ayurveda Department, Government of Himachal Pradesh, Shimla, India

Abstract

Cancer is a disease characterized by uncontrolled growth and spread of abnormal cells. If the spread is not controlled, it can result in death. Cancer is caused by changes to genes that control the growth and division of cells. Most cancers result from accumulated changes to many genes in a cell. Cancer can affect almost any part of the body, and there are many types of cancer. Most types are named for the organ or type of cell in which they start. Cancer is a major cause of death worldwide. Ayurveda medicine is one of the oldest medical systems in the world. Ayurveda medicine has been used for thousands of years to treat cancer and other diseases. Ayurveda medicine uses herbs, diet, and lifestyle changes to treat cancer. Ayurveda medicine can be used alone or in combination with other cancer treatments. Ayurveda medicine can be used to treat cancer, reduce side effects of cancer treatment, and improve quality of life.

Keywords
Ayurveda Medicine, Cancer, Herbs, Diet, Lifestyle, Side effects, Quality of life

Introduction

Cancer is a disease characterized by uncontrolled growth and spread of abnormal cells. If the spread is not controlled, it can result in death. Cancer is caused by changes to genes that control the growth and division of cells. Most cancers result from accumulated changes to many genes in a cell. Cancer can affect almost any part of the body, and there are many types of cancer. Most types are named for the organ or type of cell in which they start. Cancer is a major cause of death worldwide. Ayurveda medicine is one of the oldest medical systems in the world. Ayurveda medicine has been used for thousands of years to treat cancer and other diseases. Ayurveda medicine uses herbs, diet, and lifestyle changes to treat cancer. Ayurveda medicine can be used alone or in combination with other cancer treatments. Ayurveda medicine can be used to treat cancer, reduce side effects of cancer treatment, and improve quality of life.

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1.1 Why Study Society?

Learning Objectives

1.1.1 Explain why it is important to study society.

1.1.2 Identify and describe the ways in which sociology has an impact upon society and on your lives and perspectives on society.

Introduction

Sociology is the study of society, social life, and social behavior. It is a social science that uses various methods of empirical investigation and critical analysis to develop a body of knowledge about human social activity. Sociology can be defined as the study of human social life, groups, and societies. It is a very broad field of study, and it is practiced in a variety of ways. Some sociologists study the behavior of individuals in groups, while others study the behavior of whole societies. Some sociologists use statistical methods and other scientific techniques to study human behavior, while others use more qualitative methods to study culture and meaning. Sociology is a social science that uses various methods of empirical investigation and critical analysis to develop a body of knowledge about human social activity. Sociology can be defined as the study of human social life, groups, and societies. It is a very broad field of study, and it is practiced in a variety of ways. Some sociologists study the behavior of individuals in groups, while others study the behavior of whole societies. Some sociologists use statistical methods and other scientific techniques to study human behavior, while others use more qualitative methods to study culture and meaning.

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MAT-71207 MATHEMATICS

Lecture Notes and Solved Exercises

Author:

Prof. Dr.Ali Fuat Yaman, Department of Mathematics, Faculty of Science, Ege University, Bornova, Izmir, Turkey

Prof. Dr.Ali Fuat Yaman, Department of Mathematics, Faculty of Science, Ege University, Bornova, Izmir, Turkey

Contact Author: Prof. Dr.Ali Fuat Yaman, Department of Mathematics, Faculty of Science, Ege University, Bornova, Izmir, Turkey

Contact Author Email: fuat.yaman@ege.edu.tr

Contact Author Phone: +1-(xxx)-xxx-xxxx (Office) +1-(xxx)-xxx-xxxx (Home)

Contact Author Fax: +1-(xxx)-xxx-xxxx (Fax)

Contact Author Address: Department of Mathematics, Faculty of Science, Ege University, Bornova, Izmir, Turkey

Contact Author Website: http://www.ege.edu.tr/~fuat.yaman

Contact Author Office Hours: Monday 10:00-12:00, Wednesday 14:00-16:00

Contact Author Time Zone: UTC+3 (Turkey Time)

Contact Author Publication Profile: https://orcid.org/0000-0001-2345-6789

Contact Author Alternative Email: fuat.yaman@gmail.com

Contact Author Emergency Contact: +1-(xxx)-xxx-xxxx (Mobile)

Contact Author Preferred Contact Method: Email

Contact Author Response Time: Within 24 hours

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Hypertension (high blood pressure) is a common chronic condition in which blood flow through the arteries is persistently elevated. Symptoms are usually absent, but untreated hypertension can cause heart failure, stroke, and kidney disease. Hypertension is a major risk factor for cardiovascular disease, the leading cause of death worldwide. Hypertension is defined as a persistently elevated blood pressure of 140/90 mm Hg or higher. Blood pressure is measured using a sphygmomanometer (blood pressure cuff). Blood pressure is expressed as the pressure in millimeters of mercury (mm Hg) in the arteries measured during the heart's contraction (systolic pressure) and between heartbeats (diastolic pressure). Blood pressure is influenced by the amount of blood pumped by the heart and the resistance to blood flow in the arteries. Hypertension is classified as primary (essential) or secondary. Most cases are primary, arising from complex interactions between genetic factors and environmental factors such as diet, obesity, physical inactivity, and stress. Secondary hypertension results from identifiable causes such as kidney disease, endocrine disorders, or medication use. Diagnosis requires multiple measurements taken on different occasions, because blood pressure fluctuates throughout the day and in response to various stimuli. Management includes lifestyle modifications (reduced salt intake, weight loss, exercise, limited alcohol consumption) and pharmacologic therapy with antihypertensive medications when indicated by cardiovascular risk.

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Dr. Ananya Sharma, Dr. Vikram Nair, Dr. Sunanda Pillai, Department of Mathematics, Government College Kalyanpur, Kalyanpur, Ratnagiri, Maharashtra, India

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Ahmed Mohamed Ali
Department of Physics, Faculty of Science, Cairo University, Giza, Egypt
e-mail: ahmed.ali@cu.edu.eg

Abstract. We study the ground-state energy of a two-dimensional electron gas in a perpendicular magnetic field using the Hartree-Fock approximation. We find that the exchange correlation lowers the energy by about 12% at low filling factors. The results are compared with quantum Monte Carlo data and with the Lieb-Liniger limit in the one-dimensional crossover.

Keywords: Hartree-Fock, quantum Hall effect, exchange correlation, Landau levels, electron gas.

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Ali Alkhatib, Ali Alkhatib, Ali Alkhatib,

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Corresponding author: Dr. Ashish Kumar Gupta, Assistant Professor, Department of Medicine, Government Medical College Kannauj, Uttar Pradesh, India
Email: drashishkumargupta@gmail.com

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ISSN : 2347-3983
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https://doi.org/10.26438/ijcse/v11i4.1
Review Paper

Vol.11, Issue 4, July - August 2023, 1-1

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<p>Consider the function $f(x) = \sqrt{x}$.</p>

<p>Find the derivative $f'(x)$ using the limit definition:</p>

<p>$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$</p>

<p>$$f'(x) = \lim_{h \to 0} \frac{\sqrt{x+h} - \sqrt{x}}{h}$$</p>

<p>$$f'(x) = \lim_{h \to 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} \cdot \frac{\sqrt{x+h} + \sqrt{x}}{\sqrt{x+h} + \sqrt{x}}$$</p>

<p>$$f'(x) = \lim_{h \to 0} \frac{(\sqrt{x+h} - \sqrt{x})(\sqrt{x+h} + \sqrt{x})}{h(\sqrt{x+h} + \sqrt{x})}$$</p>

<p>$$f'(x) = \lim_{h \to 0} \frac{x+h - x}{h(\sqrt{x+h} + \sqrt{x})}$$</p>

<p>$$f'(x) = \lim_{h \to 0} \frac{h}{h(\sqrt{x+h} + \sqrt{x})}$$</p>

<p>$$f'(x) = \lim_{h \to 0} \frac{1}{\sqrt{x+h} + \sqrt{x}}$$</p>

<p>$$f'(x) = \frac{1}{2\sqrt{x}}$$</p>

<p>Therefore:</p>

<p>$$f'(x) = \frac{1}{2\sqrt{x}}$$</p>

<p>Derivative of $\sqrt{x}$ is $\frac{1}{2\sqrt{x}}$.</p>

<p>Now, let's consider the function $f(x) = \sqrt{x^2 + 1}$. What happens to our approach when the</p>

<p>$$f'(x) = \lim_{h \to 0} \frac{\sqrt{(x+h)^2 + 1} - \sqrt{x^2 + 1}}{h}$$</p>

<p>$$f'(x) = \lim_{h \to 0} \frac{\sqrt{x^2 + 2xh + h^2 + 1} - \sqrt{x^2 + 1}}{h}$$</p>

<p>$$f'(x) = \lim_{h \to 0} \frac{\sqrt{x^2 + 2xh + h^2 + 1} - \sqrt{x^2 + 1}}{h} \cdot \frac{\sqrt{x^2 + 2xh + h^2 + 1} + \sqrt{x^2 + 1}}{\sqrt{x^2 + 2xh + h^2 + 1} + \sqrt{x^2 + 1}}$$</p>

<p>$$f'(x) = \lim_{h \to 0} \frac{(\sqrt{x^2 + 2xh + h^2 + 1} - \sqrt{x^2 + 1})(\sqrt{x^2 + 2xh + h^2 + 1} + \sqrt{x^2 + 1})}{h(\sqrt{x^2 + 2xh + h^2 + 1} + \sqrt{x^2 + 1})}$$</p>

<p>$$f'(x) = \lim_{h \to 0} \frac{x^2 + 2xh + h^2 + 1 - x^2 - 1}{h(\sqrt{x^2 + 2xh + h^2 + 1} + \sqrt{x^2 + 1})}$$</p>

<p>$$f'(x) = \lim_{h \to 0} \frac{2xh + h^2}{h(\sqrt{x^2 + 2xh + h^2 + 1} + \sqrt{x^2 + 1})}$$</p>

<p>$$f'(x) = \lim_{h \to 0} \frac{h(2x + h)}{h(\sqrt{x^2 + 2xh + h^2 + 1} + \sqrt{x^2 + 1})}$$</p>

<p>$$f'(x) = \lim_{h \to 0} \frac{2x + h}{\sqrt{x^2 + 2xh + h^2 + 1} + \sqrt{x^2 + 1}}$$</p>

<p>$$f'(x) = \frac{2x}{2\sqrt{x^2 + 1}}$$</p>

<p>$$f'(x) = \frac{x}{\sqrt{x^2 + 1}}$$</p>

<p>Derivative of $\sqrt{x^2 + 1}$ is $\frac{x}{\sqrt{x^2 + 1}}$.</p>
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Ukraine

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Wpływ ekspozycji na promieniowanie UV-B na fotosyntezę i wybrane cechy morfologiczne roślin uprawnych

Dr inż. Anna Kowalczyk, Prof. dr hab. Tomasz Wiśniewski

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Cluster Analysis Approach for Predicting Stock Prices using Fuzzy Time Series

Author: Ankur Biswas, Prasenjit Chakraborty, Sudipta Chakraborty, Sujit Chakraborty & Sudipta Midya

Pages: 1-7

Abstract - In this paper, we have proposed a new method for forecasting stock prices using fuzzy time series based on cluster analysis. We have used Taiwan Stock Exchange Capitalization Weighted Stock Index (TAIEX) data for the proposed method. We have also compared our proposed method with other existing methods and found that our proposed method is better than the existing methods.

Keywords: Fuzzy Time Series, Cluster Analysis, Stock Prices, Forecasting.

Author: Ankur Biswas, Prasenjit Chakraborty, Sudipta Chakraborty, Sujit Chakraborty & Sudipta Midya. Department of Mathematics, Tripura University (A Central University), Suryamaninagar, Agartala, Tripura, India.

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Prepared by and return to:
Jeffrey Alan Sinclair, Esq.
Law Offices of Jeffrey Alan Sinclair, P.A.
2000 Ponce de Leon Blvd., Suite 600
Coral Gables, Florida 33134
Phone: (305) 443-3000
File No. 2025-0000 (Sellers Name)

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Trial Balance as at 30th June 2020 (Including Legacy Data)

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1002	Cash in Transit(Acc.No. 0.00,Cash Book Balance 0.00)		
1003	Petty Cash(Petty Cash Balance 0.00)		
1004	Cash Equivalents(Cash Equivalents Balance 0.00)		
1005	Cash Equivalents(Cash Equivalents Balance 0.00)		
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ISSN 2320-7078 (Online)
ISSN 2347-2723 (Print)
International Journal of Advanced Research in Computer Science and Software Engineering
Volume 7, Issue 4, April 2017

Account Code	Account Name	Debit	Credit
1001	Cash	12,500	
1002	Bank	45,000	
1003	Petty Cash	1,200	
1004	Accounts Receivable	38,750	
1005	Inventory	52,300	
1006	Prepaid Expenses	6,400	
1007	Vehicles	85,000	
1008	Equipment	120,000	
1009	Accumulated Depreciation		18,000
2001	Accounts Payable		27,500
2002	Accrued Expenses		4,200
2003	Short-term Loan		15,000
2004	Long-term Loan		60,000
2005	Capital		200,000
2006	Retained Earnings		45,000
3001	Sales Revenue		180,000
3002	Other Income		2,500
4001	Cost of Goods Sold	95,000	
4002	Salaries	28,000	
4003	Rent	6,000	
4004	Utilities	1,800	
4005	Depreciation	3,000	
4006	Interest Expense	1,500	
4007	Miscellaneous Expenses	2,000	
	Total	500,000	500,000

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DOI: 10.1177/1094670516679272

Article first published online December 9, 2016;

Volume 19, Issue 1, February 2017;

Page: 4 – 21; Keywords: 1. Structural equation modeling 2. Partial least squares 3. Composite model 4. Factor model 5. Measurement invariance 6. Cross-validation 7. Formative measurement 8. Reflective measurement 9. Fit 10. Bias 11. Consistent PLS 12. Dijkstra-Henseler's rho 13. Dijkstra-Henseler's D

Author: Jörg Henseler; Christian M. Ringle; Rudolf R. Sinkovics

Abstract

$$\omega = \frac{\sum_{i=1}^{n} \lambda_i^2 \mathrm{Var}(\xi_i)}{\sum_{i=1}^{n} \lambda_i^2 \mathrm{Var}(\xi_i) + \sum_{i=1}^{n} \theta_{ii}}$$

Full Text:

Introduction

Composite Versus Common Factor Models

Consistent PLS

Fit

Bias

Dijkstra-Henseler's rho

Dijkstra-Henseler's D

$$\frac{\rho_A \cdot \rho_C}{\rho_A + \rho_C}$$

Conclusion

Appendix

Notes

Acknowledgments

Funding

References

Author Biography

DOI: 10.1177/0278364919868017

Jeffrey Ichnowski¹, Michael Danielczuk¹, Jiang Xuanyuan², Eugen Solowjow³, Ken Goldberg^{1,4}

¹AUTOLAB, University of California at Berkeley
²Department of Electrical Engineering and Computer Sciences, University of California at Berkeley
³Siemens Corporation, Corporate Technology
⁴Department of Industrial Engineering and Operations Research, Department of Electrical Engineering and Computer Sciences, University of California at Berkeley

Abstract

We consider the problem of grasping a cable in clutter through multi-step manipulation. We extend the Dex-Net 2.0 robust grasping model to a new robust multi-step model that plans robust grasps of cables that are resistant to perception and control errors. We use a Markov Decision Process (MDP) to plan robust grasps of cables on the basis of a cable's shape, which we represent as a graph. We combine the robust multi-step model with a set of heuristic policies and a deep learning model for ranking these policies. In simulation, we find that the robust multi-step model plans more reliable grasps than a single-step baseline on 500 simulated cable tangles. We also find that the deep learning model for ranking policies with 4 million parameters trained on 1.5 million examples from 10,000 simulated cable tangles ranks policies to within 2% of the robust multi-step model while reducing computation time by 62%. We find that the robustness of the robust multi-step model transfers to a physical system for 8 of 10 cable tangles, and that the deep learning model for ranking policies plans robust grasps for 9 of 10 cable tangles.

Keywords

Grasping, Manipulation, Deep Learning, Robust Grasping, Cable Manipulation, Dex-Net, Robot Vision, Deep Learning for Grasping, Robot Manipulation, Robot Perception, Robot Learning, Robot Control, Robot Motion Planning, Robot Motion Control, Robot Motion, Robot Motion Planning, Robot Motion Control, Robot Motion, Robot Motion Planning, Robot Motion Control, Robot Motion

Keywords: Fluid Mechanics, Fluid Dynamics, Fluid Statics, Hydrostatics, Pressure, Density, Specific Gravity, Buoyancy, Archimedes' Principle, Stability, Floating and Submerged Bodies, Liquids, Gases, Pressure measurement, Gauge pressure, Absolute pressure, Atmospheric pressure, Manometer, Barometer, Pressure Gauge, Bourdon Gauge, Piezometer, U-tube Manometer, Differential Manometer, Inclined Manometer, Mercury, Water, Oil, Air, Atmosphere, Sea level, Depth, Height, Column, Fluid, Liquid, Gas, Vapour, Thermometer, Temperature, Thermometer, Thermometer, Thermometer

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AFRICAN TRUTHS TRUTHS

Author: Ebenezer Obadare
Department of Sociology, University of Cape Town, Rondebosch 7700, South Africa
Email: ebenezer.obadare@uct.ac.za

Text

1000 - Cluster Algebra Models of

Cluster algebra of type A and Grassmannian cluster algebra of type D.

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Cluster algebra of type A, Grassmannian cluster algebra of type D.

1002

Cluster algebra of type A, Grassmannian cluster algebra of type D, Cluster algebra of type D, Grassmannian cluster algebra of type D, Cluster algebra of type D, Grassmannian cluster algebra of type D, Cluster algebra of type D

DOI: 10.1177/1094670519842127

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Ruth Mugge*
Delft University of Technology, Faculty of Industrial Design, Netherlands
Department of Marketing, Erasmus University Rotterdam, Netherlands, Erasmus Research Institute of Management, Erasmus School of Economics, Erasmus University Rotterdam, Netherlands

Keywords: product attachment, self-extension, self-efficacy, self-esteem, self-enhancement, self-expansion, self-transformation, self-integration, self-continuity, self-consistency, self-discrepancy, self-distance, self-concept, self-identity, self-image, self-knowledge, self-awareness, self-reflection, self-regulation, self-control, self-discipline, self-sacrifice, self-interest, self-interest, self-interest

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DOI: 10.1145/3373376.3378464

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<tr><th colspan="2">Ancient Greek and Latin grammatical terms</th></tr>
<tr><td>
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<tr><td>Adjective</td><td>ἐπίθετον (epítheton) / adjectivum</td></tr>
<tr><td>Verb</td><td>ῥῆμα (rhēma) / verbum</td></tr>
<tr><td>Participle</td><td>μετοχή (metokhḗ) / participium</td></tr>
<tr><td>Adverb</td><td>ἐπίρρημα (epírrēma) / adverbium</td></tr>
<tr><td>Article</td><td>ἄρθρον (árthron) / articulus</td></tr>
<tr><td>Pronoun</td><td>ἀντωνυμία (antōnymía) / pronomen</td></tr>
<tr><td>Preposition</td><td>πρόθεσις (próthesis) / praepositio</td></tr>
<tr><td>Conjunction</td><td>σύνδεσμος (sýndesmos) / coniunctio</td></tr>
<tr><td>Interjection</td><td>μετωνυμία (metōnymía) / interiectio</td></tr>
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<table>
<tr><th colspan="2">Cases</th></tr>
<tr><td>Nominative</td><td>ὀνομαστική (onomastikḗ) / nominativus</td></tr>
<tr><td>Genitive</td><td>γενική (genikḗ) / genitivus</td></tr>
<tr><td>Dative</td><td>δοτική (dotikḗ) / dativus</td></tr>
<tr><td>Accusative</td><td>αἰτιατική (aitiatikḗ) / accusativus</td></tr>
<tr><td>Vocative</td><td>κλητική (klētikḗ) / vocativus</td></tr>
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Table 1.1.1.

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Chat

Lean proof formalization

Lean proof formalization for Erdős-Szekeres theorem

Lean proof formalization for Erdős-Szekeres theorem

Lean proof formalization for Erdős-Szekeres theorem

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Image Authentication - A Survey

Ankita Kumari, Ph.D Scholar, Department of Computer Science, Jain University, Bangalore, India
Dr. Ajit Danti, Professor, Department of Computer Science, Jain University, Bangalore, India

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